

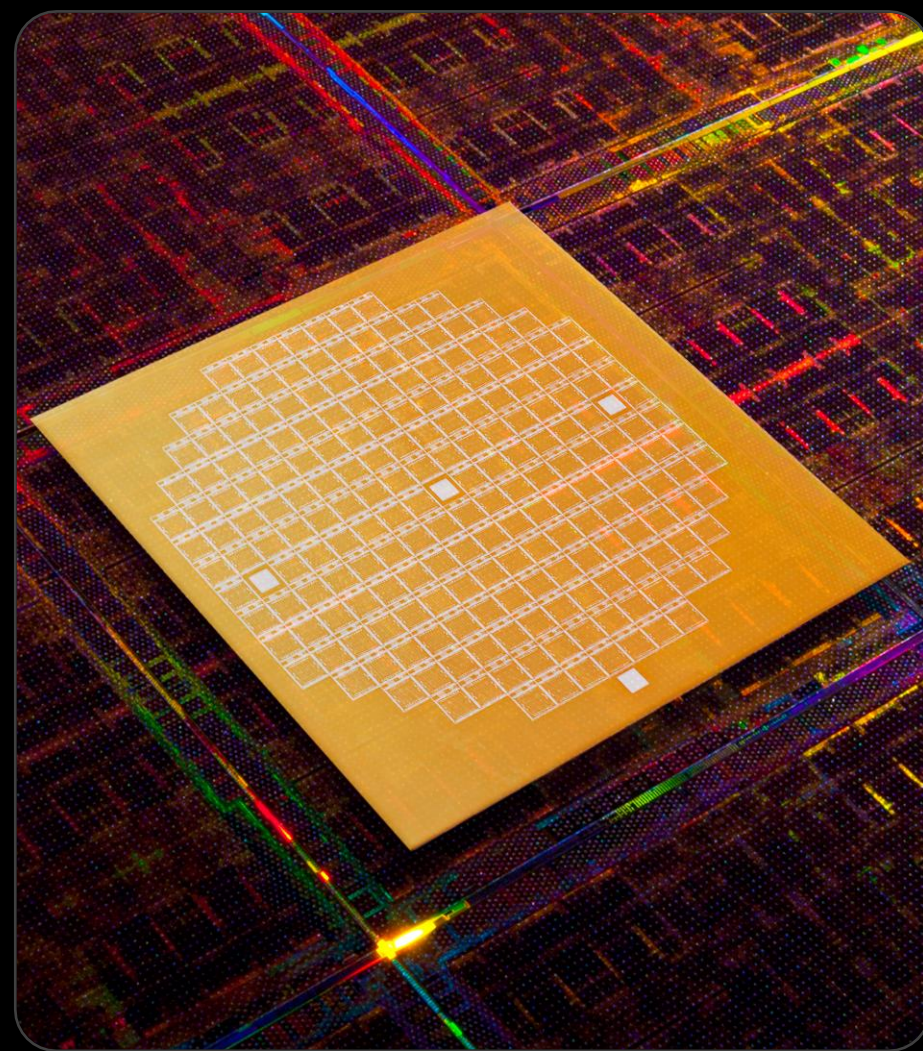


NCHC AI-Powered Physics Bootcamp - NVIDIA Earth-2 Overview

Jay Chen | Solution Architect, Mar 2026

NVIDIA Platform and CUDA-X Accelerates Every Industry

6M+ Developers & 900+ SDKs/Models



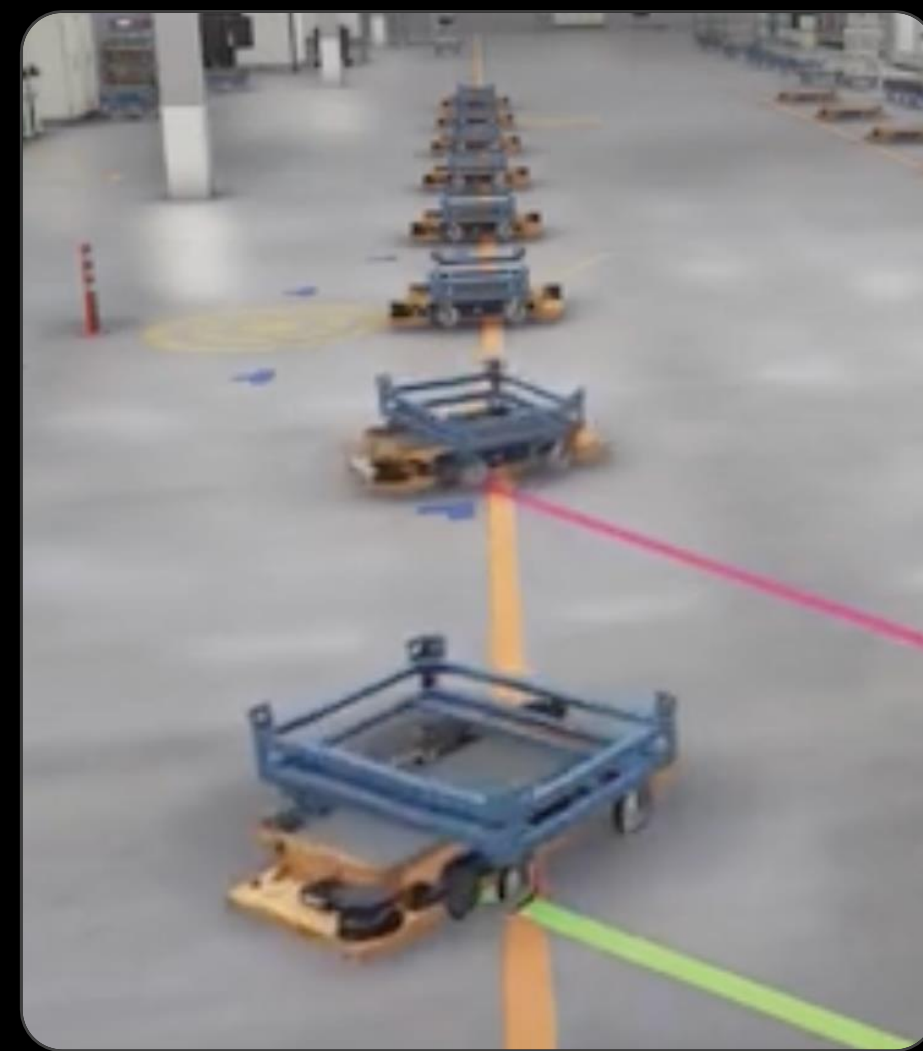
cuLitho

Computational
Lithography



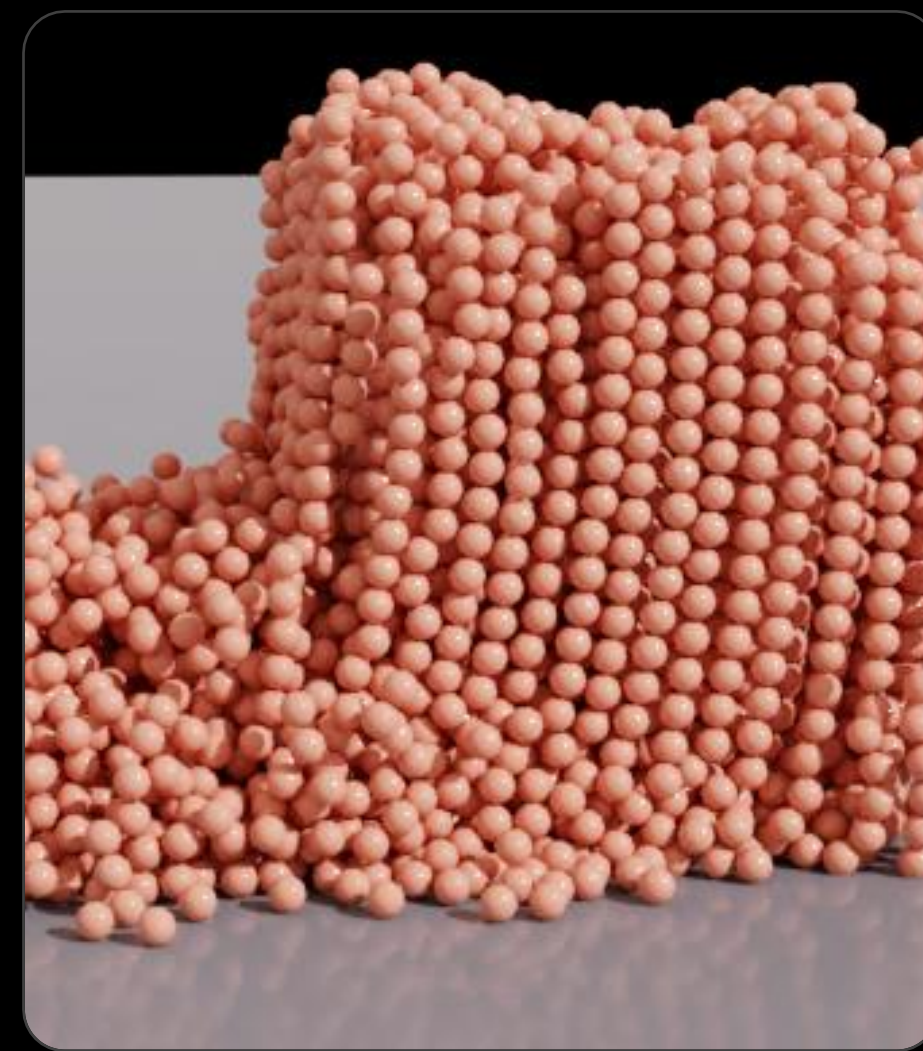
cuDSS

CAE



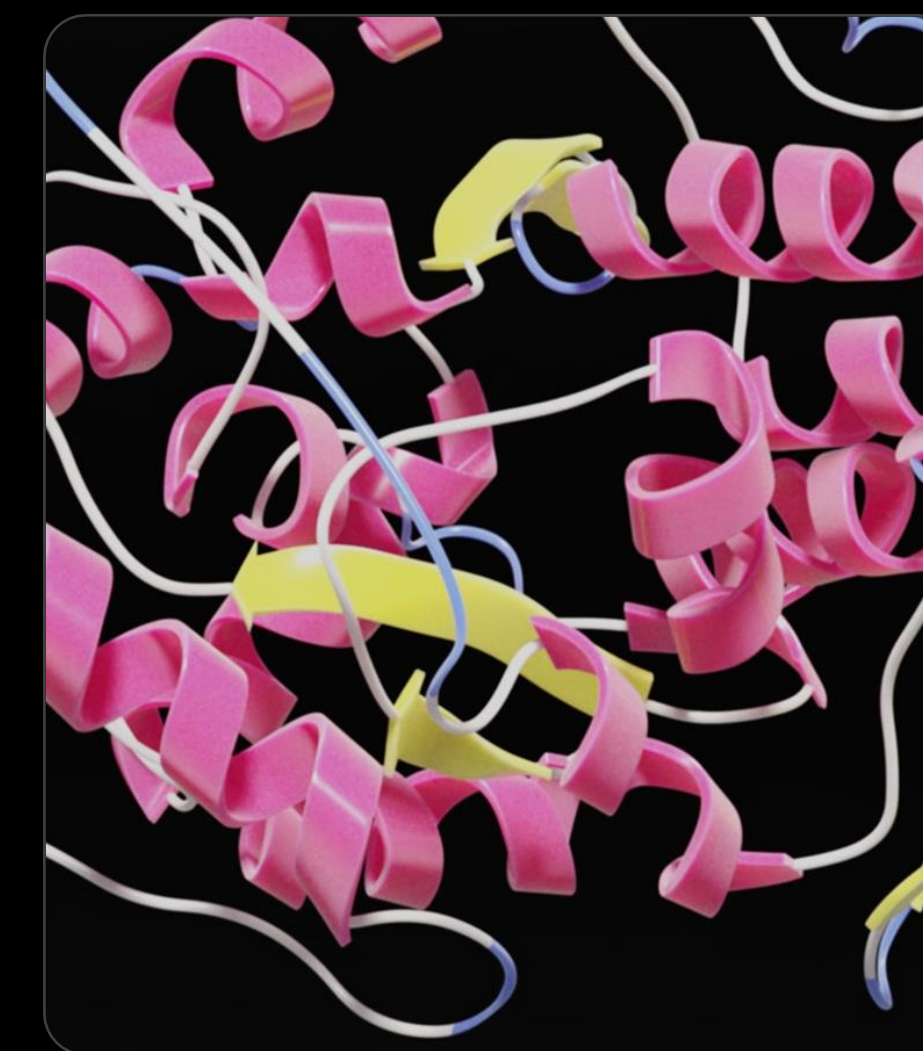
cuOpt

Decision Optimization



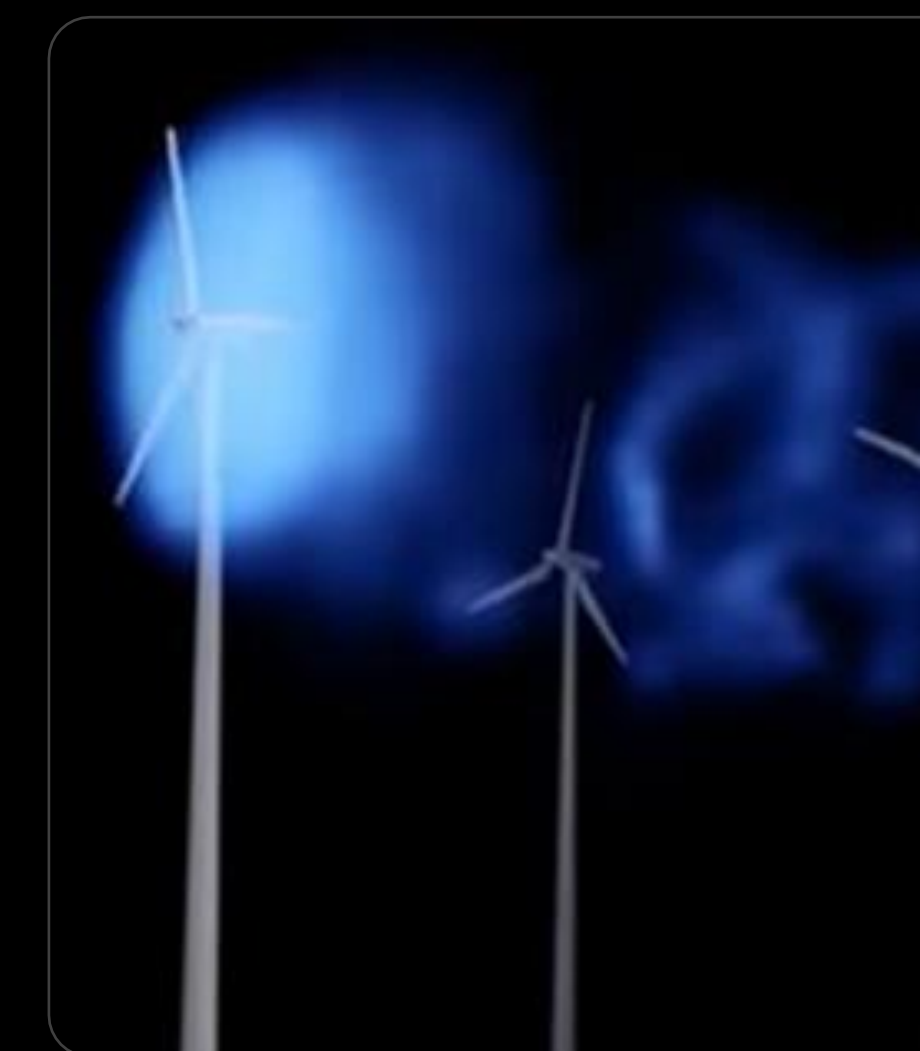
Warp

Physical Simulation



cuDF

Data Processing



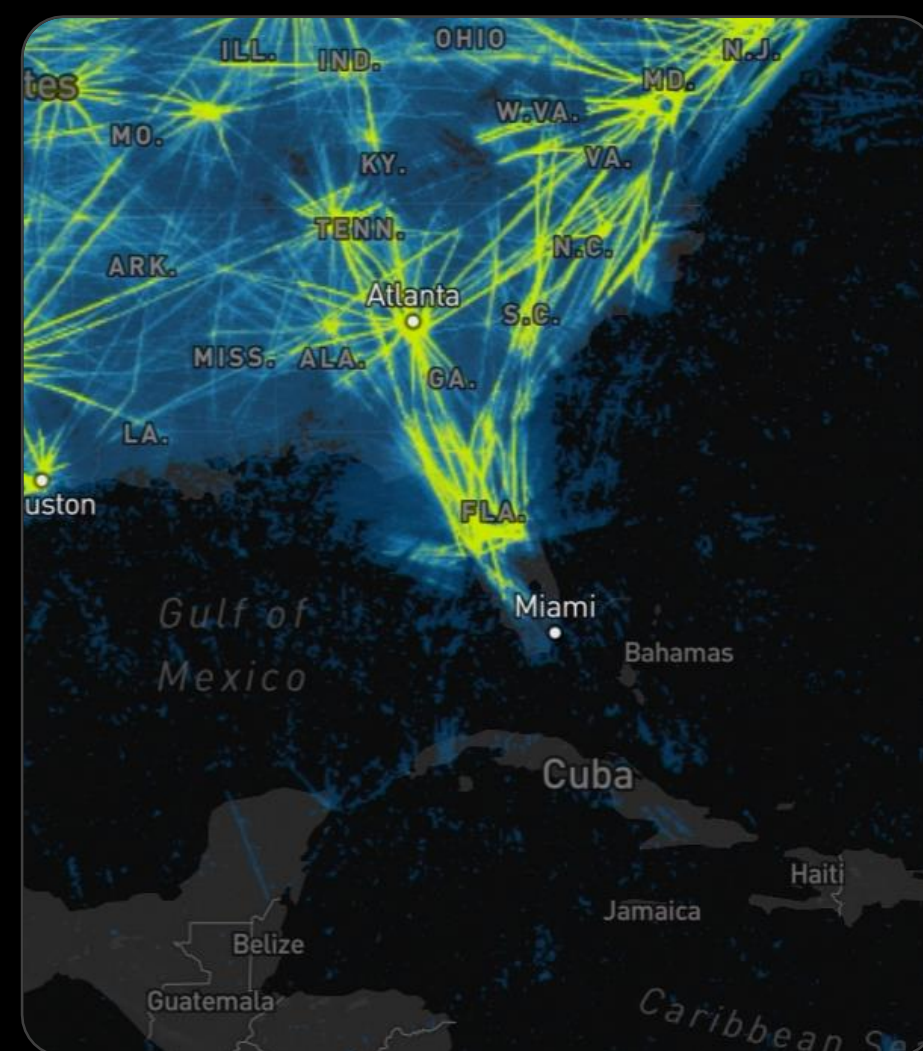
PhysicsNeMo

AI Physics



CUDA-Q

Quantum Computing



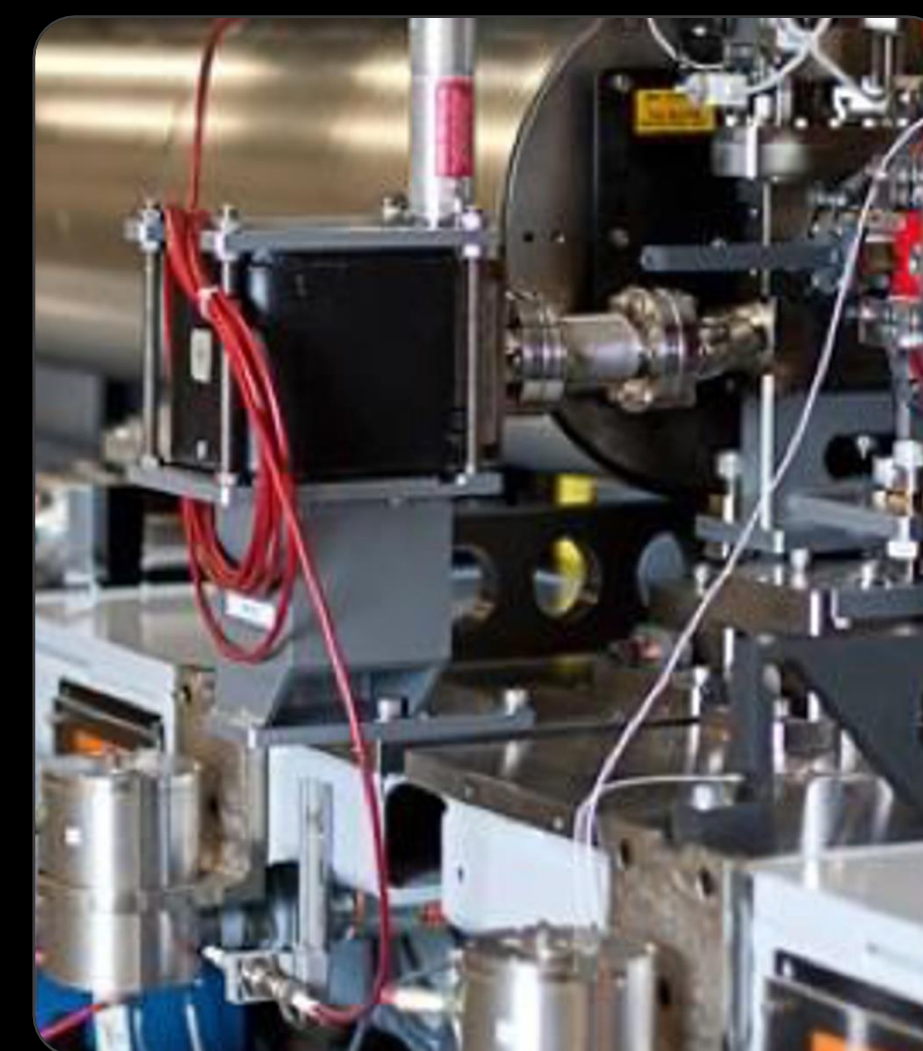
cuEquivariance

Drug & Materials
Discovery



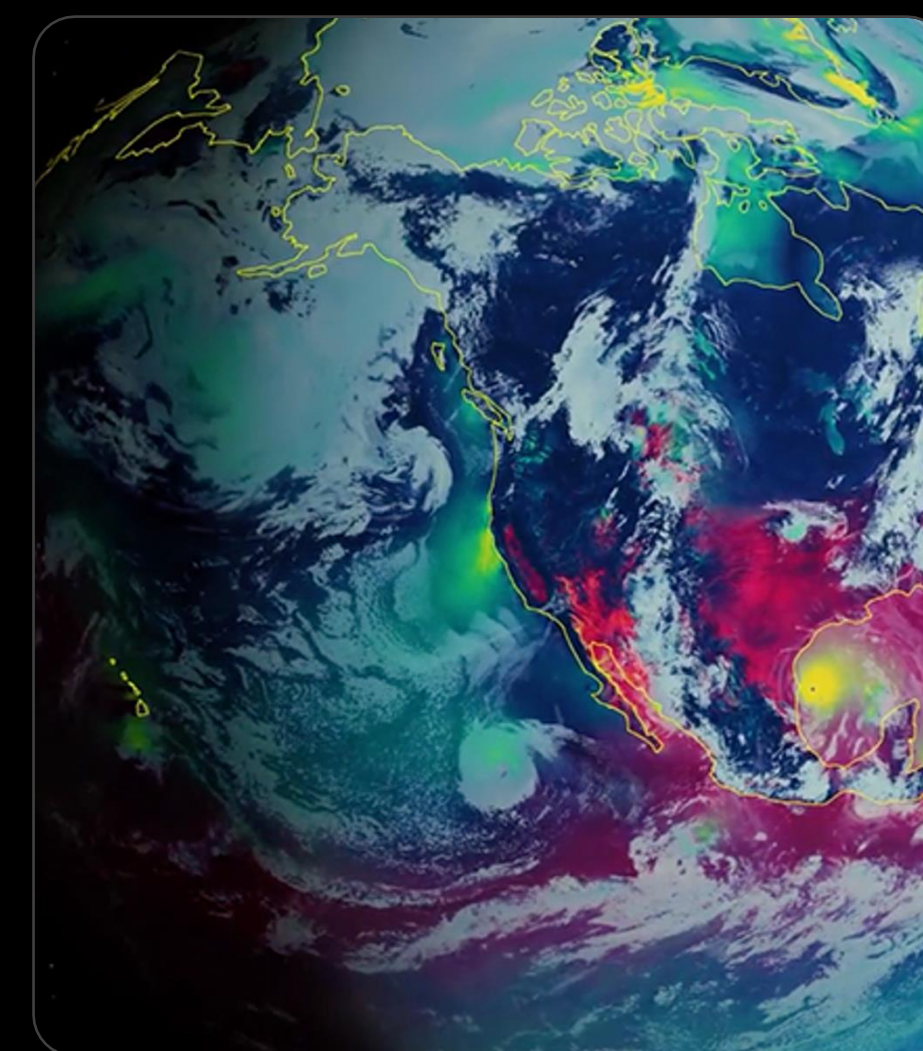
ALCHEMI

AI Materials Science



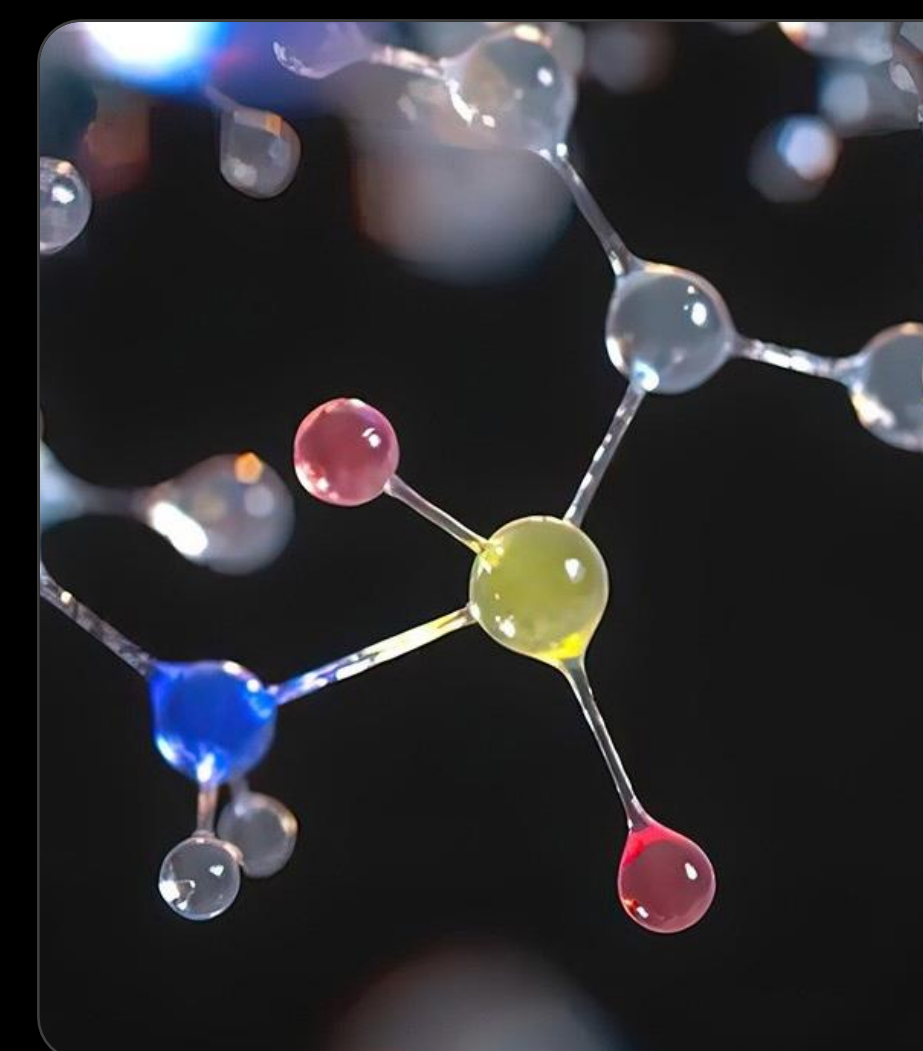
Holoscan

Edge HPC



Earth-2

Weather Analytics



Parabricks

Gene Sequencing



cuPyNumeric

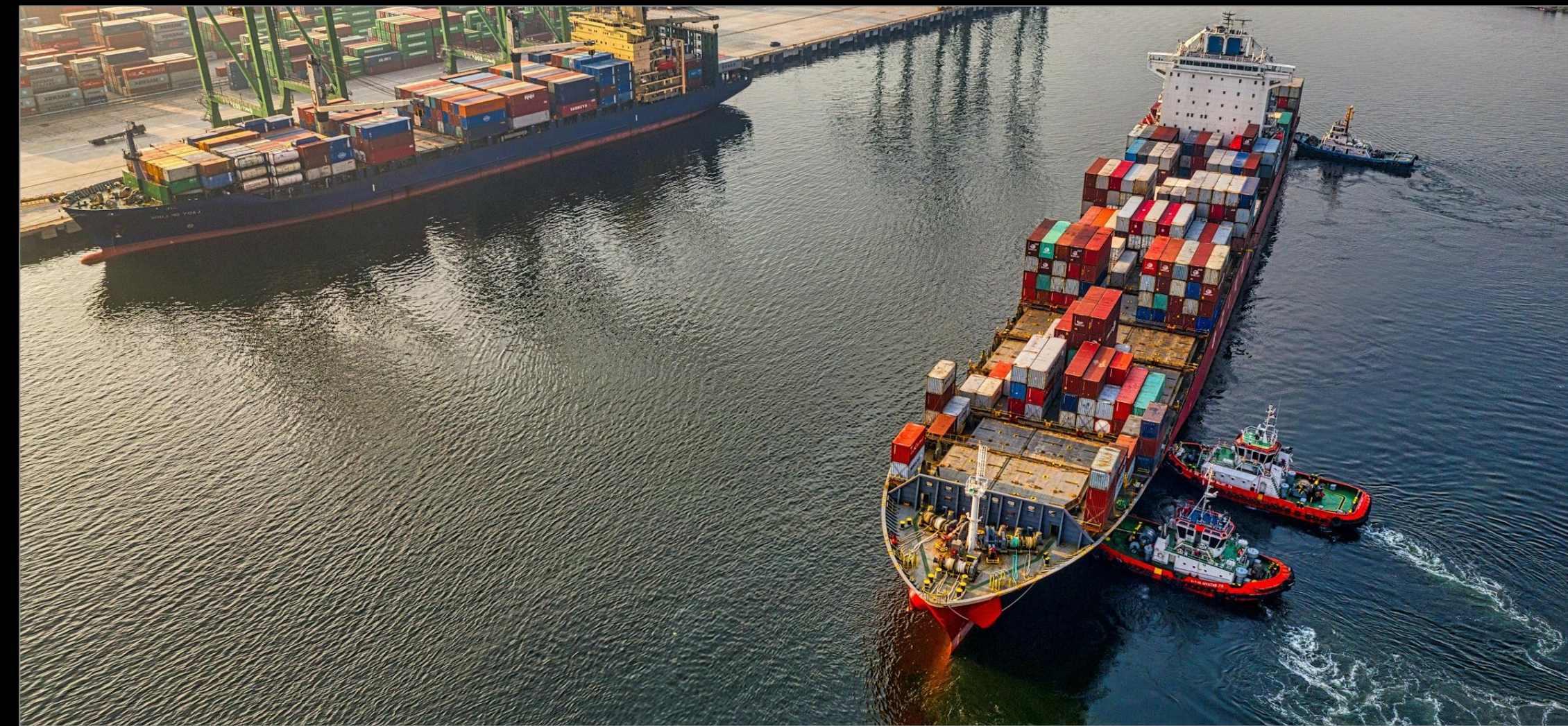
Numerical Computing

Weather Prediction is Integral to Modern Society

Simulations and forecasts drive planning and decision-making



Agriculture & Forestry



Transport & Logistics



Energy



Risk management



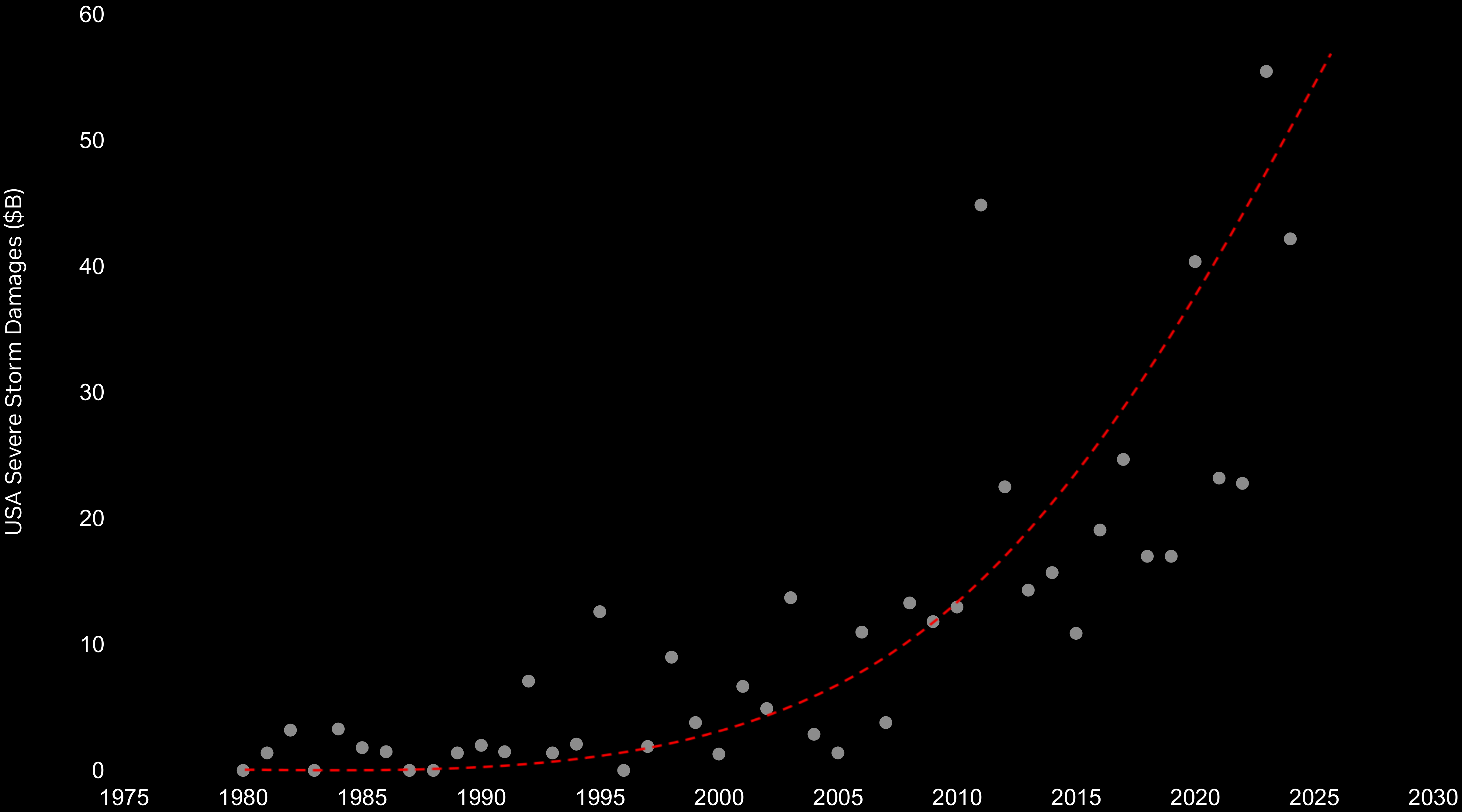
Land development



Recreation

Weather And Climate Impact Every Human Being And Every Business

25X Increase in Disaster Costs in Last 40 Years and Severe Storms Are Increasing Every Year



Climate and Weather Simulation is a Massive Computation Challenge

AI models are skilled enough to drive more accessible and efficient forecasting

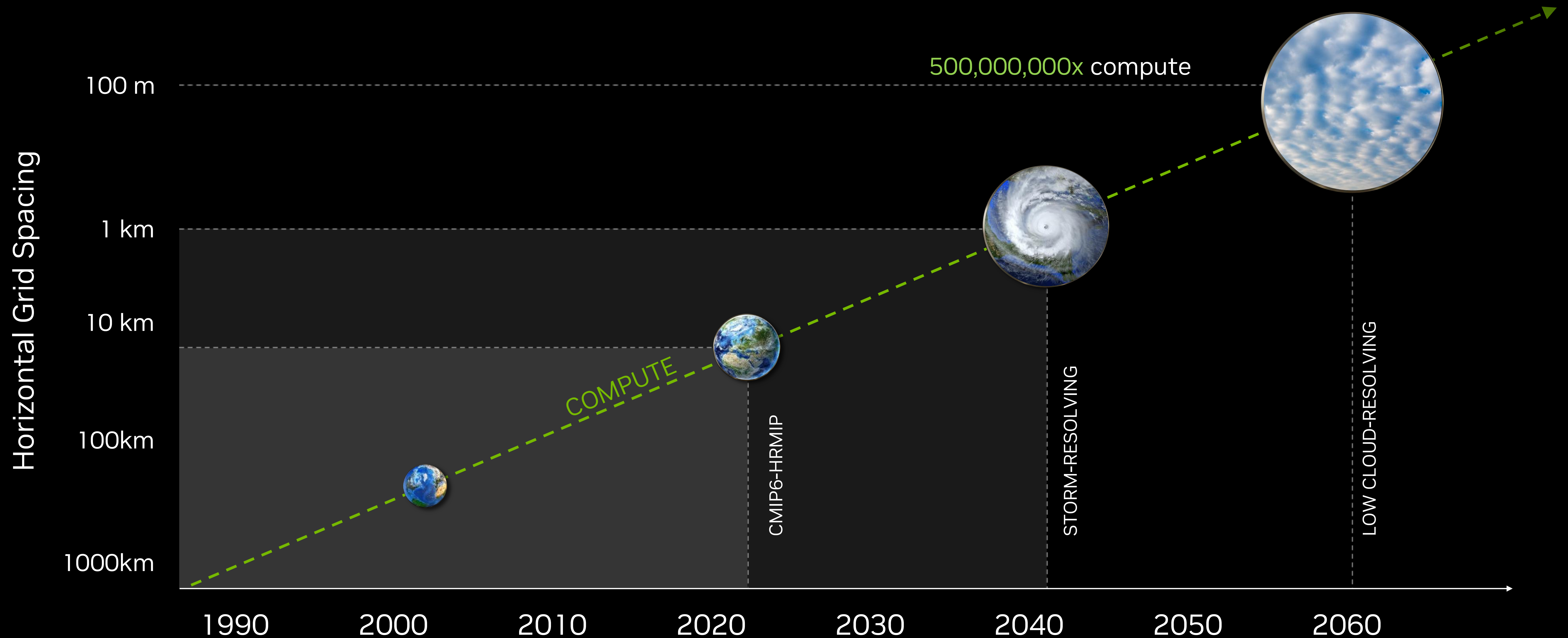
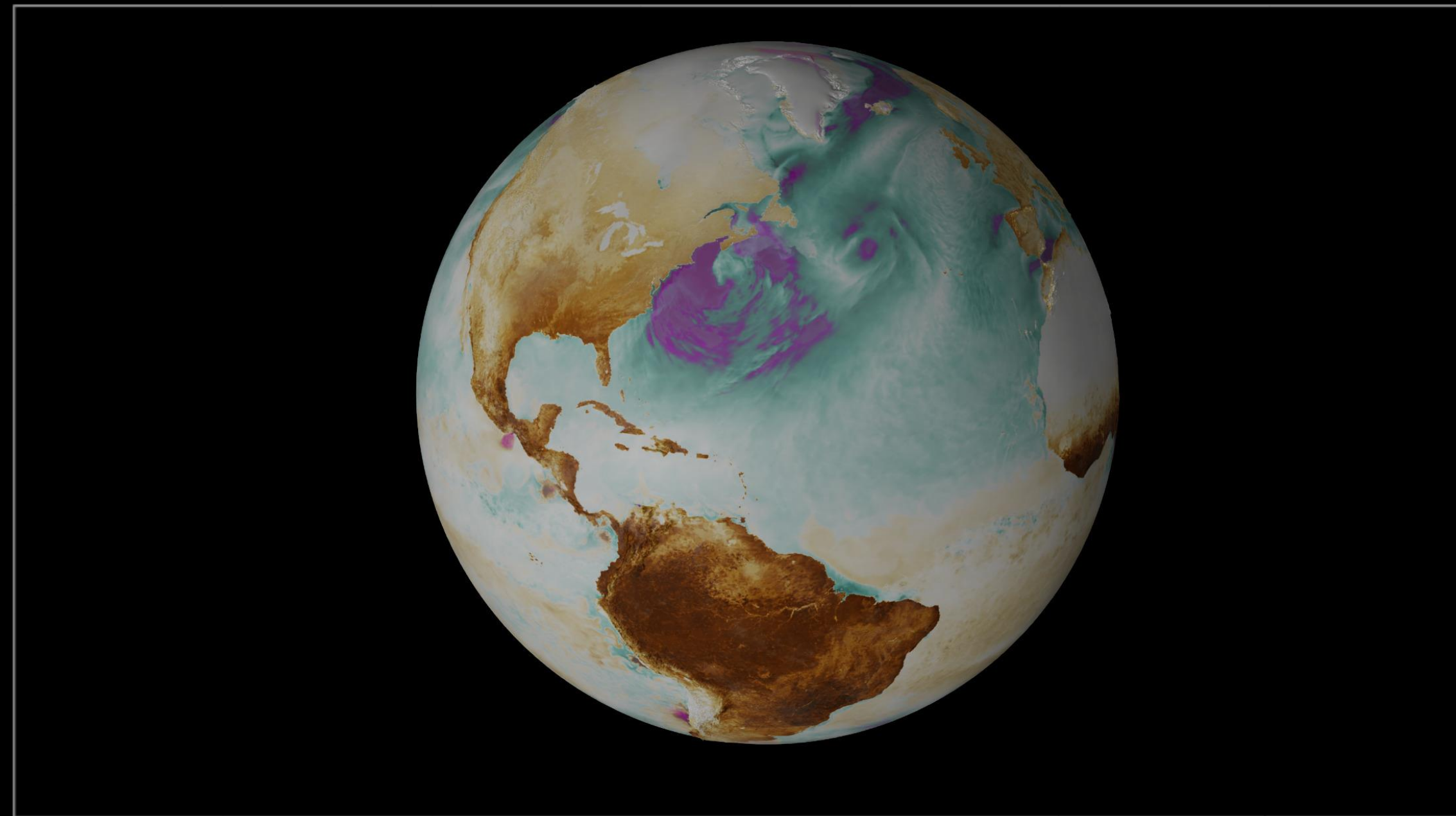


Figure adapted from: Schneider, T., Teixeira, J., Bretherton, C. et al. "Climate goals and computing the future of clouds". *Nature Climate Change* 7, 3–5 (2017)

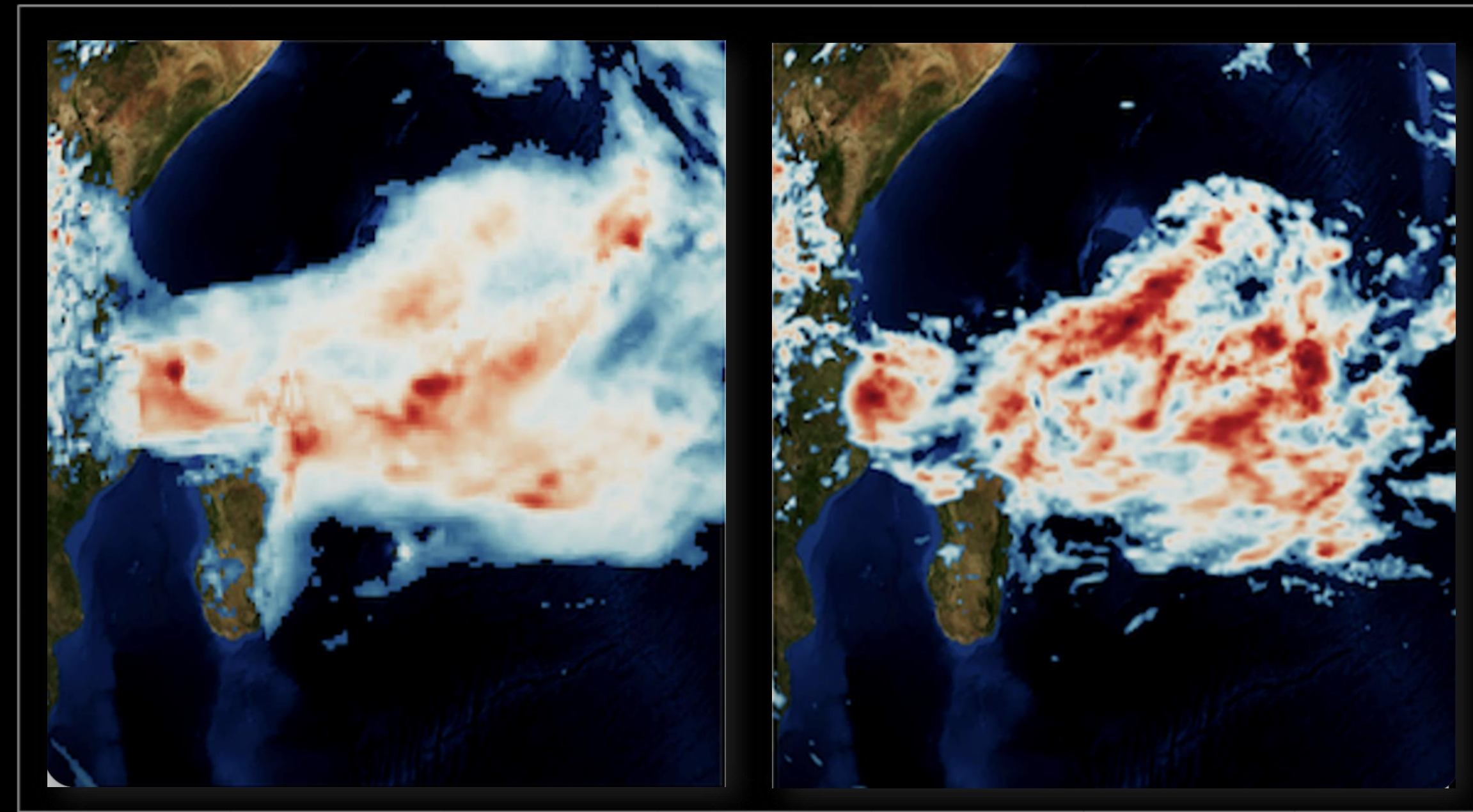
Spotlight on Scientific Excellence: The 2025 Gordon Bell Finalists

Five Gordon Bell Finalists pushing Open Science boundaries tapping NVIDIA GH200 Supercomputers

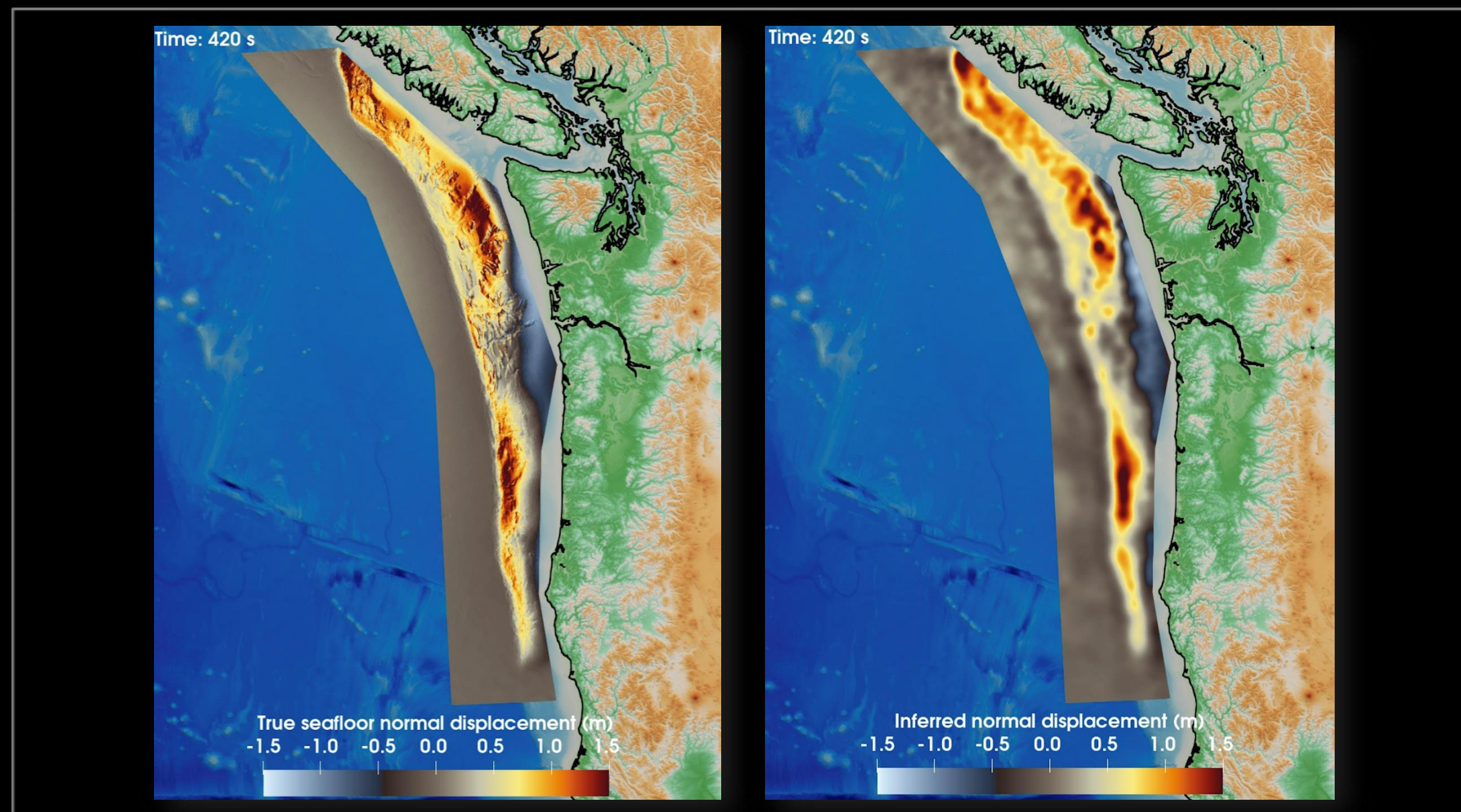
FULL EARTH SYSTEM AT 1KM
Daniel Klocke (MPI-M) et al.



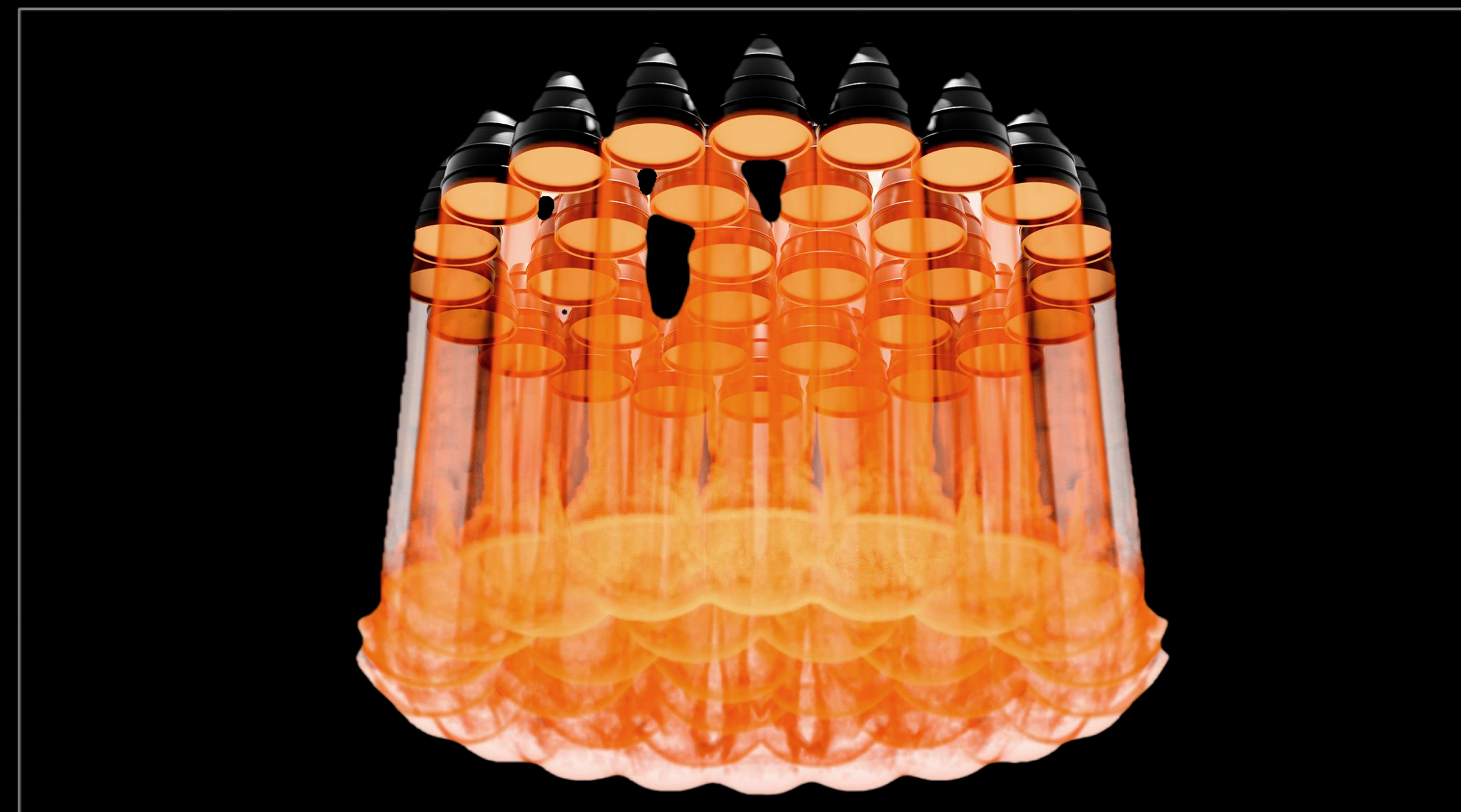
ORBIT-2 SCALING WEATHER & FORECAST FOUNDATION MODELS
Prasanna Balaprakash (ORNL) et al.



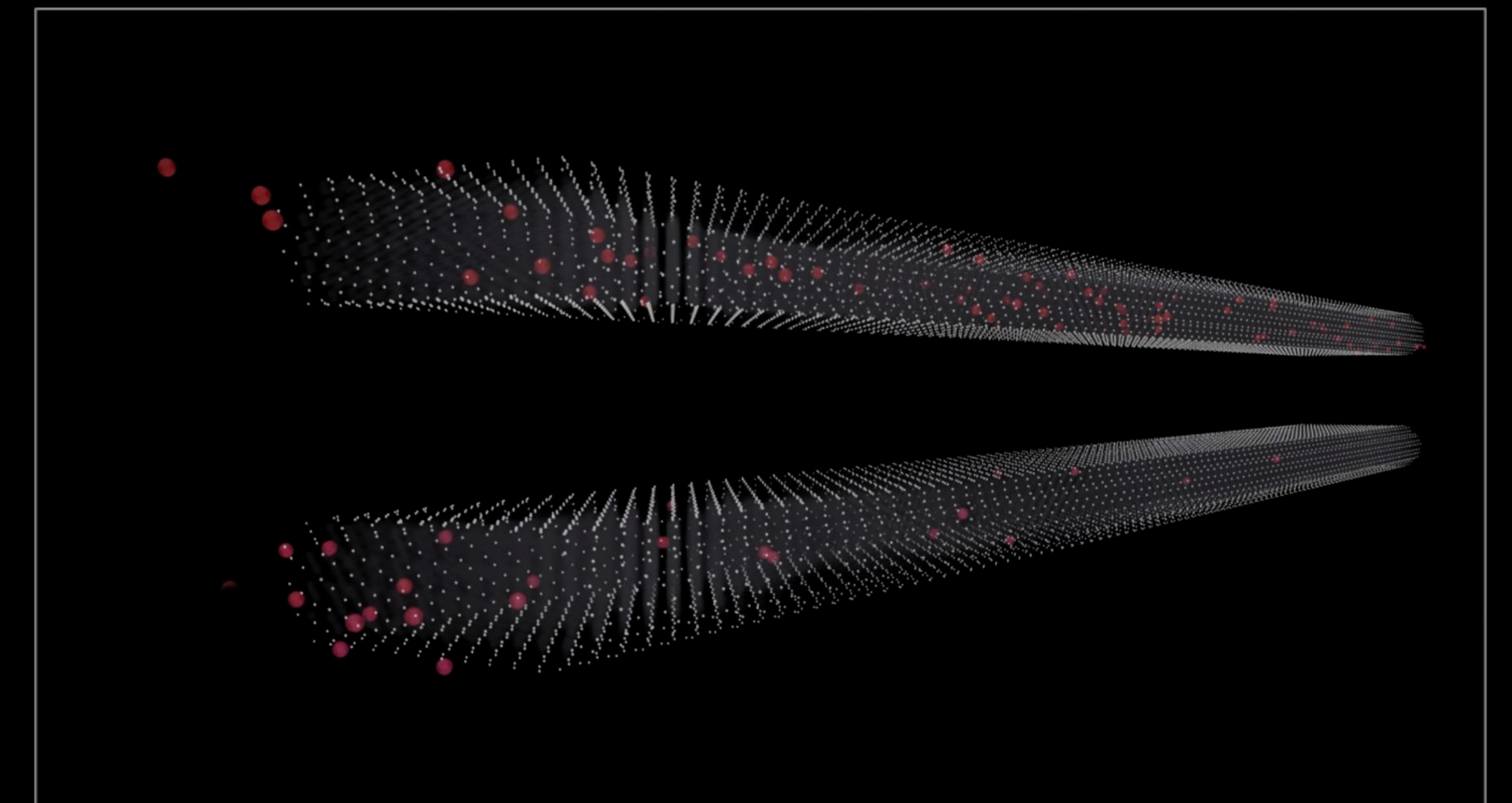
DIGITAL TWIN FOR TSUNAMI FORECAST
Omar Ghattas (University of Texas Austin) et al.



SIMULATING MANY ENGINES SPACE-CRAFT
Spencer H. Bryngelson (Georgia Institute of Technology) et al.



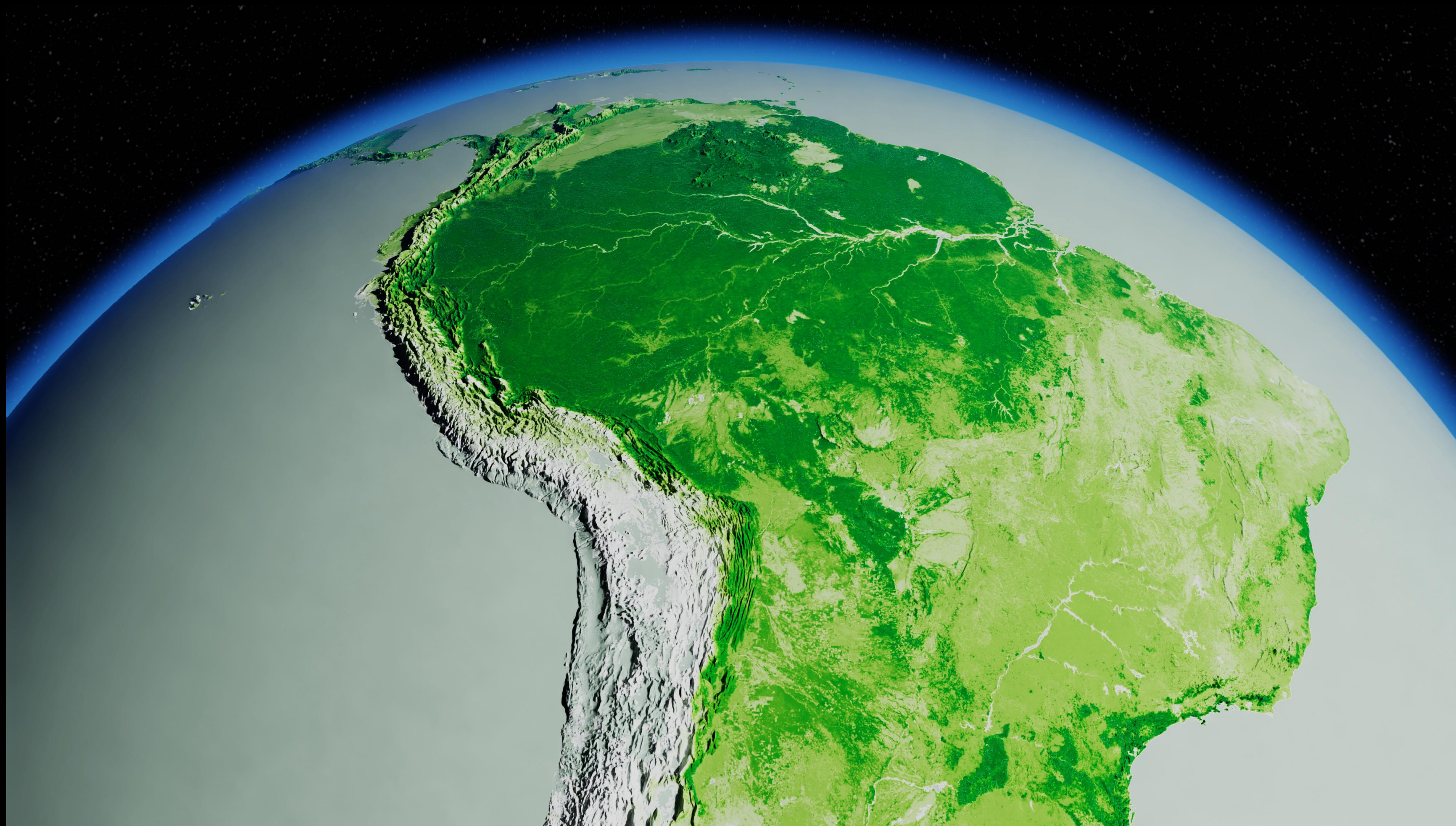
REVOLUTIONIZING NEXT GEN TRANSISTOR DESIGN
Mathieu Luisier (ETHZ) et al.



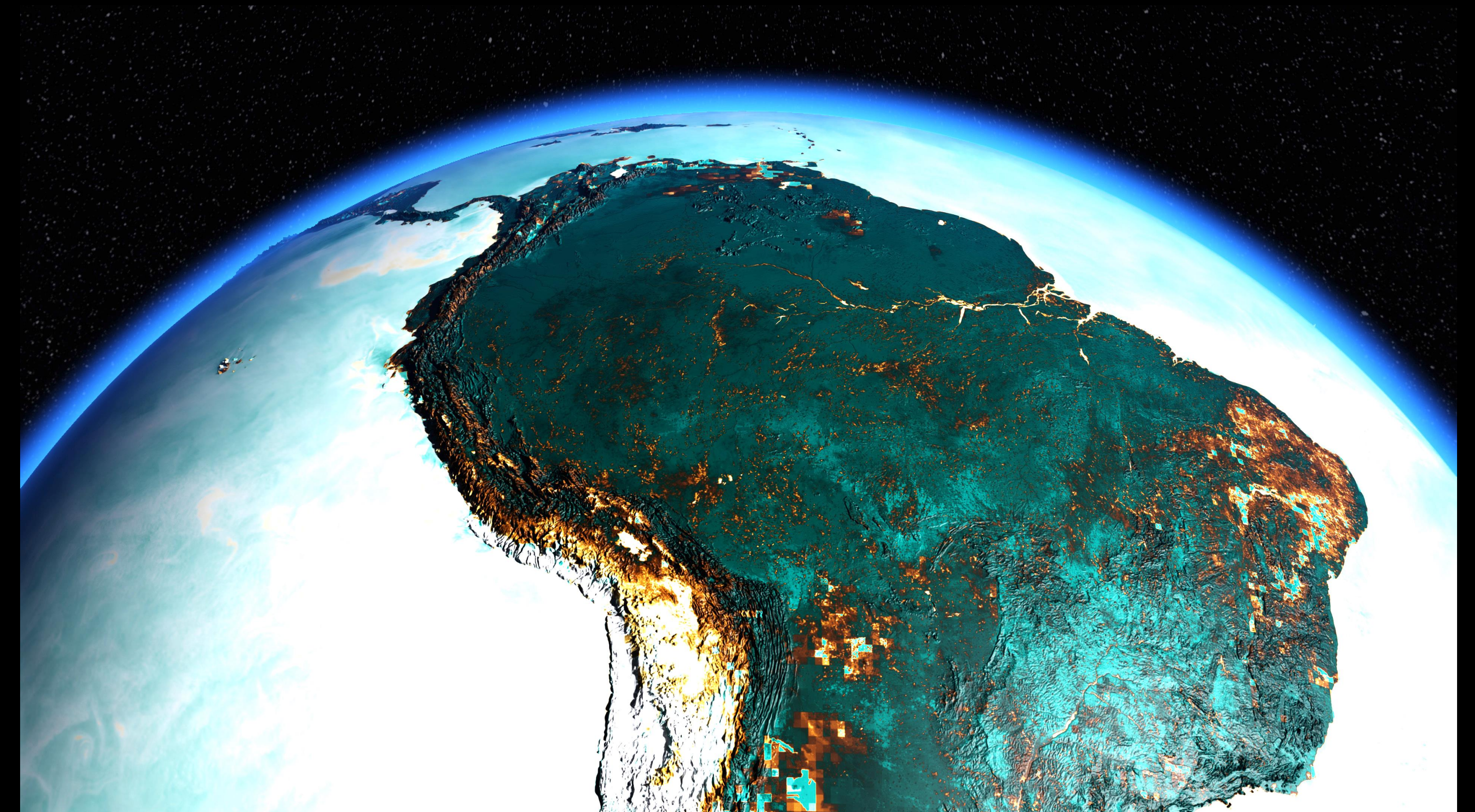
High Resolution Full Earth Climate Simulation on Jülich JUPITER

Achieves 146 Simulated Days per Day on 20,480 Grace Hoopers

Land Model with Vegetation



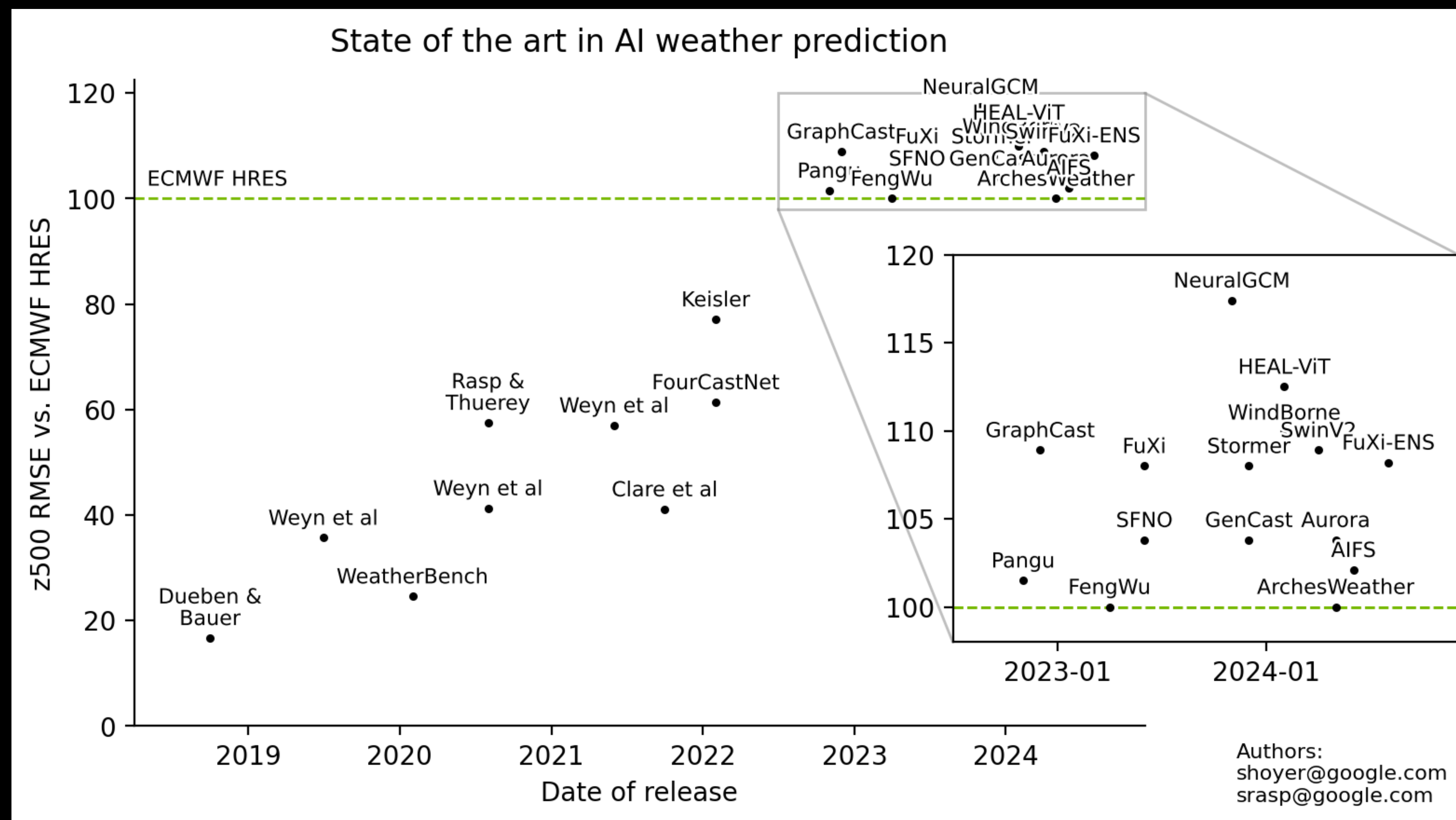
Land Model with CO₂ Flux



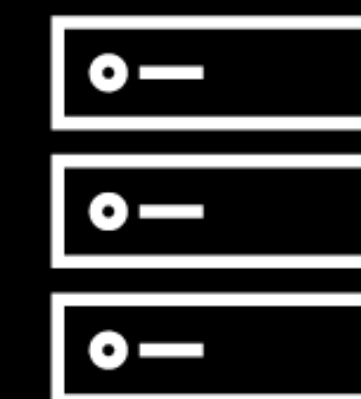
HPCWire Readers Choice award for Top HPC-Enabled Scientific Achievement.

Advantages of AI Weather Models

Compelling skill, resource requirements, and accessibility



High skill



Less hardware



Faster inference

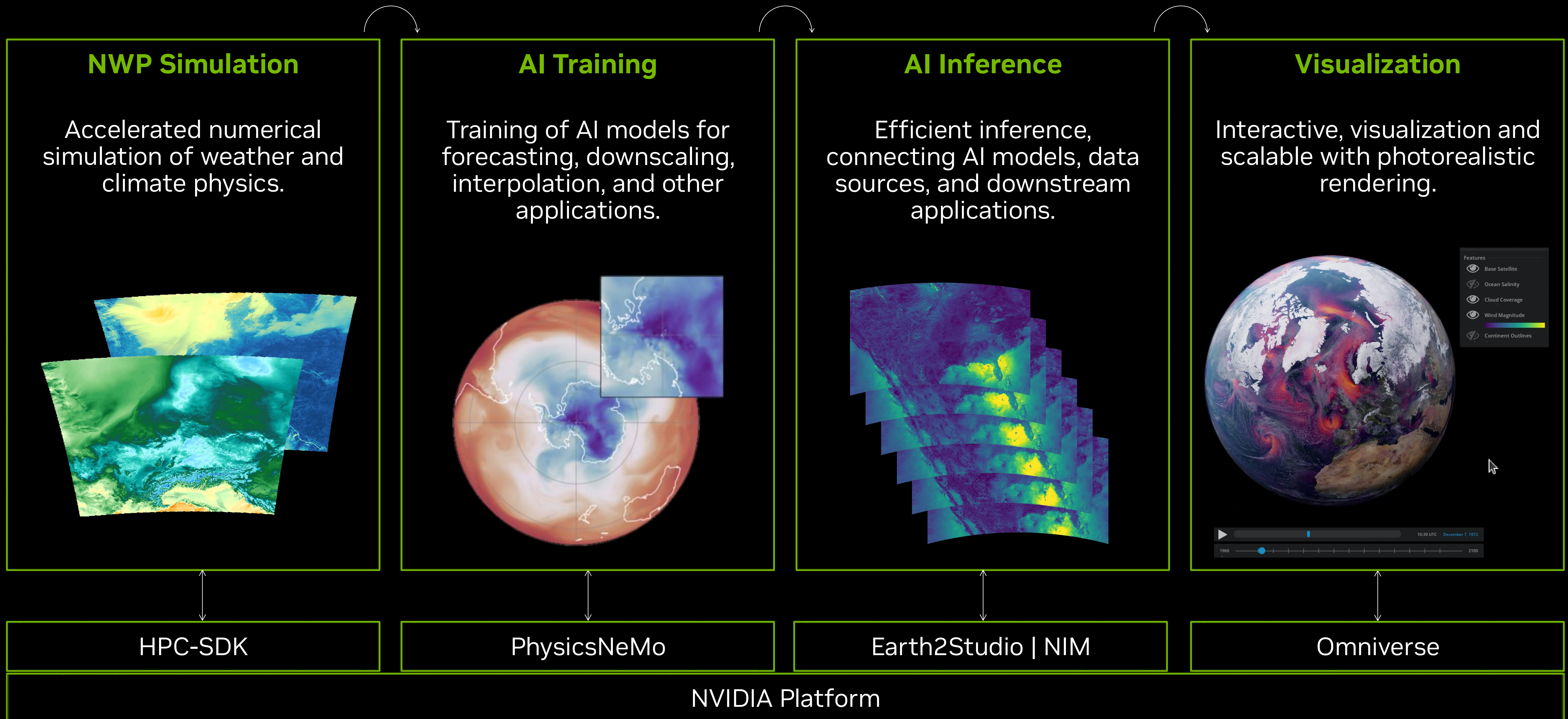


More accessible

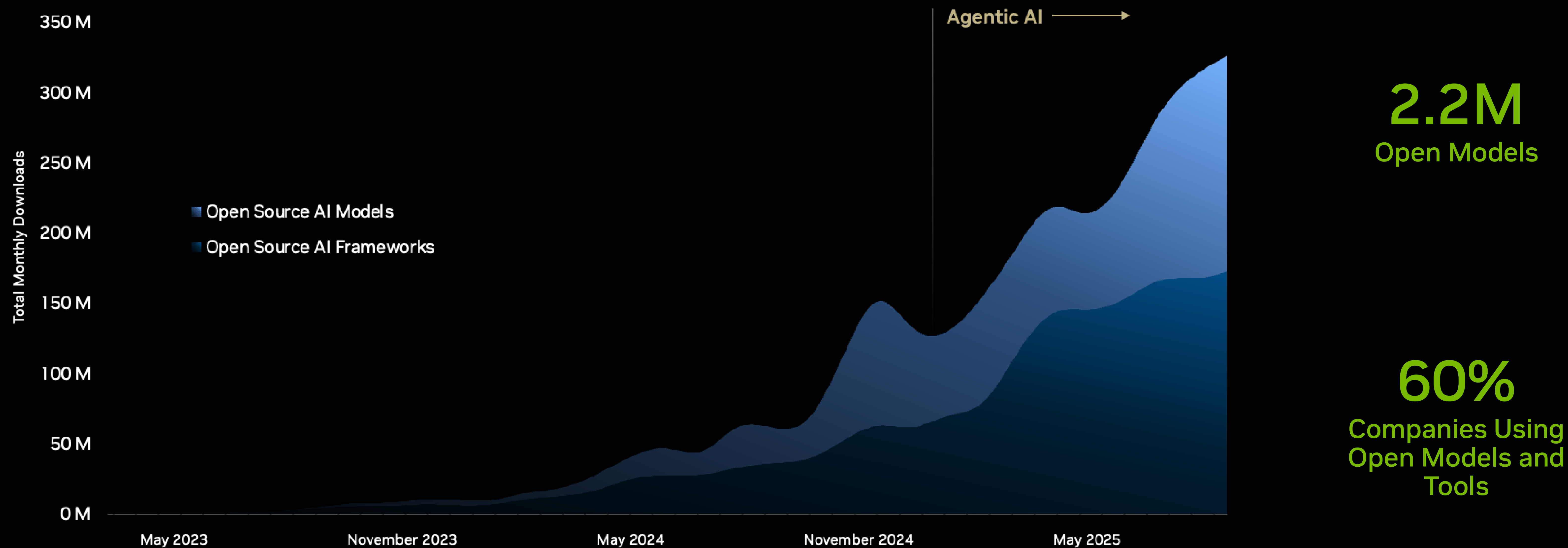
Rasp, Stephan (2024). AI-Weather SotA vs. Time.
<https://doi.org/10.6084/m9.figshare.28083515.v1>

NVIDIA Earth-2

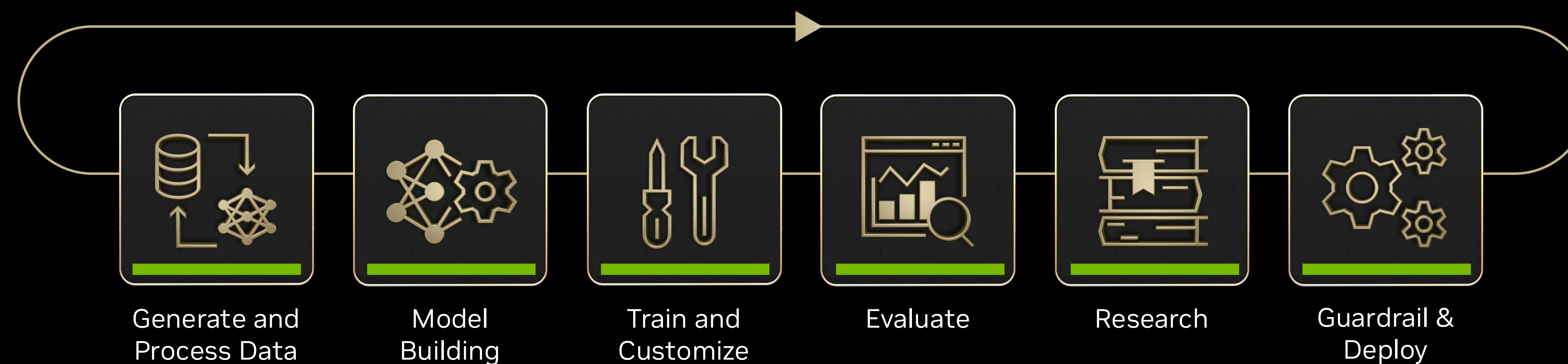
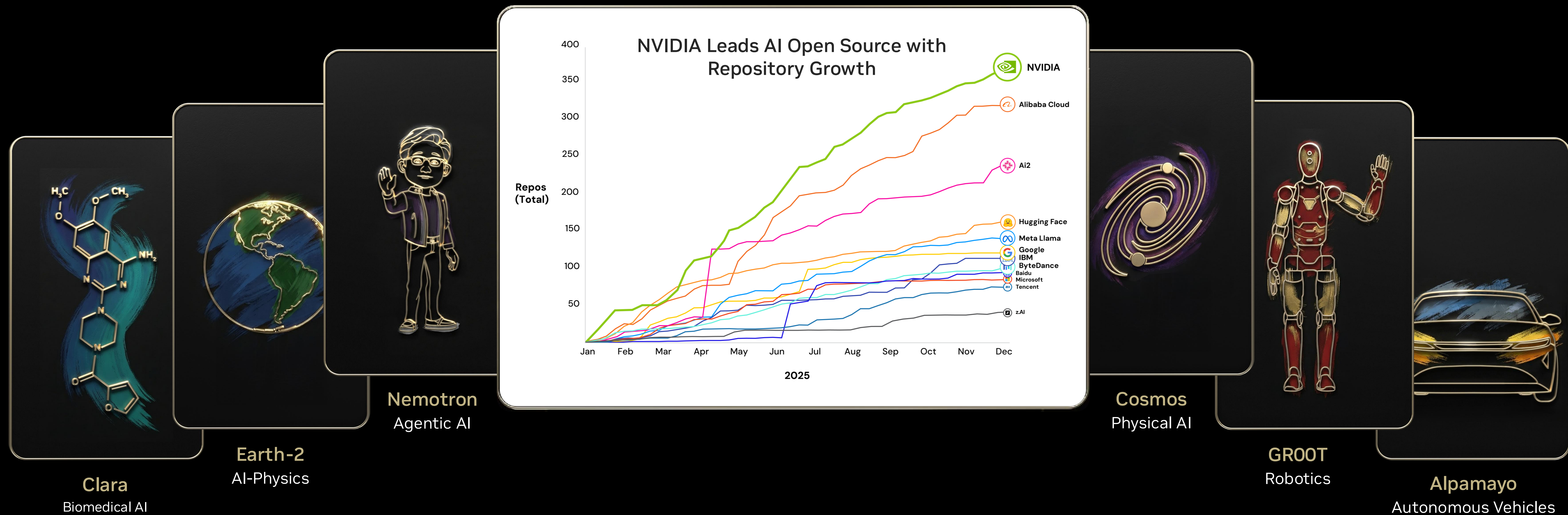
Accelerated weather and climate modeling and with high-resolution interactive visualization



Open Source is the Backbone of AI Boom

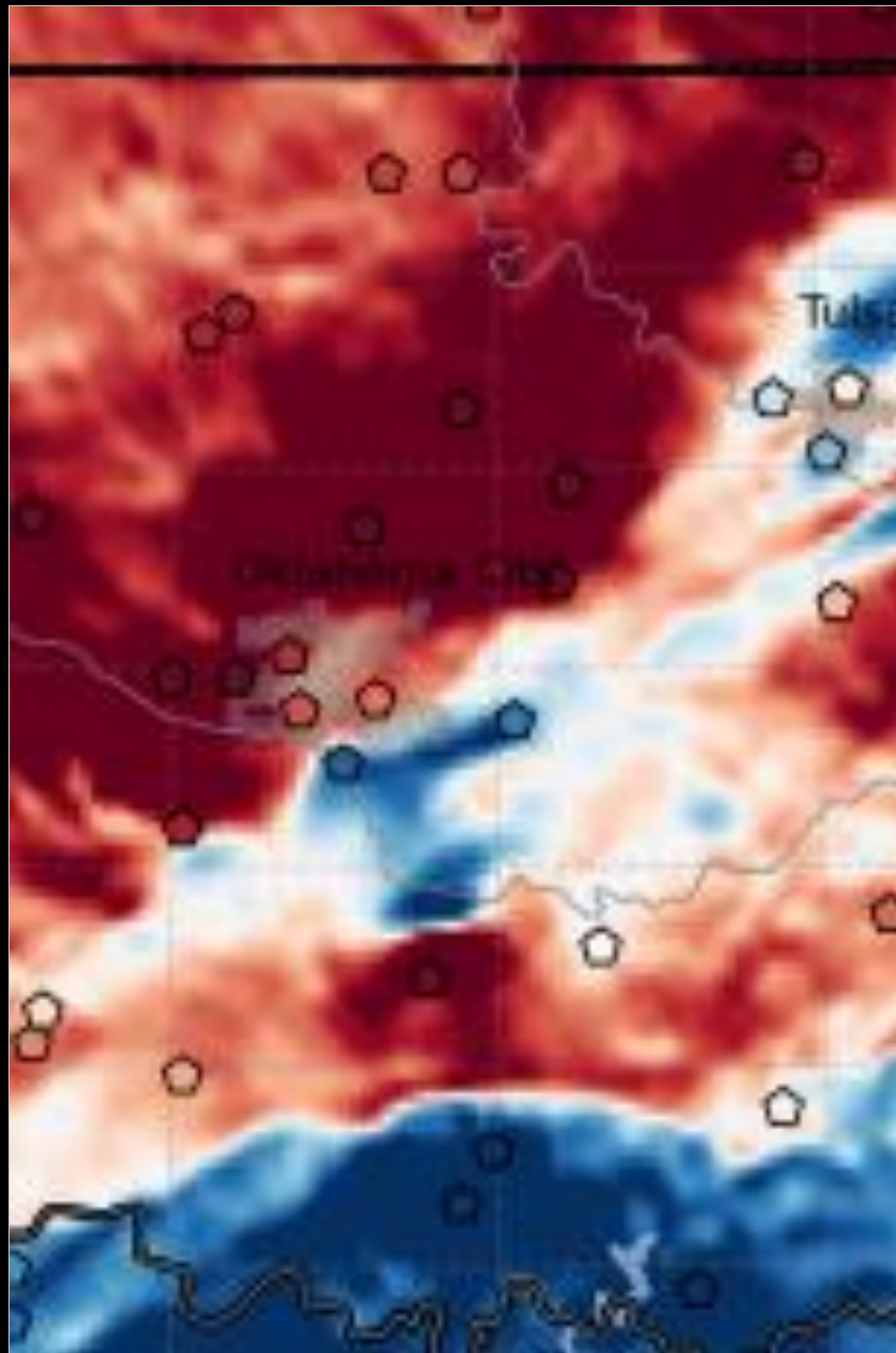


NVIDIA Leads Open Model Community



NVIDIA Earth-2: Open Models

New techs are optimized by PhysisNeMo, Earth2Studio



Stochastic Data Assimilation (SDA)

19 Jun 2024

<https://arxiv.org/abs/2406.16947>

Oxford, NVIDIA



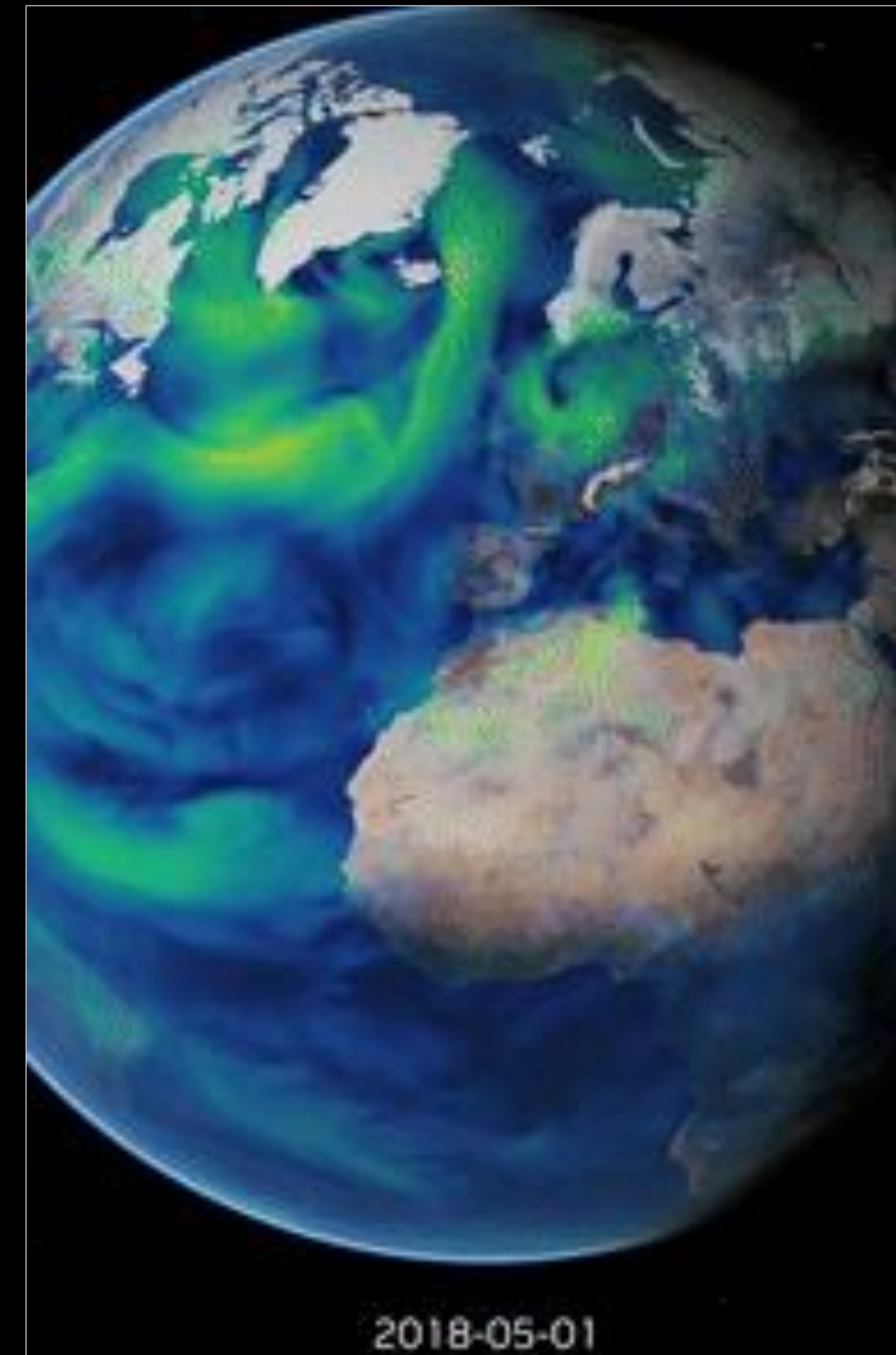
Huge Ensembles (HENS)

6 Aug 2024

<https://arxiv.org/pdf/2408.03100>

<https://arxiv.org/abs/2408.01581>

LBL, Berkeley, NVIDIA



StormCast, FourCastNet3

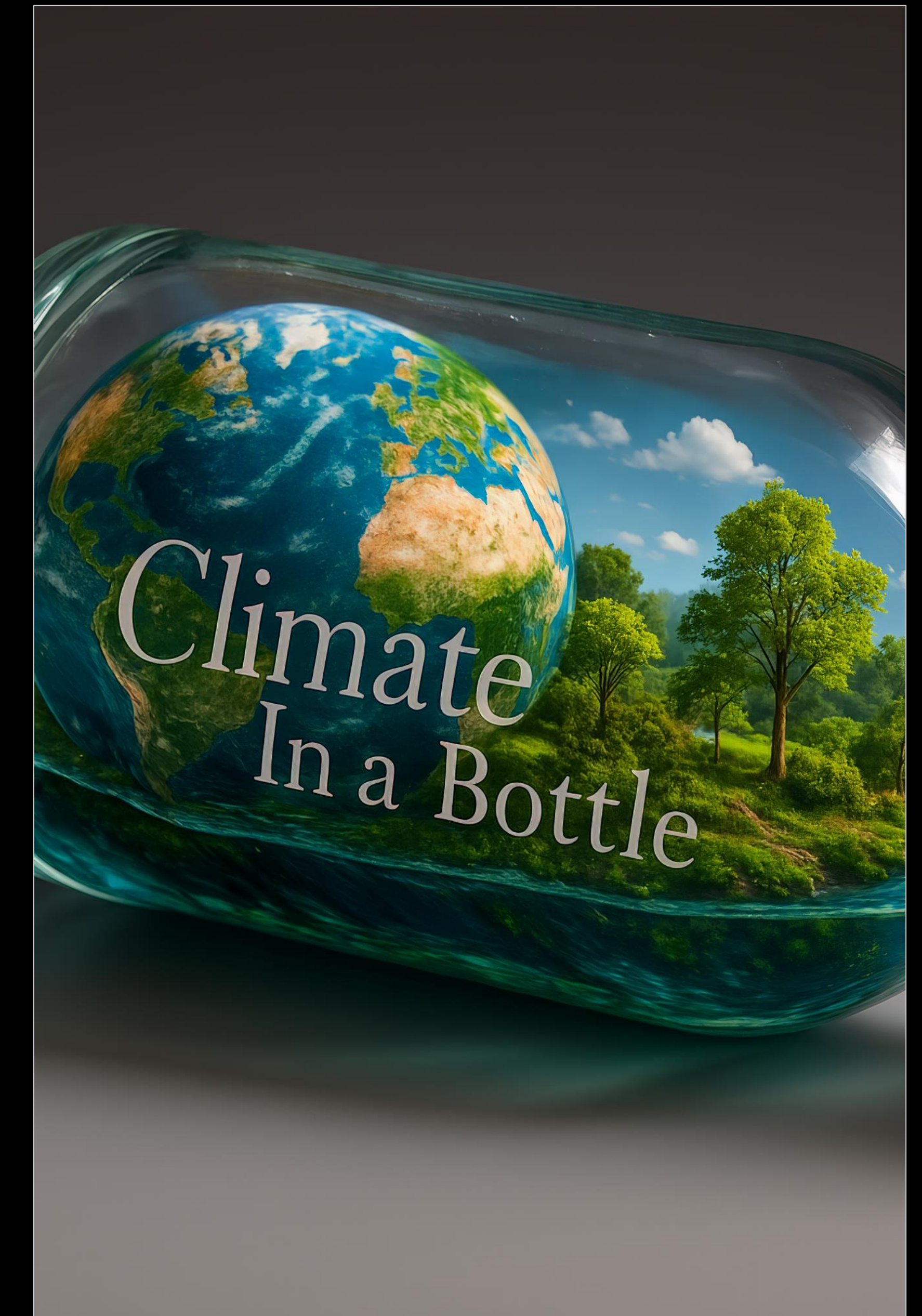
19 Aug 2024, 18 Jul 2025

<https://arxiv.org/html/2408.10958v1>

NVIDIA, LBNL, Washington

<https://arxiv.org/abs/2507.12144>

NVIDIA, LBNL,



cBottle

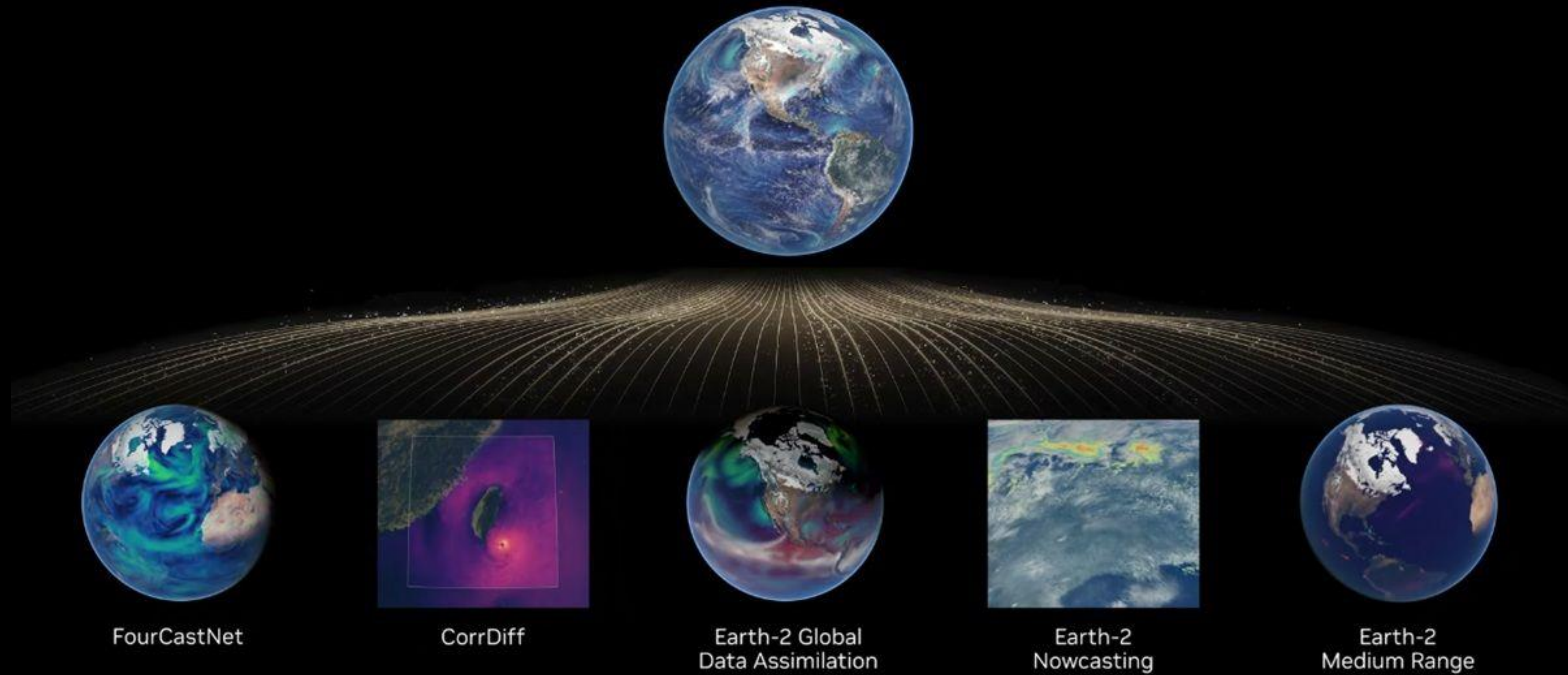
10 May 2025

<https://arxiv.org/abs/2505.06474>

NVIDIA

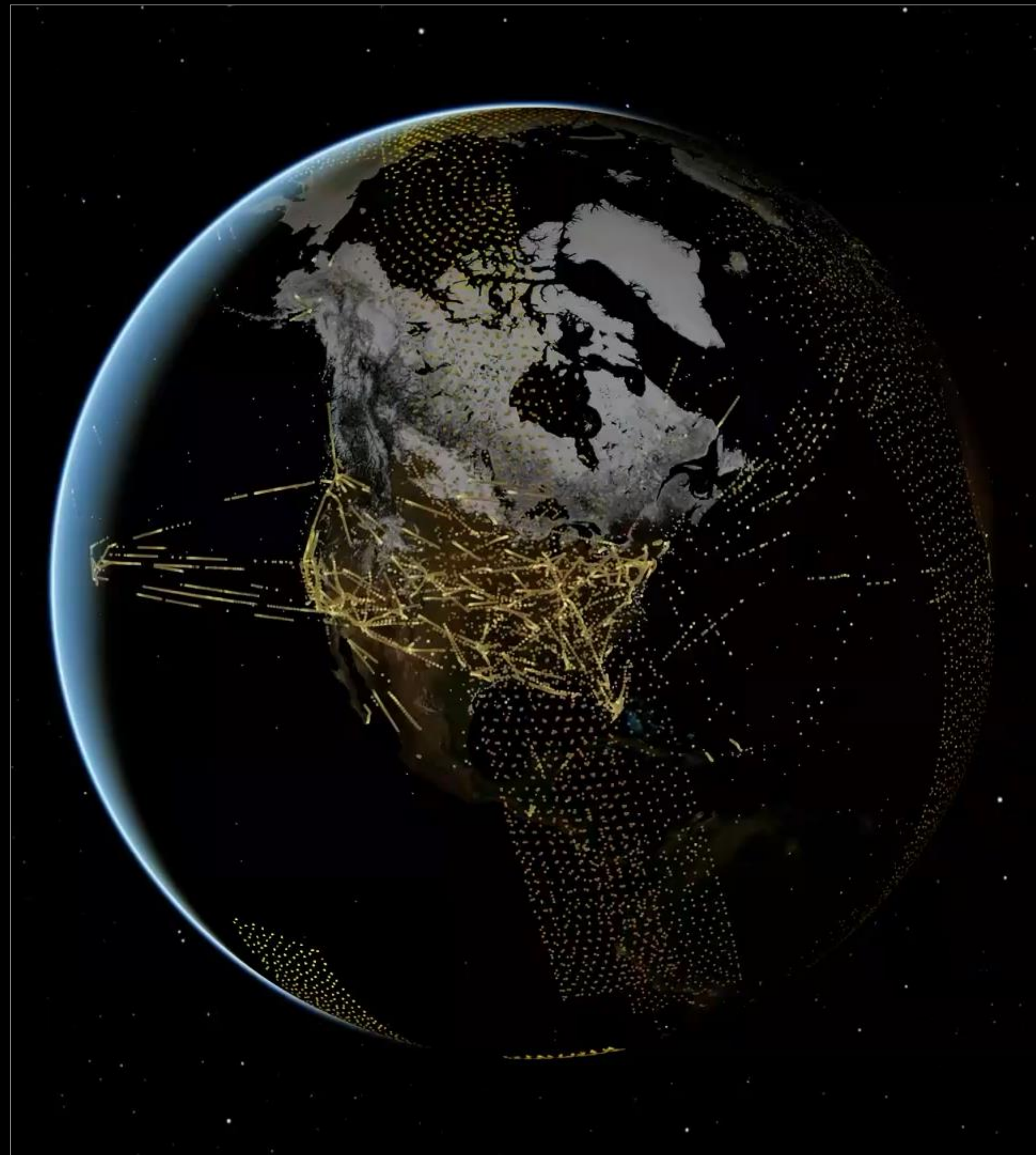
AMS 2026 Earth-2 Family of Open Models

Across Data Assimilation, Medium Range, and Nowcasting



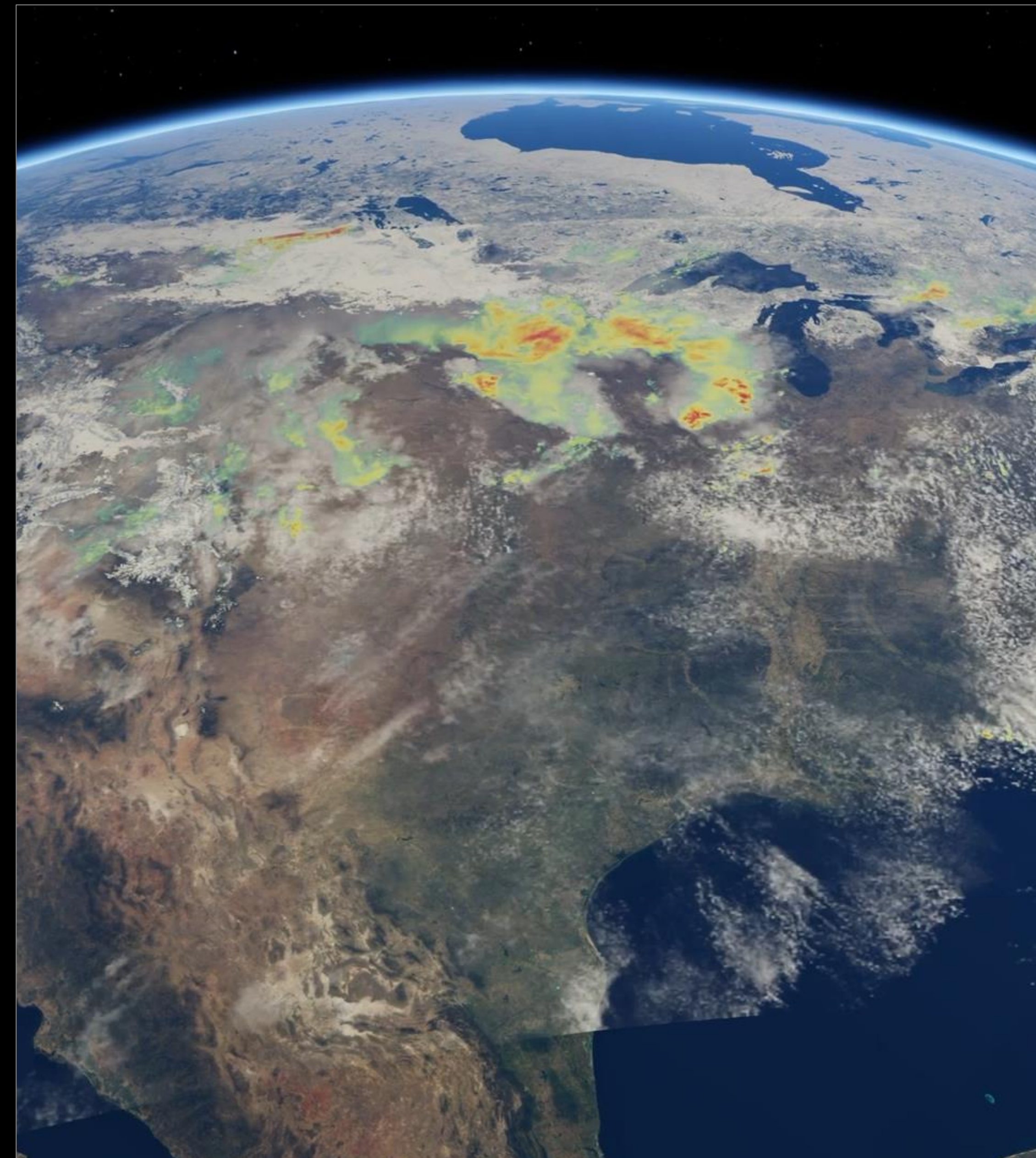
AMS 2026 Earth-2 Family of Open Models

<https://huggingface.co/blog/nvidia/earth-2-open-models>



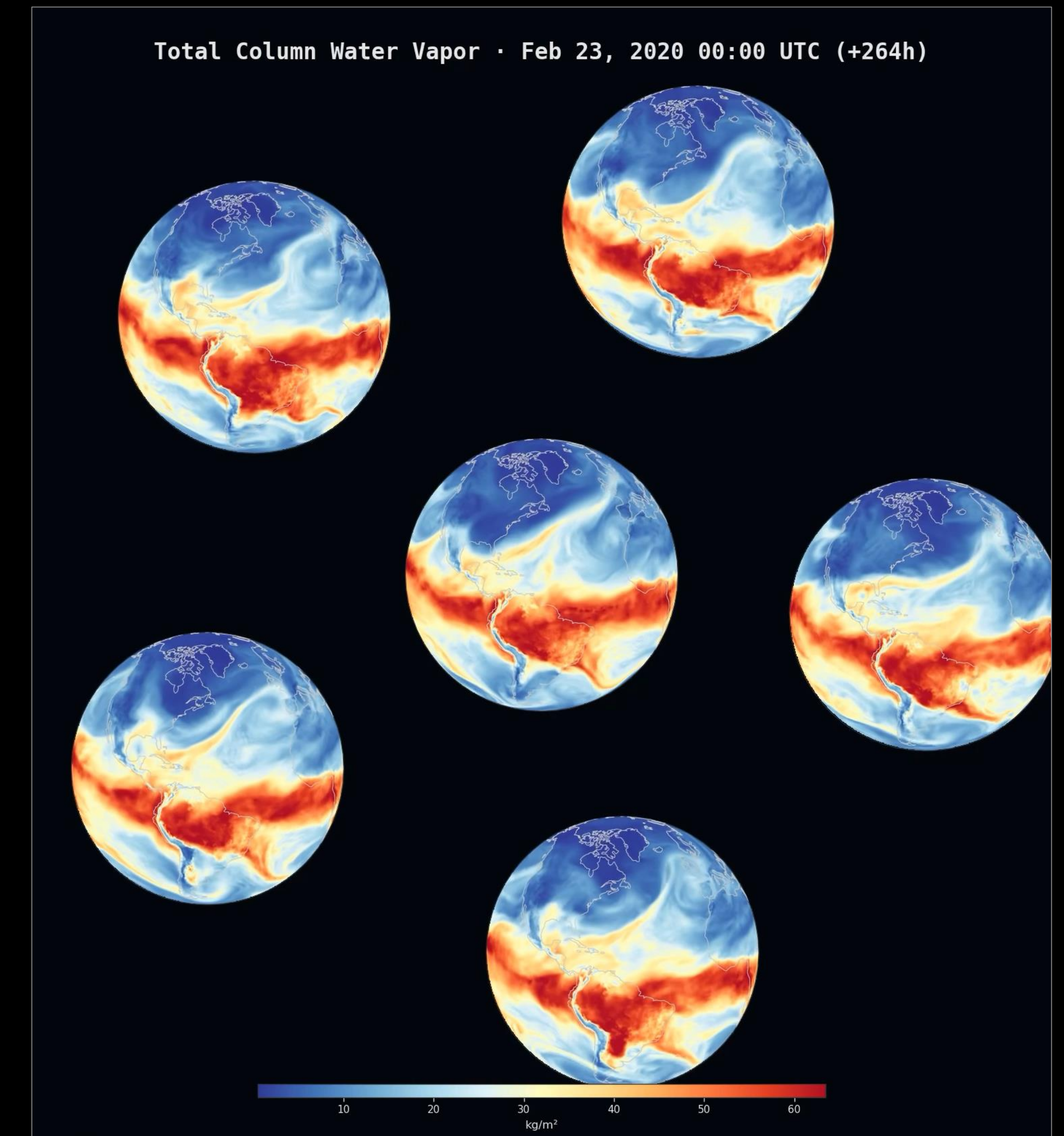
Earth-2 Global DA - HealDA

https://research.nvidia.com/publication/2026-01_healda-highlighting-importance-initial-errors-end-end-ai-weather-forecasts



Earth-2 Nowcasting - StormScope

https://research.nvidia.com/publication/2026-01_learning-accurate-storm-scale-evolution-observations

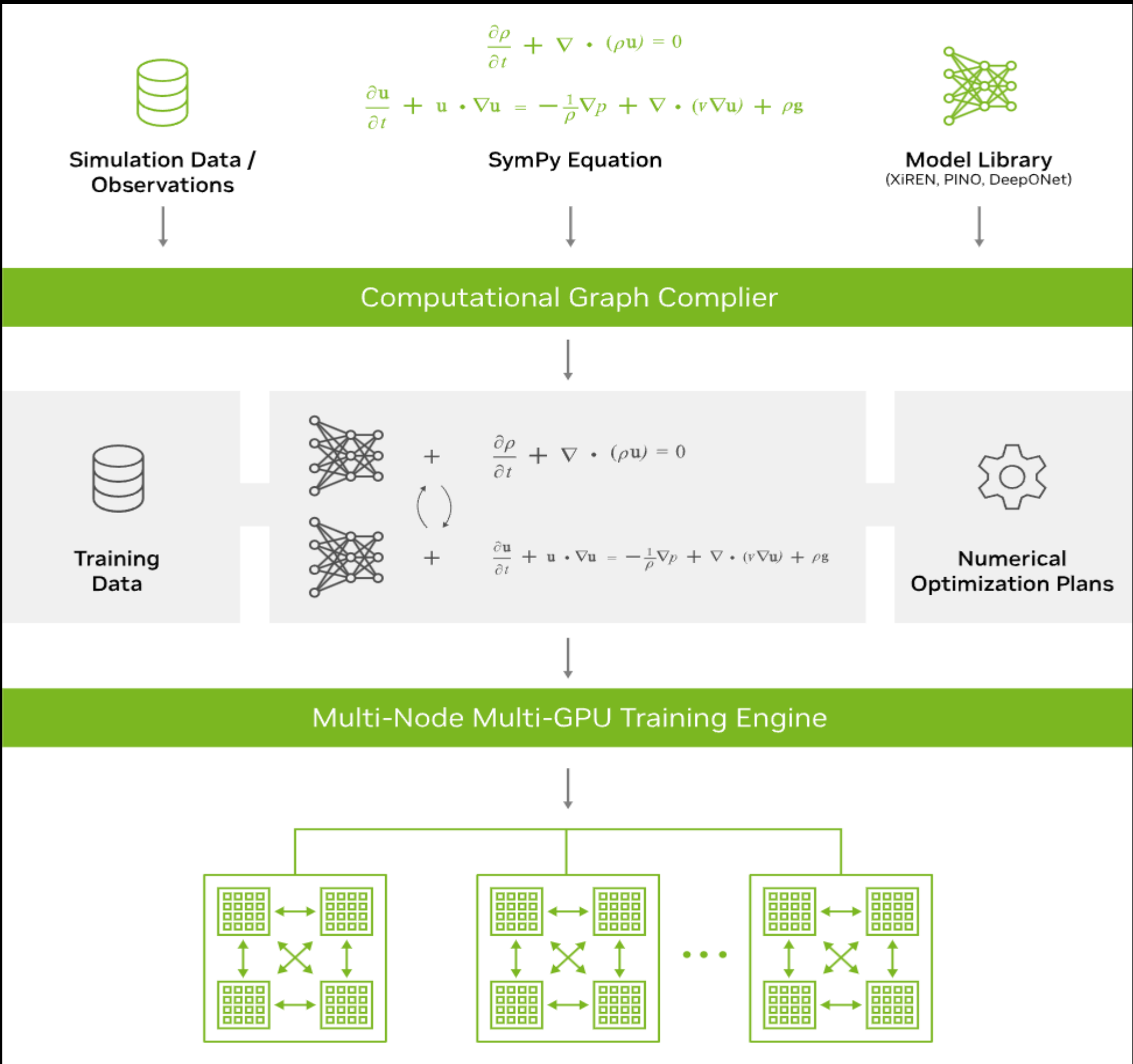


Earth-2 Medium Range - Atlas

https://research.nvidia.com/publication/2026-01_demystifying-data-driven-probabilistic-medium-range-weather-forecasting

Earth-2: AI Training & Inference Enhancement

Best technology to accelerated DLWPs



PhysicsNeMo - Training

<https://github.com/NVIDIA/physicsnemo>

Earth2Studio

Python 3.10+ | License Apache 2.0 | Code Style Black | coverage 91%

Earth2Studio is a Python-based package designed to get users up and running with AI weather and climate models *fast*. Our mission is to enable everyone to build, research and explore AI driven meteorology.

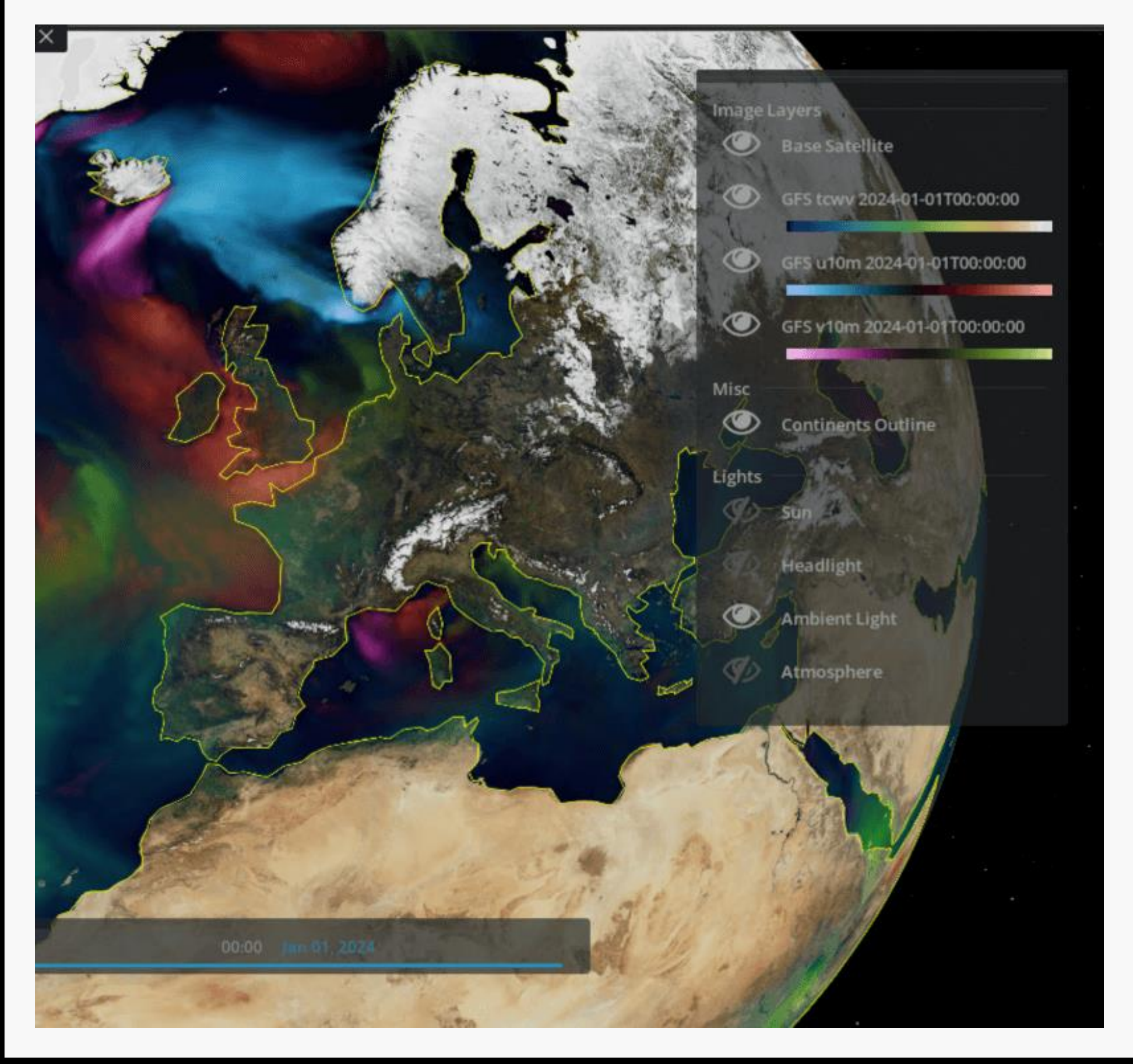
- Earth2Studio Documentation -

[Install](#) | [User-Guide](#) | [Examples](#) | [API](#)

Pre-Trained Model Zoo | Cloud Datasources | GPU Accelerated Utilities | Composable APIs

Earth2Studio - Exploration

<https://github.com/NVIDIA/earth2studio>



New Blueprints – Reference App

<https://build.nvidia.com/nvidia/ai-weather-forecasting>

PhysicsNeMo

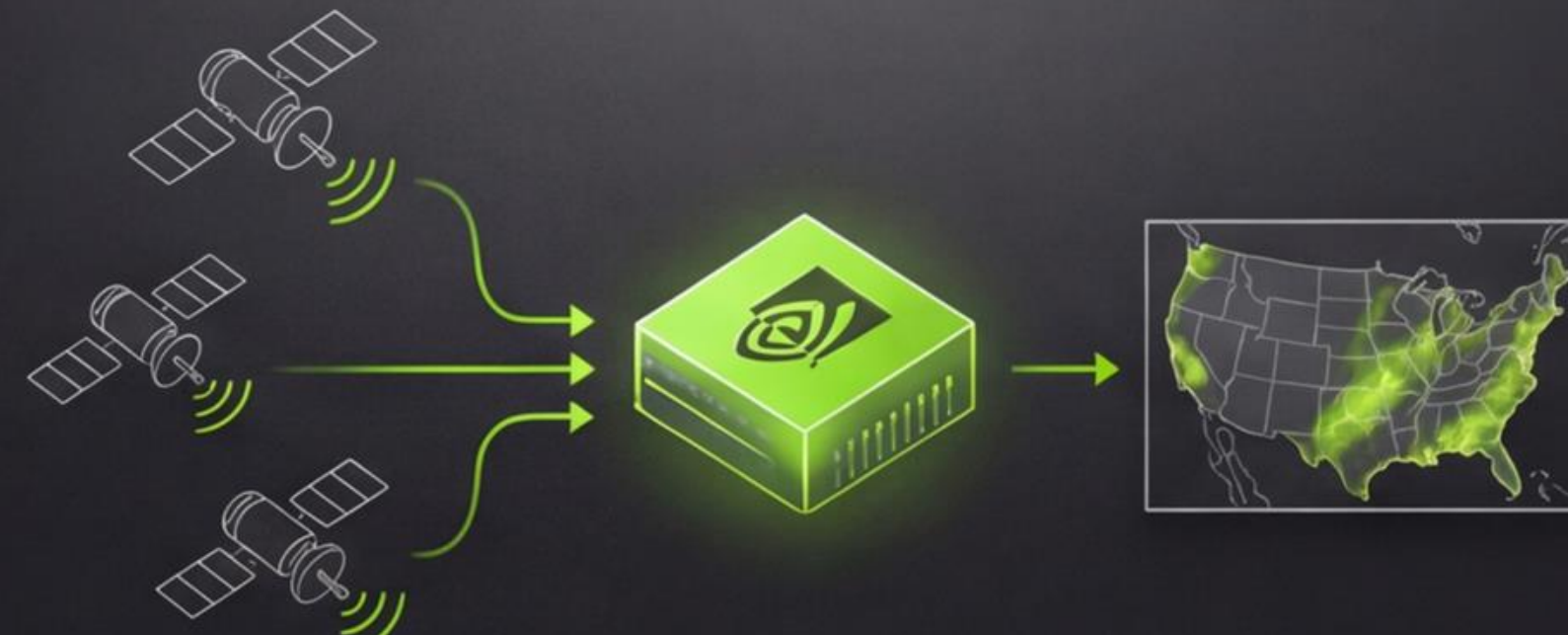
A high-performance open-source AI training framework for physics-ML models including weather and climate
https://docs.nvidia.com/physicsnemo/latest/physicsnemo/examples_weather.html



FourCastNet3

1000+ Member AI Ensembles

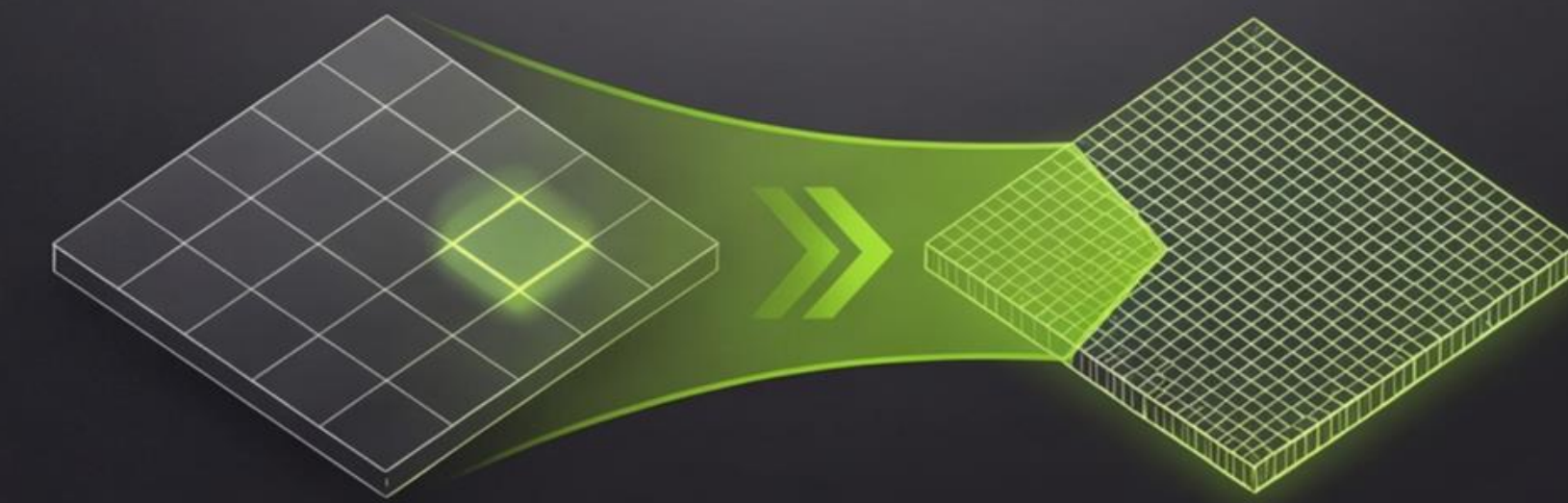
Generates thousands of well-calibrated forecast members.



SDA

AI Data Assimilation

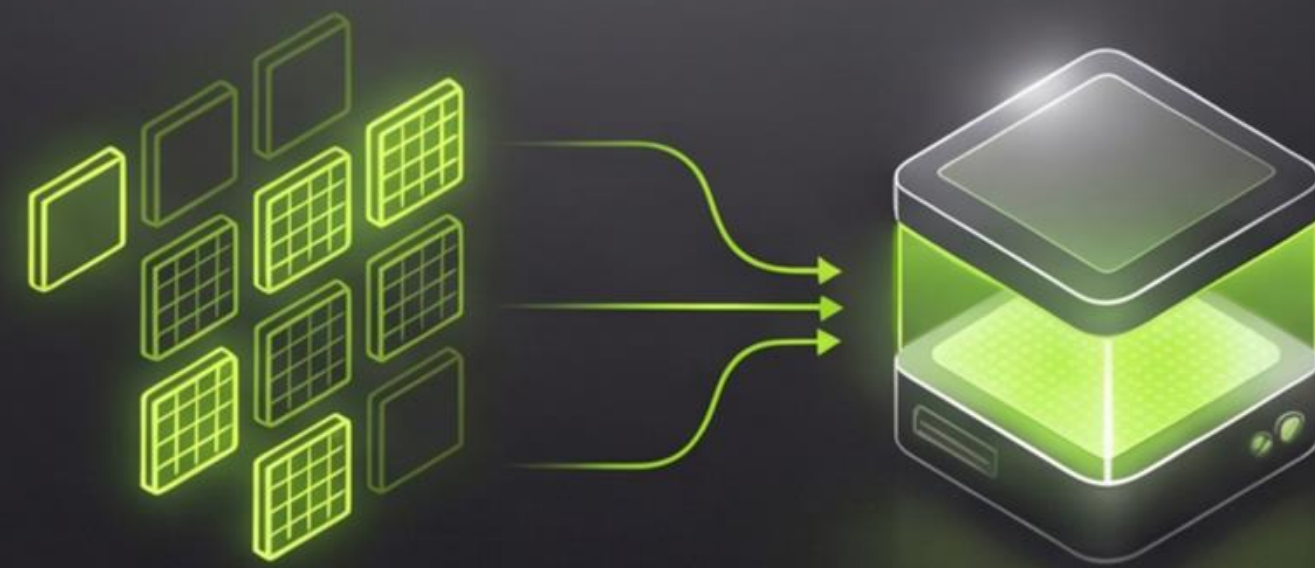
Enables real-time forecasting through accelerated data assimilation.



CorrDiff

AI Downscaling 25km → 3km

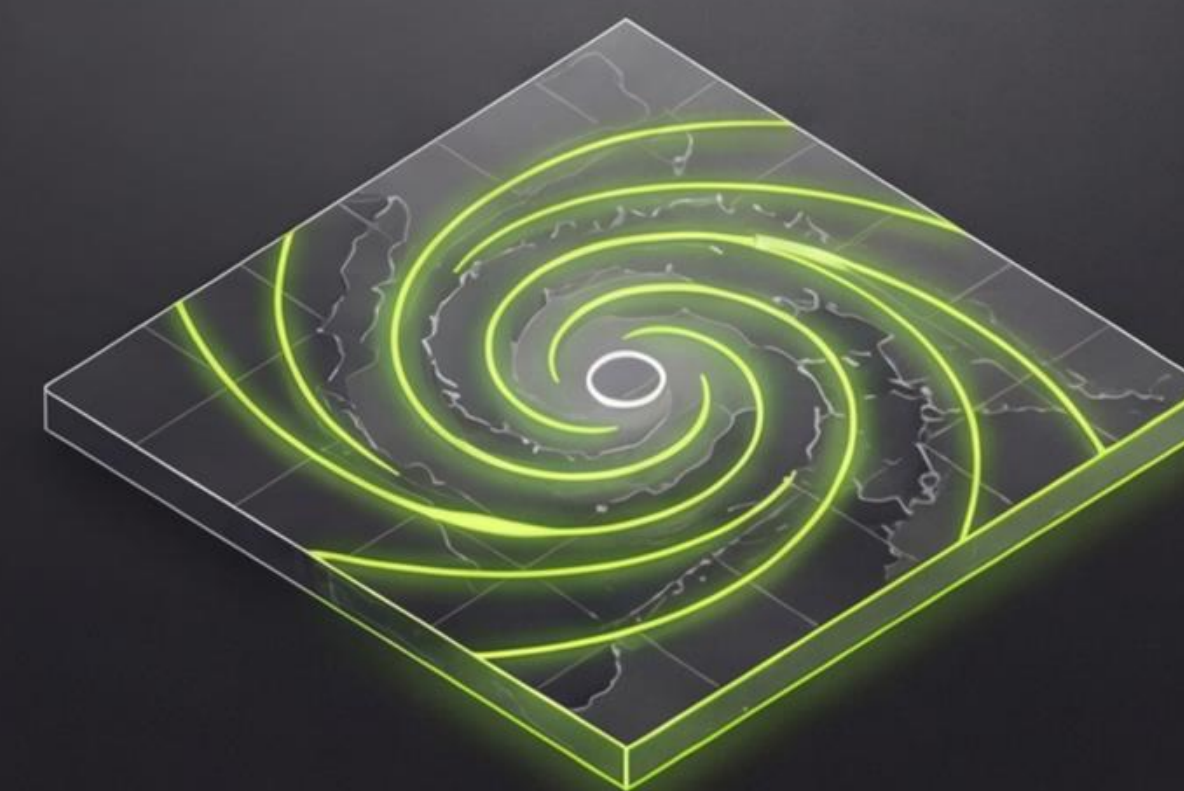
Performs km-scale downscaling in seconds.



Climate in a Bottle

Dataset compressed into portable AI model

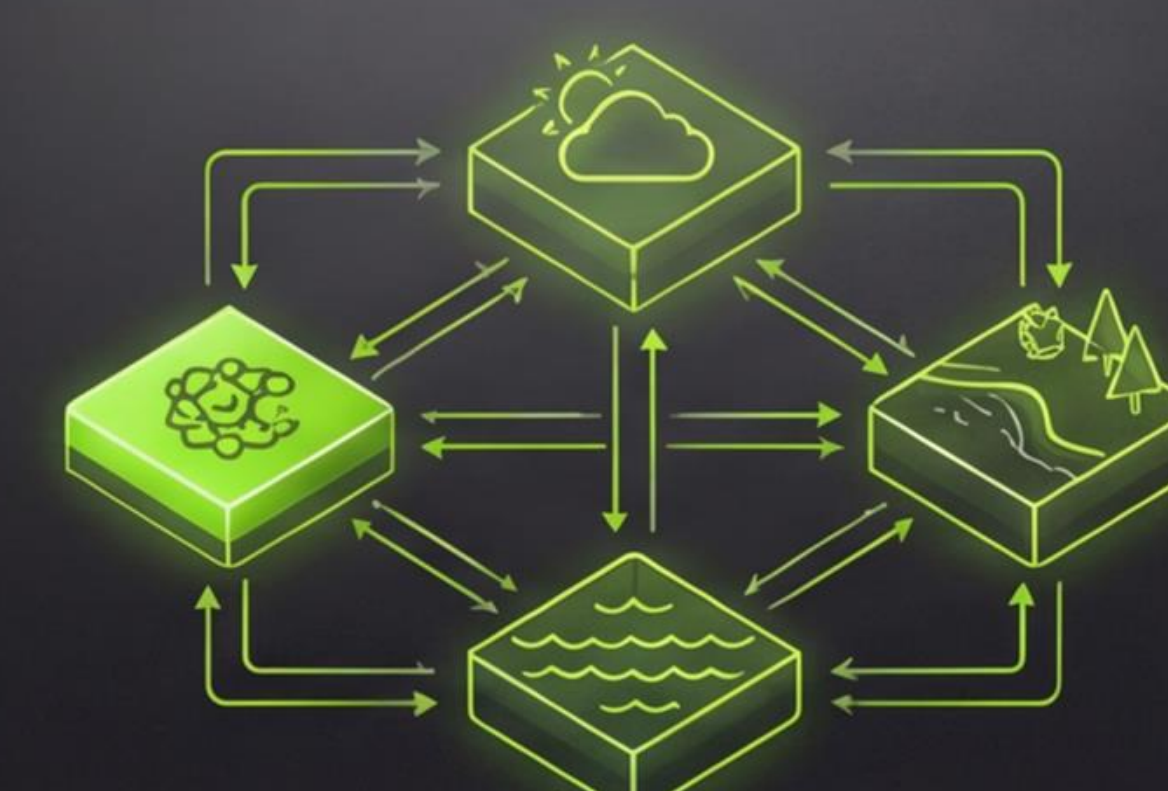
Compresses massive datasets for accessibility.



StormCast

3km Regional AI Forecasts

Provides nearly instantaneous regional forecasts.



DLESyM

AI Earth System Model Prototype

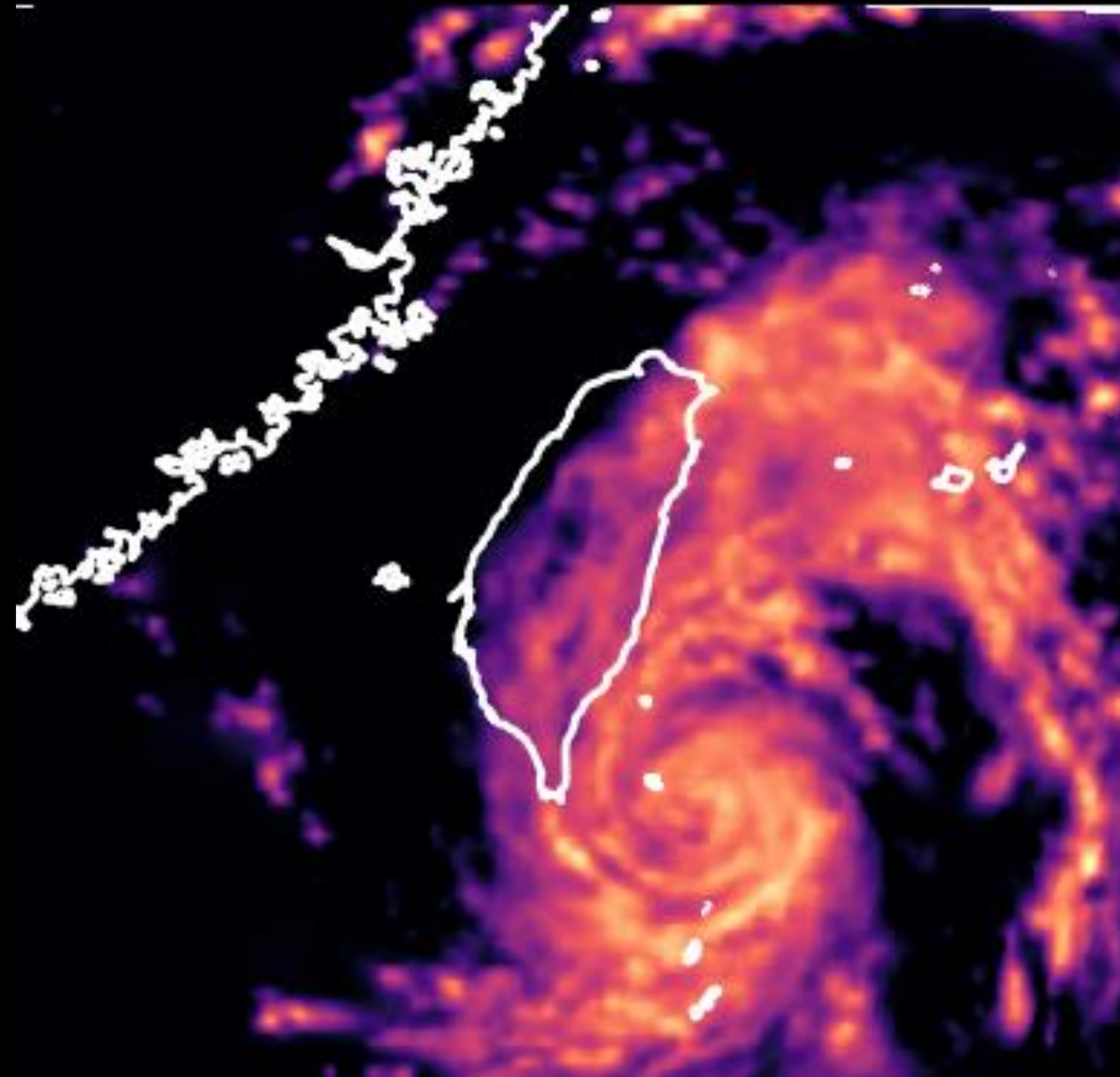
A prototype for a coupled AI Earth-System climate model.

An exmaple: CorrDiff

Generative Super-Resolution Architecture Powered by Residual Diffusion

[Input]

ERA5 consists of 20 channels at a
resolution of 36×36
(25 KM resolution)



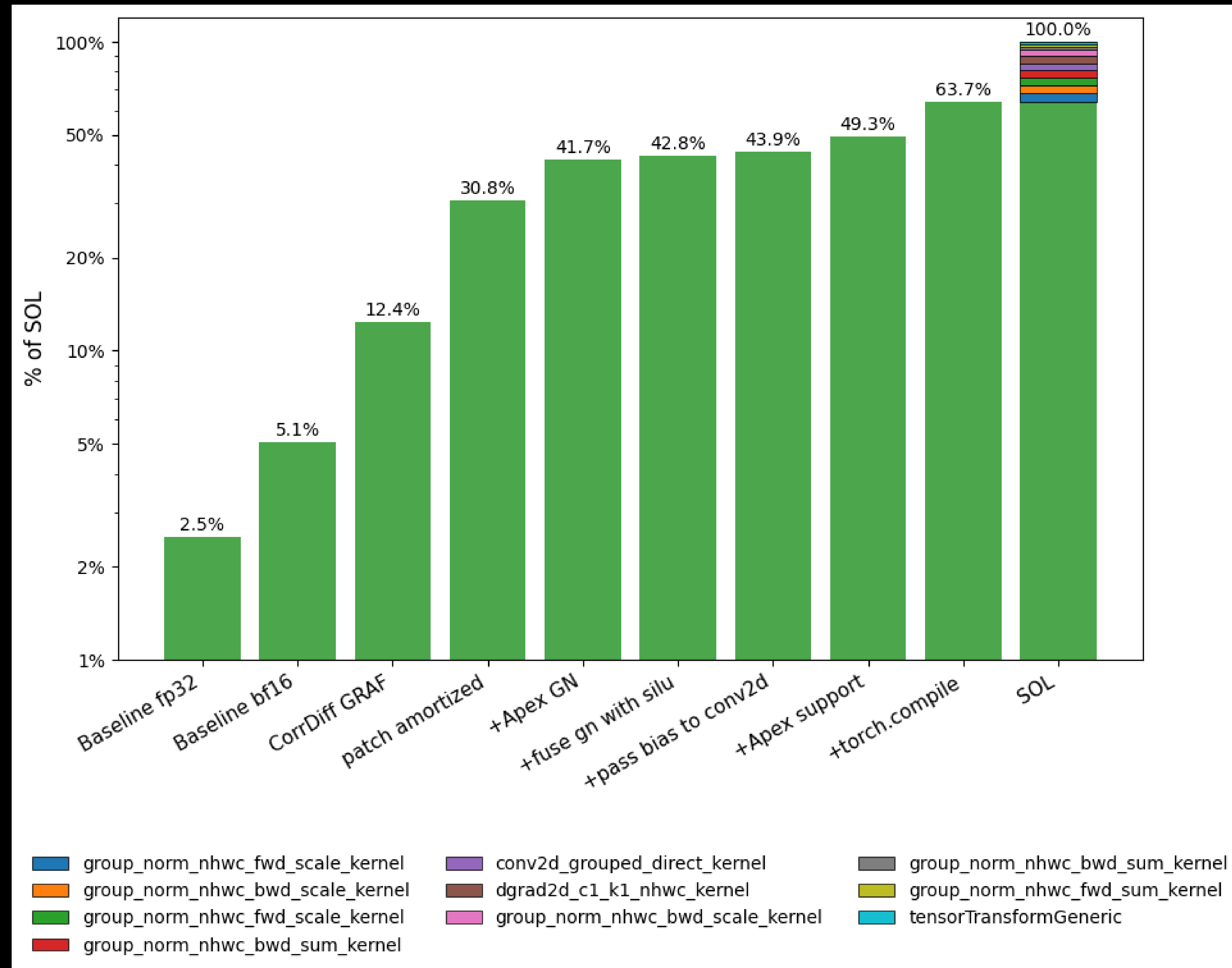
[Output]

CWA-RWRF consists of 4 channels at a
resolution of 448×448
(2 KM resolution)

CorrDiff: Super-Resolution & Channel Synthesis

CorrDiff Performance Optimization: 25x speedup

Lowers bar to wide adoption and enables scalability to the entire planet



Patch-Based Multi-Diffusion

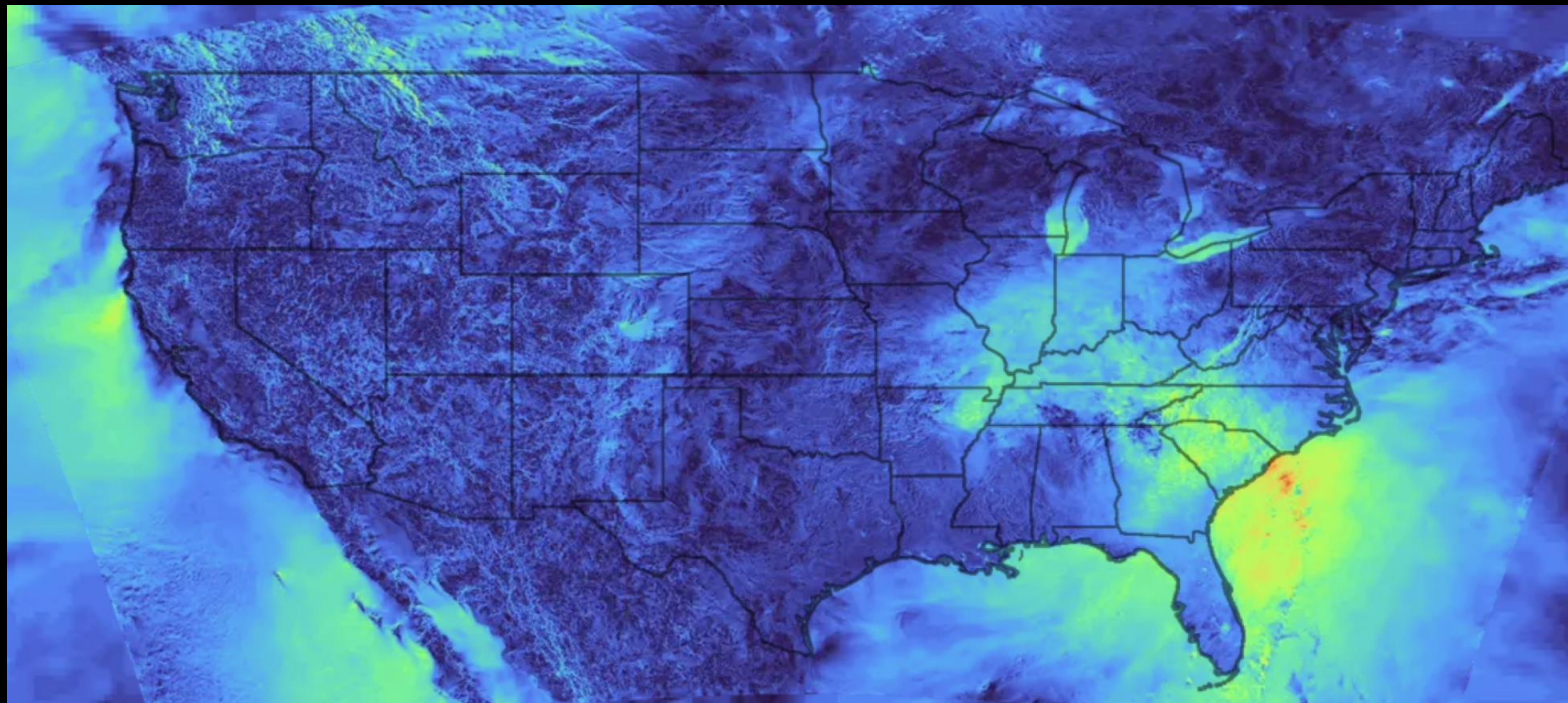
- Bar-Tal, O., Yariv, L., Lipman, Y., & Dekel, T. (2023). MultiDiffusion: Fusing Diffusion Paths for Controlled Image Generation. In *arXiv [cs.CV]*. arXiv. <http://arxiv.org/abs/2302.08113>



Figure 4. Text-to-Panorama comparison to Blended Latent Diffusion (BLD) (Avrahami et al., 2022a) and Stable Inpainting (SI) (Rombach et al., 2022). Our framework produces seamless and diverse content whereas the baselines either contain repetitive content, visible seams or artifacts.

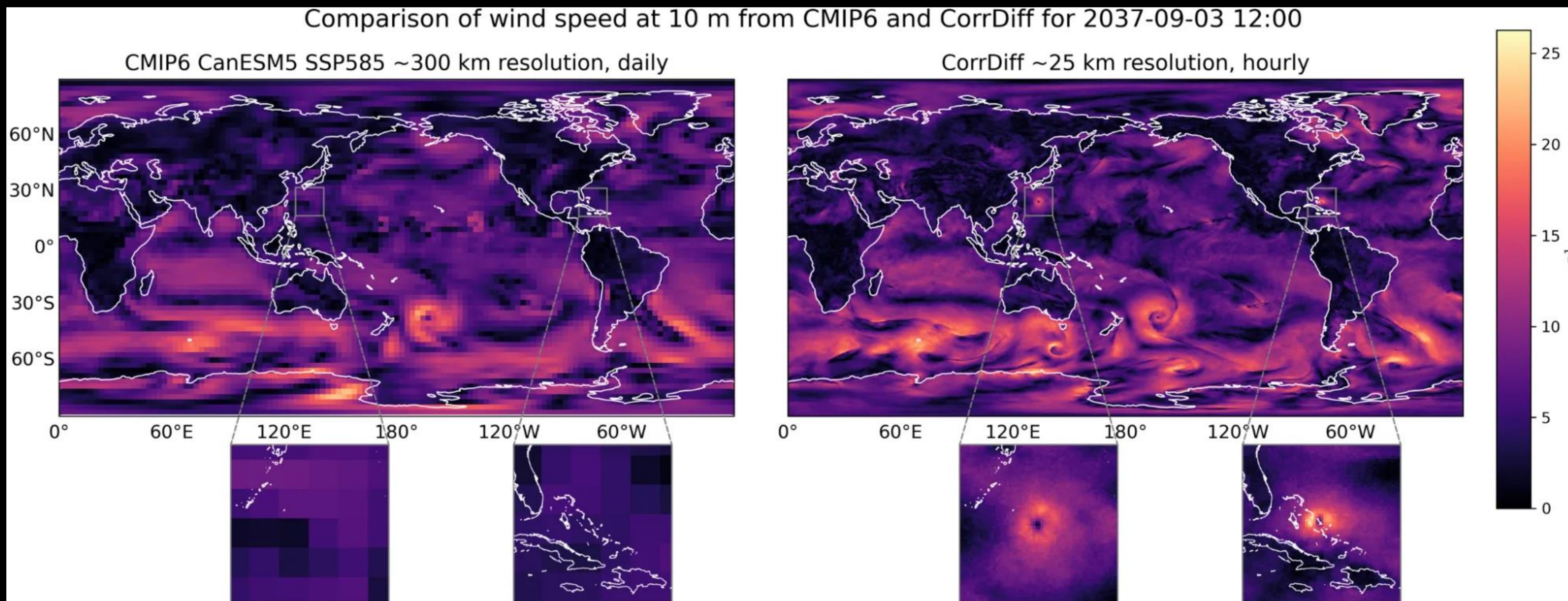
CorrDiff with Patch-Based Multi-Diffusion

Unlocks Scaling to Large Domains Powered by Patch-Based Multi-Diffusion



S&P Global Energy: Use CorrDiff for Climate Downscaling

Unlock Local Detail in Coarse Climate Projections

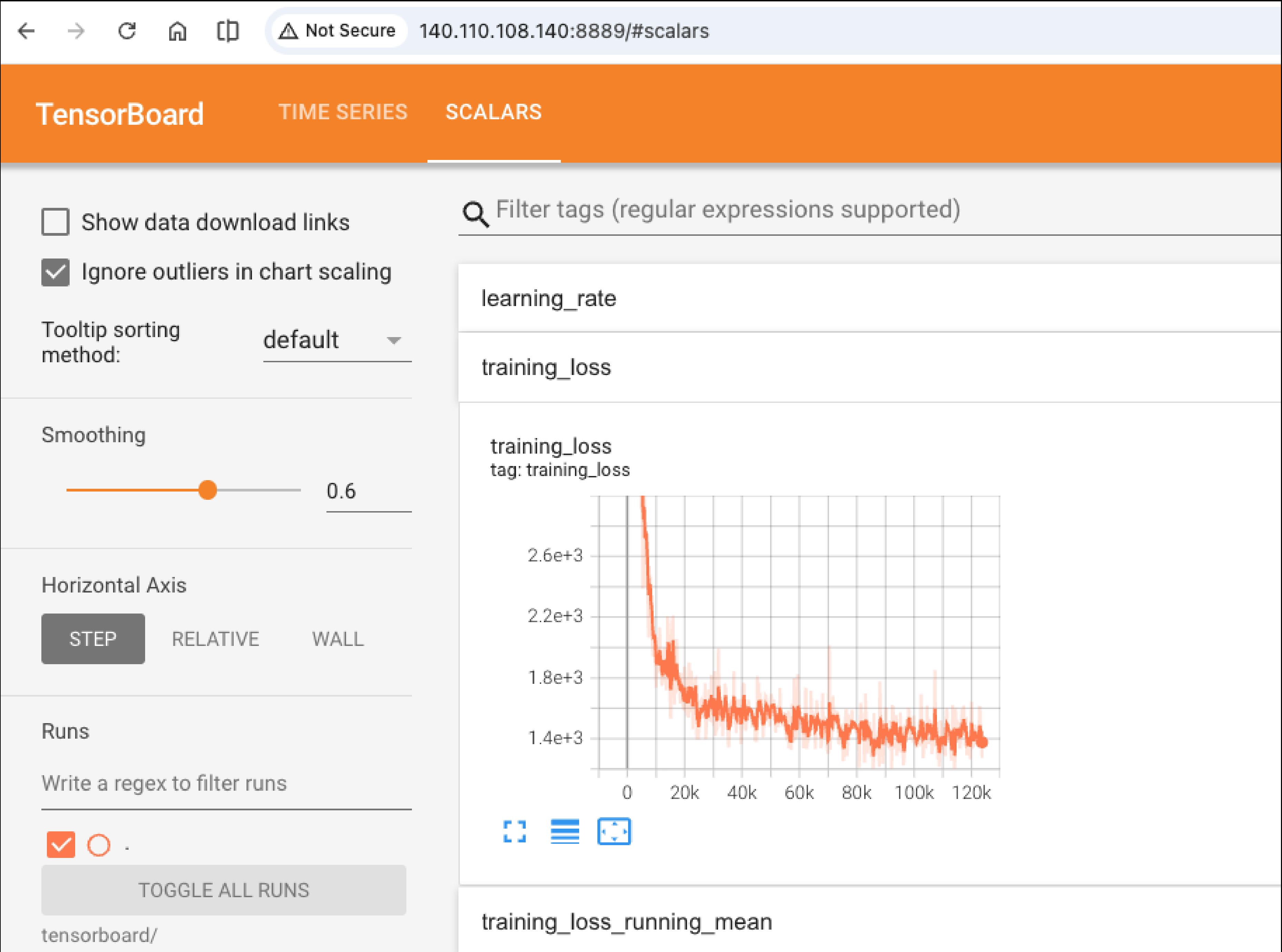


10 m wind speed on Sept. 3, 2037 (SSP585).

The coarse input (left) misses local detail, while AI downscaling to hourly resolution (right) reveals two tropical cyclones

DEMO - Train CorrDiff with PhysicsNeMo

Train CorrDiff on HRRR-Mini dataset



DEMO - Train CorrDiff with PhysicsNeMo

Train CorrDiff on HRRR-Mini dataset

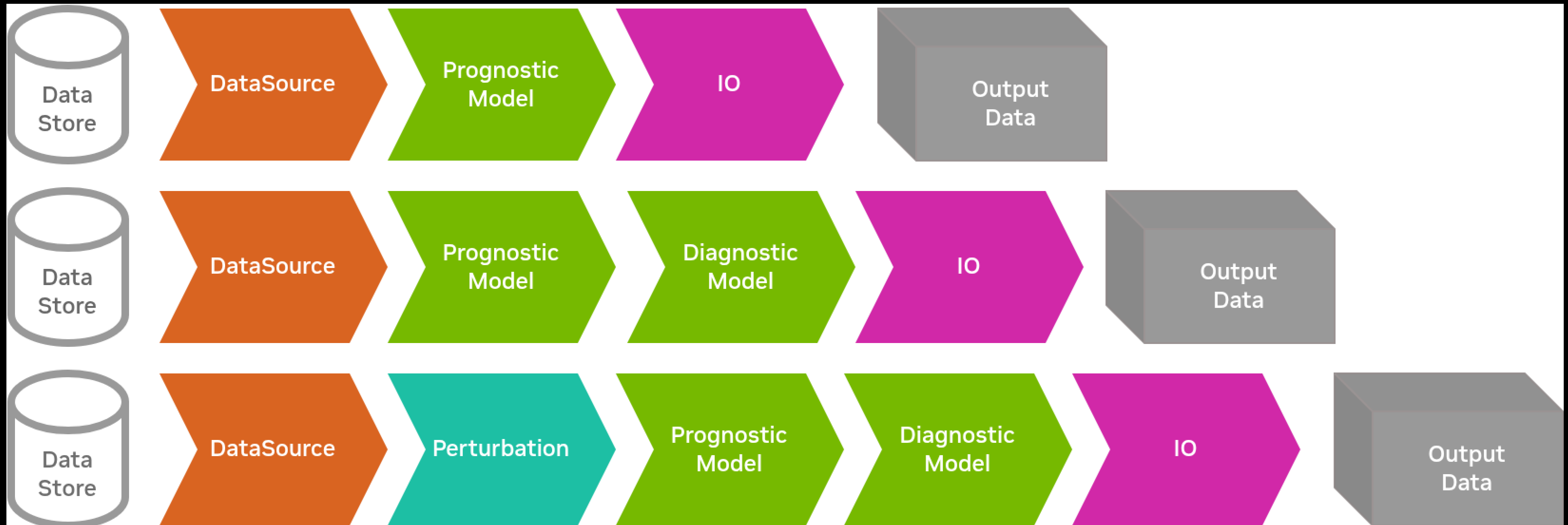
```
cd ~/AI-Powered-Physics-Bootcamp
ngc registry resource download-version "nvidia/modulus/modulus_datasets-hrrr_mini:1"
git clone https://github.com/NVIDIA/physicsnemo
docker run --gpus all -it --rm --net host --shm-size=1g --ulimit memlock=-1 --ulimit stack=67108864 -v $PWD:$PWD -w $PWD
  nvcr.io/nvidia/physicsnemo/physicsnemo:25.11 bash
ln -s $PWD/modulus_datasets-hrrr_mini_v1 physicsnemo/examples/weather/corrdiff/data
cd physicsnemo
pip install .
cd examples/weather/corrdiff/
pip install -r requirements.txt
git config --global --add safe.directory /home/ubuntu/AI-Powered-Physics-Bootcamp/physicsnemo
nohup tensorboard --logdir tensorboard/ --port 8889 &
python train.py --config-name=config_training_hrrr_mini_regression.yaml
```


Earth2Studio

An AI inference pipeline toolkit
<https://github.com/NVIDIA/earth2studio>

Prognostic (Forecasting)	Diagnostics	Datasources	Data Structures
SFNO/HENS (PhysicsNeMo Makani)	Climate in a Bottle (PyTorch)	ARCO ERA5 (GCS)	Zarr 3.0
StormCast (PhysicsNeMo)	CorrDiff (PhysicsNeMo)	GFS / GEFS (AWS, NOMADS)	NetCDF
AIFS (Anemoi)	Precipitation AFNO (PhysicsNeMo)	HRRR (AWS, GCS)	Xarray
GraphCast (JAX)	TC Trackers (RAPIDS)	GOES (AWS)	PyTorch Tensors
Pangu (ONNX)	Solar Radiation AFNO (PhysicsNeMo)	IFS (ECMWF Opendata)	
...	...	...	

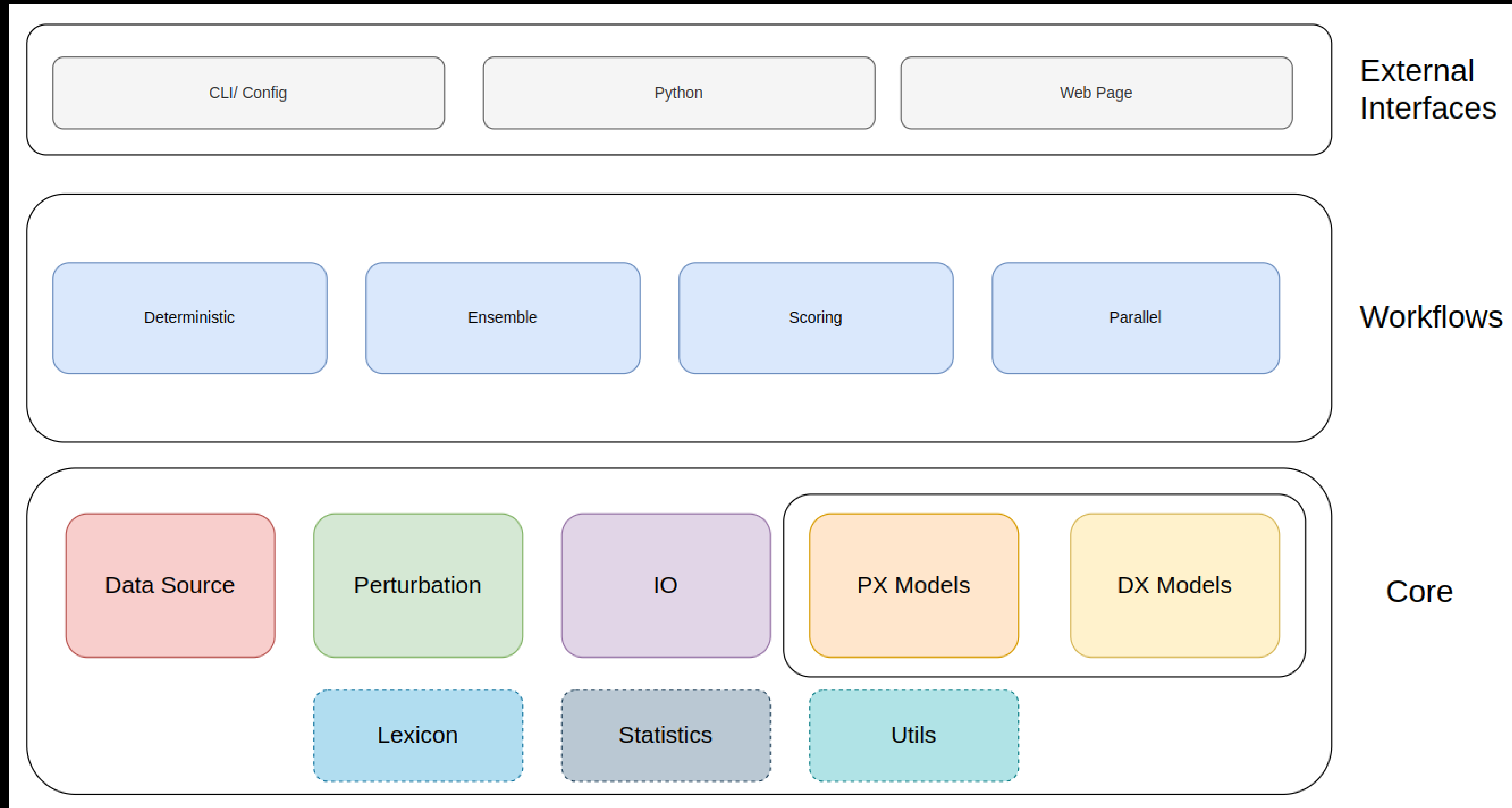
- Earth2Studio is an AI inference pipeline toolkit focused on weather and climate applications that is designed to ride on top of different AI frameworks, model architectures, data sources and SciML tooling while providing a unified API.



- The composability of the different core components in Earth2Studio easily allows the development and deployment of increasingly complex pipelines that may chain multiple data sources, AI models and other modules together.
- The unified ecosystem of Earth2Studio provides users the opportunity to rapidly swap out components for alternatives. In addition to the largest model zoo of weather/climate AI models, Earth2Studio is packed with useful functionality such as optimized data access to cloud data stores, statistical operations and more to accelerate your pipelines.

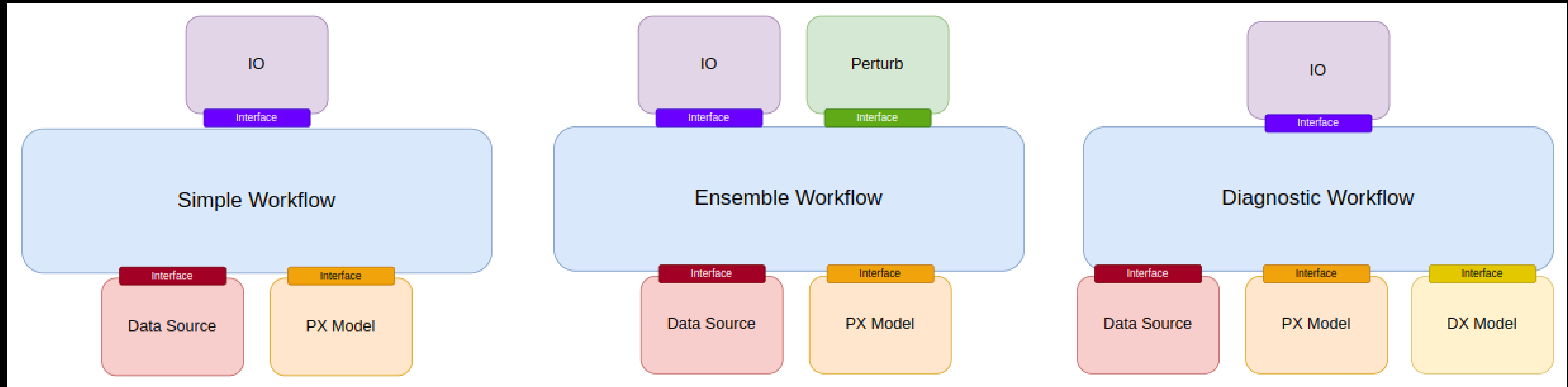
Earth2Studio- Package Design

modular architecture



Earth2Studio- Package Design

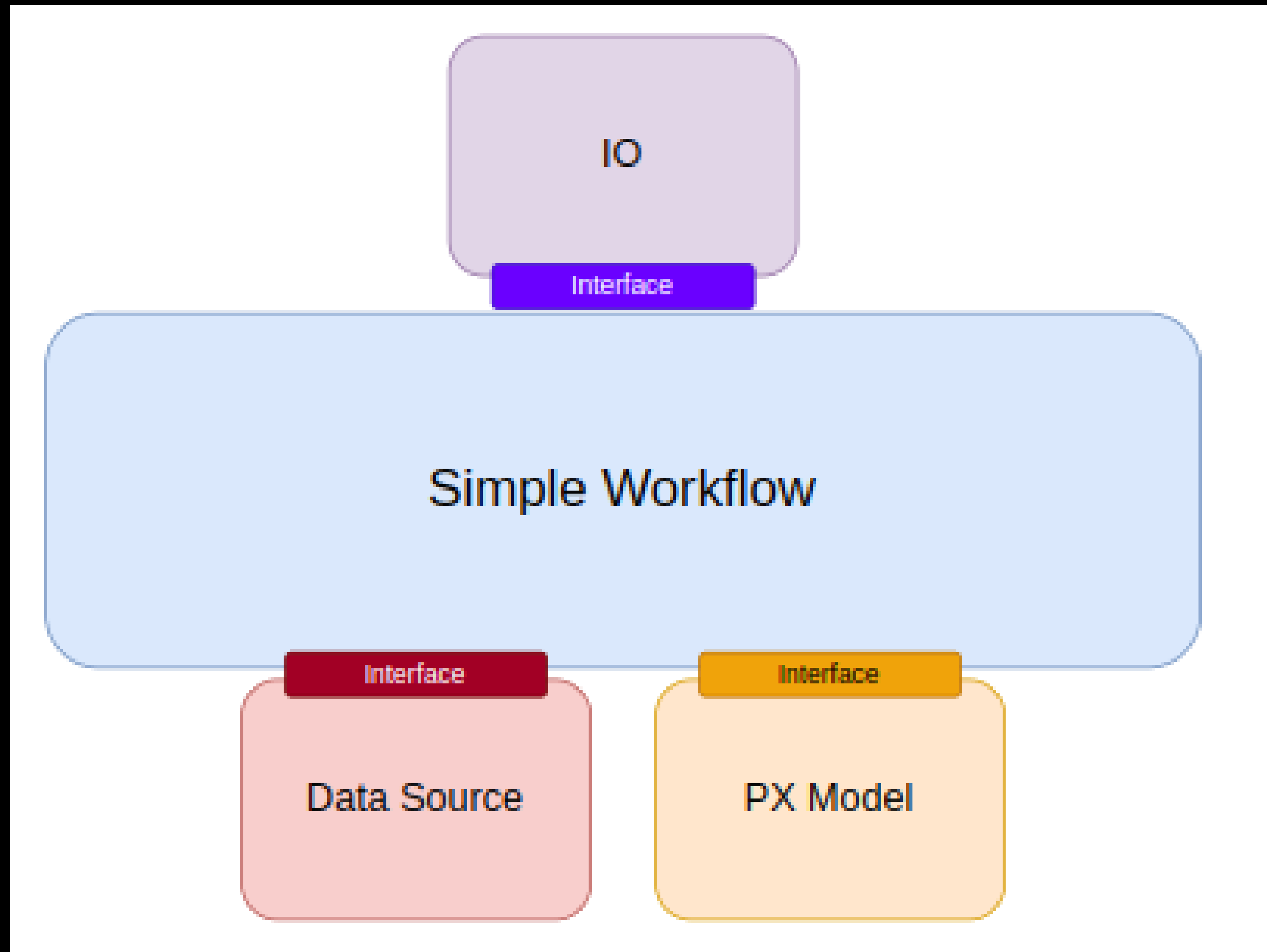
modular architecture



Key Features:

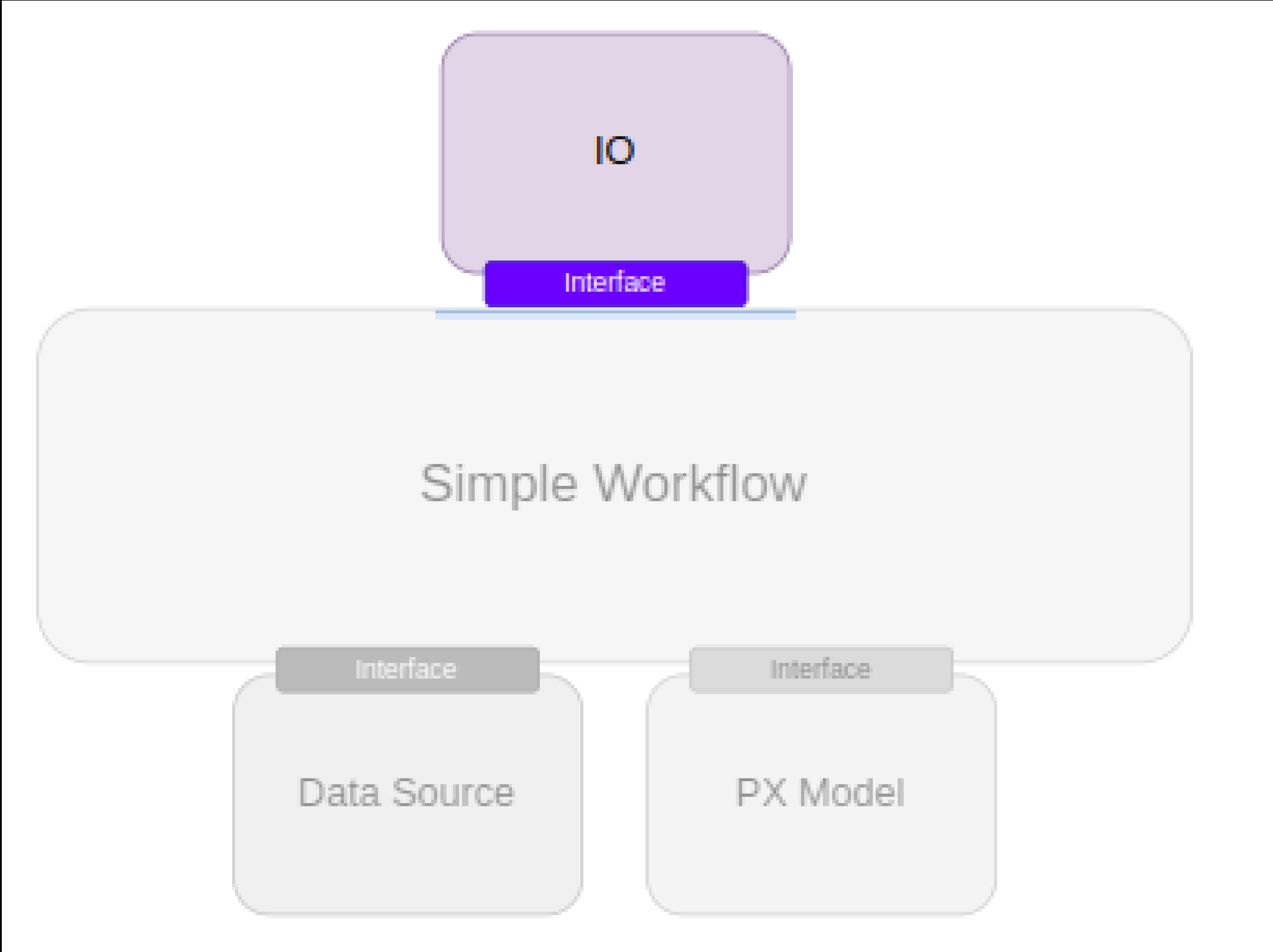
- **Built-in Workflows:** Multiple built-in inference workflows to accelerate your development and research.
- **Prognostic Models:** Support for the latest AI weather forecast models offered under a coherent interface.
- **Diagnostic Models:** Diagnostic models for mapping to other quantities of interest.
- **Datasources:** Datasources to connect on-prem and remote data stores to inference workflows.
- **IO:** Simple, yet powerful IO utilities to export data for post-processing.
- **Statistical Operators:** Statistical methods to fuse directly into your inference workflow for more complex uncertainty analysis.

Earth2Studio - A Simple Workflow



Earth2Studio - A Simple Workflow

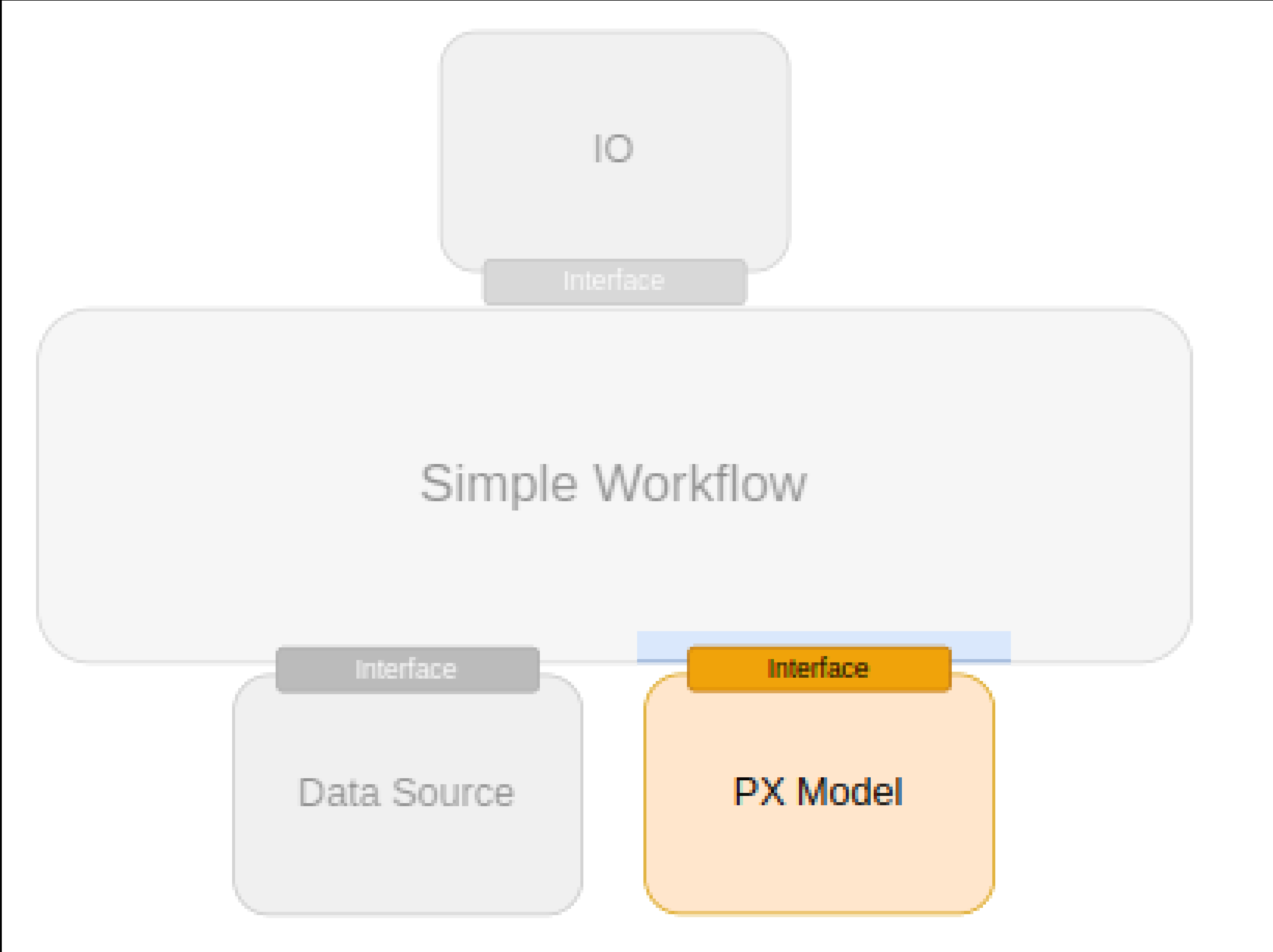
Data Sources



IO Backend	Description
<code>io.ZarrBackend</code>	Zarr format backend (used in this demo)
<code>io.NetCDF4Backend</code>	NetCDF4 format backend
<code>io.XarrayBackend</code>	Xarray backed IO object
<code>io.KVBackend</code>	Key-value (dict) backend

Earth2Studio - A Simple Workflow

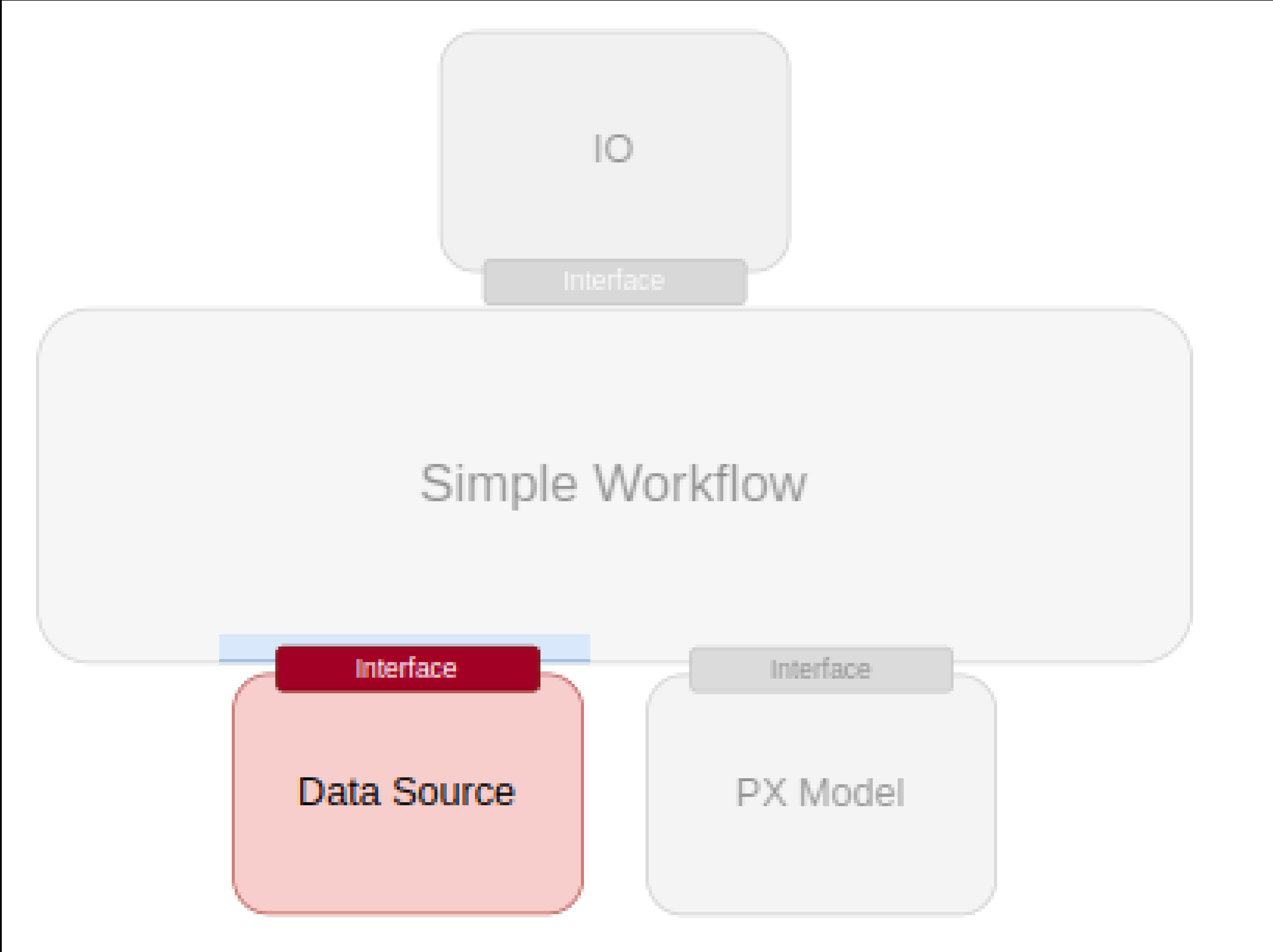
PX Models



Model	Description
<code>models.px.FCN</code>	FourCastNet - AFNO-based global weather forecasting model (used in this demo)
<code>models.px.SFNO</code>	Spherical Fourier Operator Network global prognostic model
<code>models.px.Pangu24</code>	Pangu Weather 24 hour model
<code>models.px.FuXi</code>	FuXi weather model with three auto-regressive U-net transformer models
<code>models.px.Aurora</code>	Aurora transformer-based weather model

Earth2Studio - A Simple Workflow

Data Sources



Data Source	Description
data.GFS	Global Forecast System initial state data source (used in this demo)
data.ARCO	Analysis-Ready, Cloud Optimized ERA5 re-analysis data curated by Google
data.CDS	Climate Data Store serving ERA5 re-analysis data
data.HRRR	High-Resolution Rapid Refresh North-American weather forecast model
data.IFS	Integrated Forecast System initial state data source

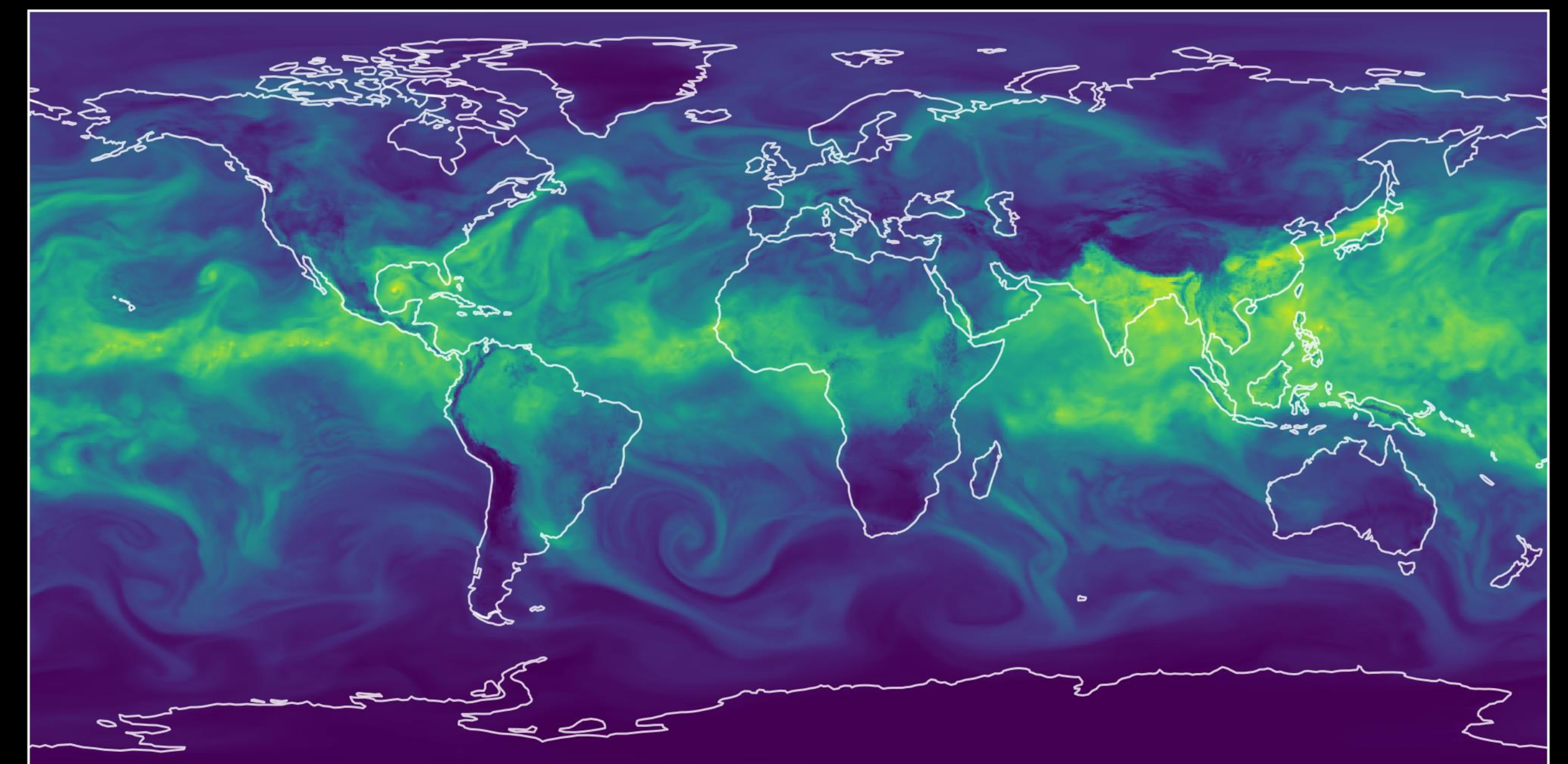
ERA5 - ECMWF Reanalysis v5

one of the most popular data source



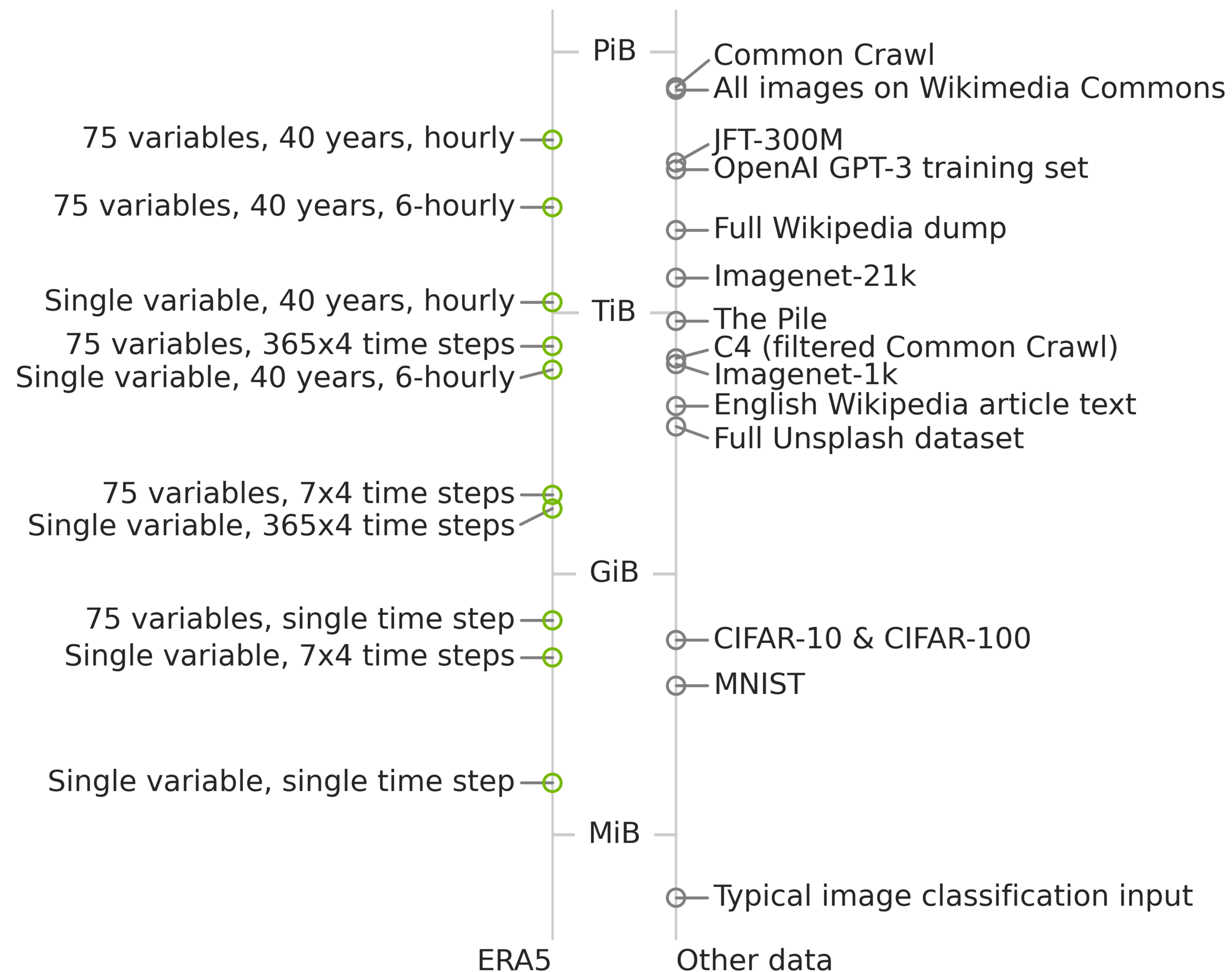
- Global reanalysis produced by ECMWF, distributed by the Copernicus Climate Service (C3S)
- **Hourly estimates** of atmospheric, land, and oceanic variables **going back to 1940**
 - Atmospheric variables on 37 pressure levels interpolated from 137 model levels
 - Land, oceanic, and additional atmospheric variables on single levels
 - Also provides three-hourly uncertainty estimates and aggregate statistics
- Available on interpolated **0.25° grid** (ca. 28 km at equator) from the Copernicus Data Store
 - Calculated on irregular grid (spectral coefficients or reduced Gaussian grid)
 - Interpolation provides rectangular grid with 721x1440 equally-spaced grid points
- Updated daily, with lag time of ca. 5 days
 - Based on fixed version of the ECMWF IFS data assimilation algorithm
 - Final release after data verification with lag time of ca. 2-3 months
- Important asset for **training AI weather models**

ERA5 total column water vapor
at one point in time



Weather Data is Big

Subsets of ERA5 compared to other data



Specialized Data Formats Make Weather Data Manageable

Handling large, multi-dimensional arrays requires targeted solutions

Zarr



- Format for storing chunked, compressed, multi-dimensional, typed arrays
- Optimized for parallel computing and distributed systems
- Hierarchical data organization via groups divided into individually compressed chunks
- Extension: .zarr
- Examples: WeatherBench 2

NetCDF4

Network Common Data Form v4



- Suite of software libraries and data formats for multi-dimensional data
- Hierarchical format implemented on top of HDF5 comprising groups, datasets, and attributes
- Tight integration of Climate and Forecast (CF) metadata conventions
- Extensions: .nc, .cdf
- Examples: GFS, CMIP6

GRIB

General Regularly-distributed Information in Binary form

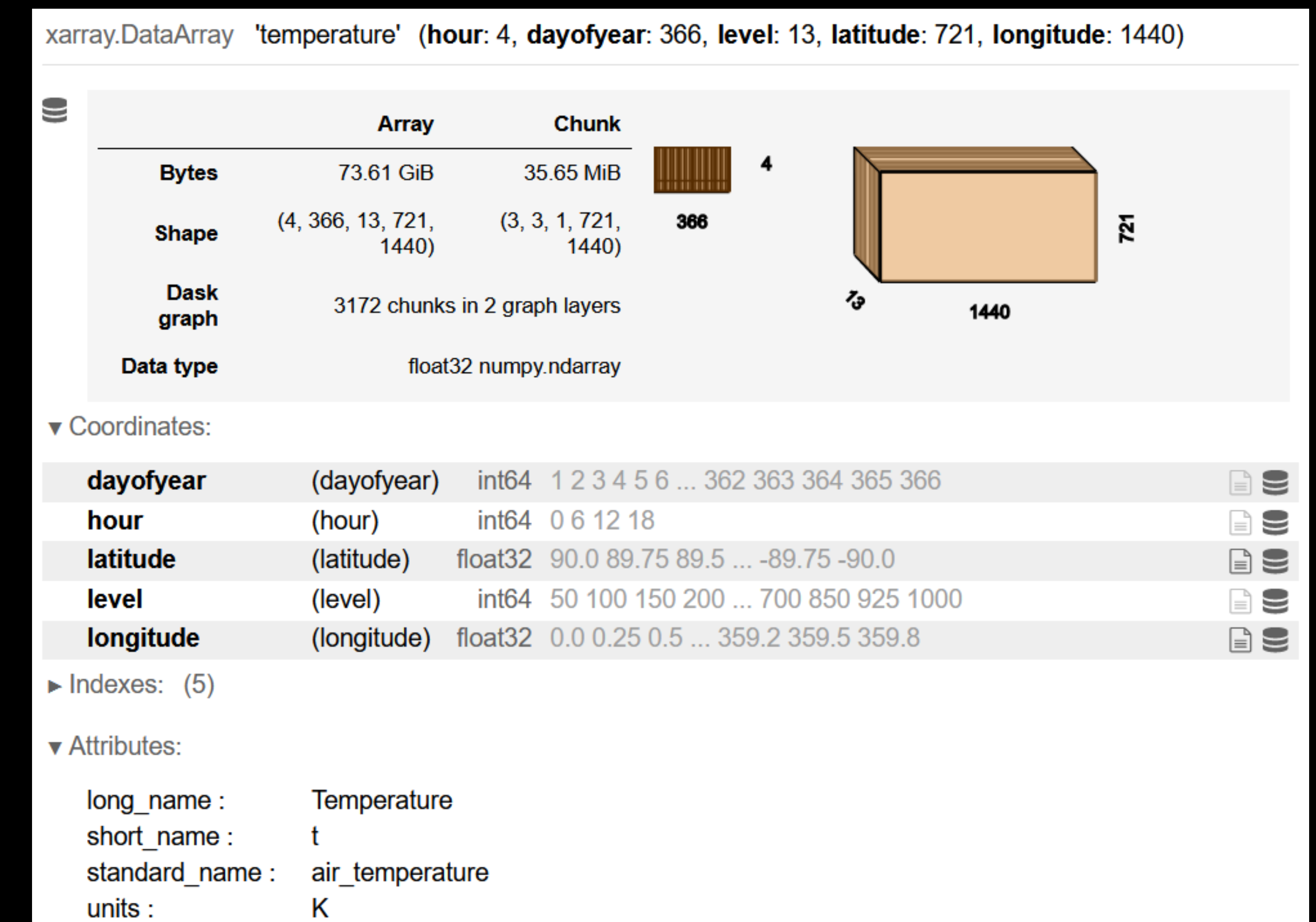


- Format for storing meteorological data
- Collection of independent, self-contained messages (2D data records)
- Each message with metadata, grid definition, and binary data
- Extensions: .grib, .grb, .gb
- Examples: ERA5, IFS

Xarray Handles Large Arrays

Navigate multiple dimensions without getting lost

- Python library for handling **multi-dimensional, labeled arrays**
- Built on top of numpy/pandas with powerful extensions for named dimensions
- Multiple variables can be aligned into coherent datasets
- Reads and writes **multiple file formats** like Zarr, NetCDF4/HDF, and GRIB
- Keeps track of metadata and attributes



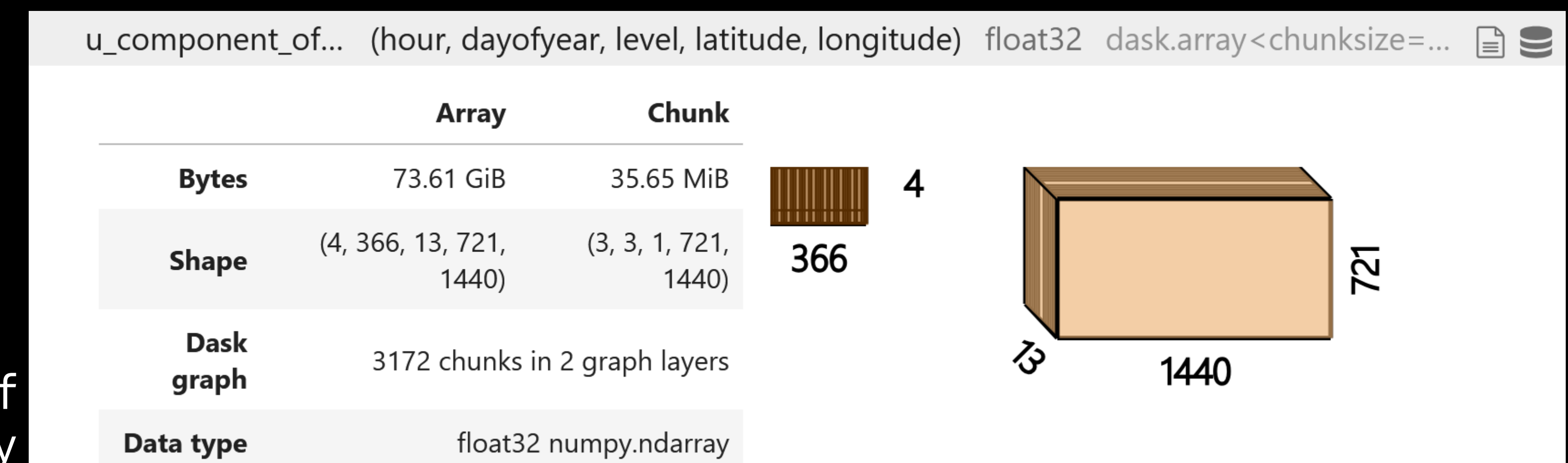
Interactive visualization of
temperature climatology

Data Chunking

Individual data elements can be accessed and transferred easier

- **Dividing large datasets** into chunks can make their handling easier
 - Transfer of compressed chunks instead of the entire dataset
 - Concurrent reads from and writes to independent chunks
- Optimal chunk structure depends on the **data access pattern**
 - Looking at small region over long period of time favors small chunks in space, large chunks in time
 - Looking at the entire globe for individual points in time favors large chunks in space, small chunks in time
- Example: WeatherBench 2 climatology data
 - Uses large chunks in space, small chunks in time, and individual chunks per variable
 - Individual chunks are ca. 36 MiB with shape (3, 3, 1, 721, 1440) for (hour, day of year, pressure level, latitude, longitude)

WeatherBench 2 chunking of
u10m as shown by xarray



Earth2Studio Makes Data Management Easy

Different data sources and unified IO backends are available

Data sources

earth2studio.data

- Simple loading of remote data sources
- ERA5, GFS, IFS, ...
- Access by date/time and variable
- Automatic caching
- Consistent alignment of output structure

```
import numpy as np
from earth2studio.data import GFS

source = GFS()
data = source(
    np.datetime64("2020-01-01").reshape(1),
    ["t2m", "tcwv", "u10m", "v10m"],
)
```

IO backends

earth2studio.io

- Backend-agnostic input/output operations
- Zarr, NetCDF4, Xarray, etc.
- Aligned interface for downstream applications like inference and evaluation
- Automatic tracking of coordinate metadata

```
from earth2studio.io import ZarrBackend

io = ZarrBackend("output.zarr")
# Pass 'io' to downstream processes
```


Earth2Studio - A Simple Workflow

Only 5 Lines of code to run

```
import os
import earth2studio.run as run
from earth2studio.data import GFS
from earth2studio.io import ZarrBackend
from earth2studio.models.px import FCN

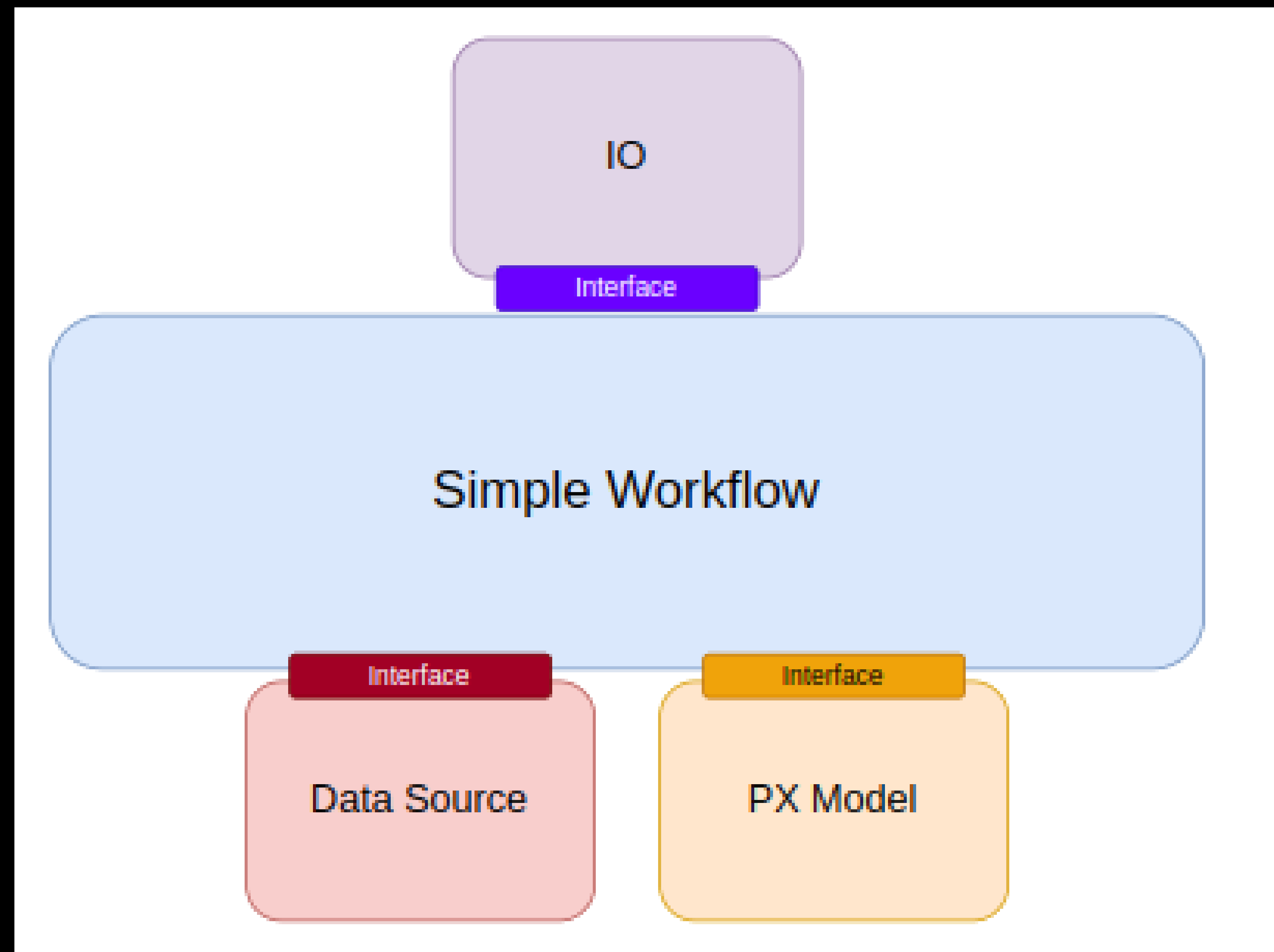
# Set cache directory
os.environ['EARTH2STUDIO_CACHE'] = os.getcwd() + "/outputs/cache"

# Prognostic Model - Load from NGC
(ngc://models/nvidia/modulus/modulus_fcn@v0.2)
package = FCN.load_default_package()
model = FCN.load_model(package)

# Data Source - Create the data source
data = GFS()

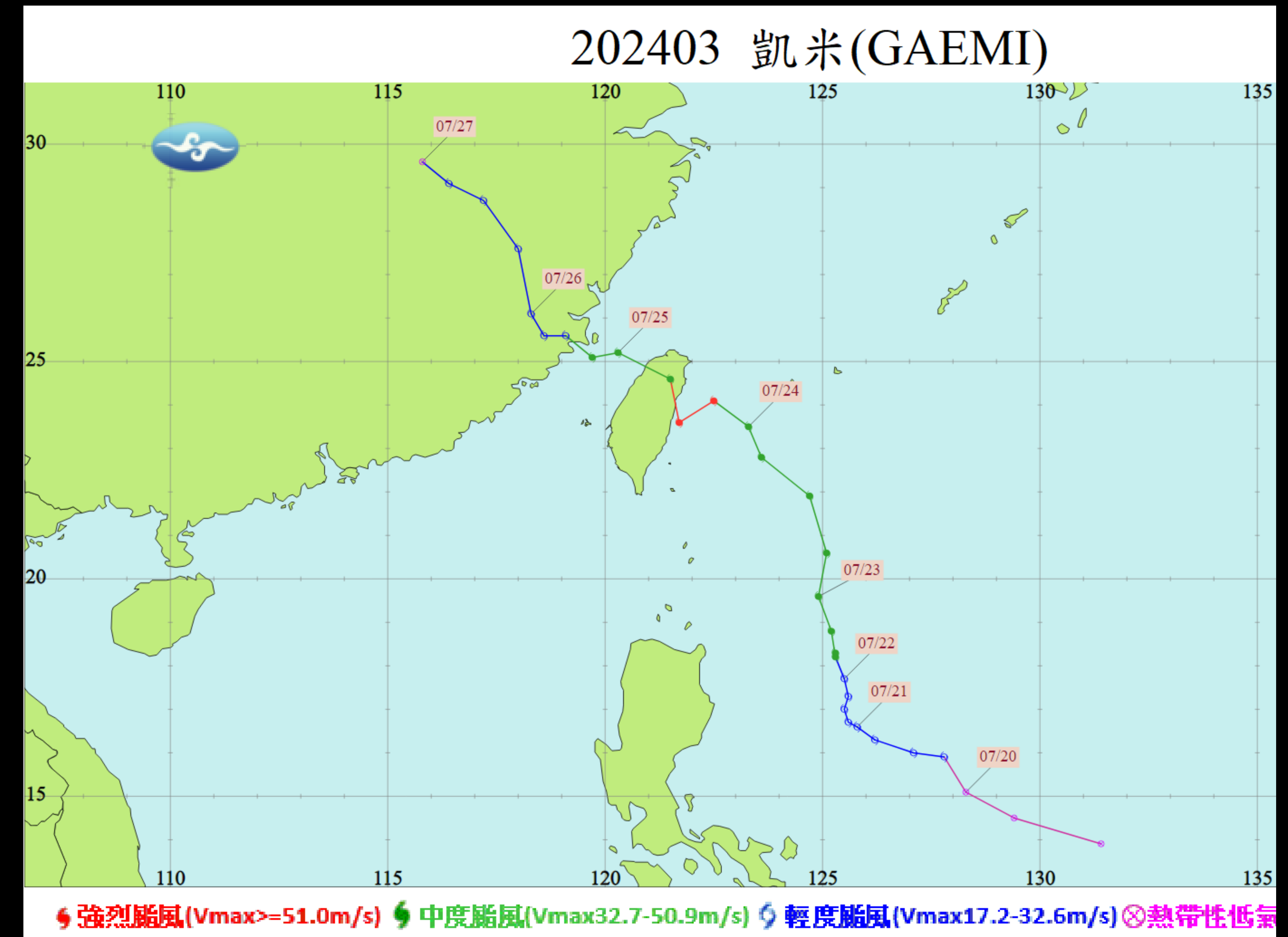
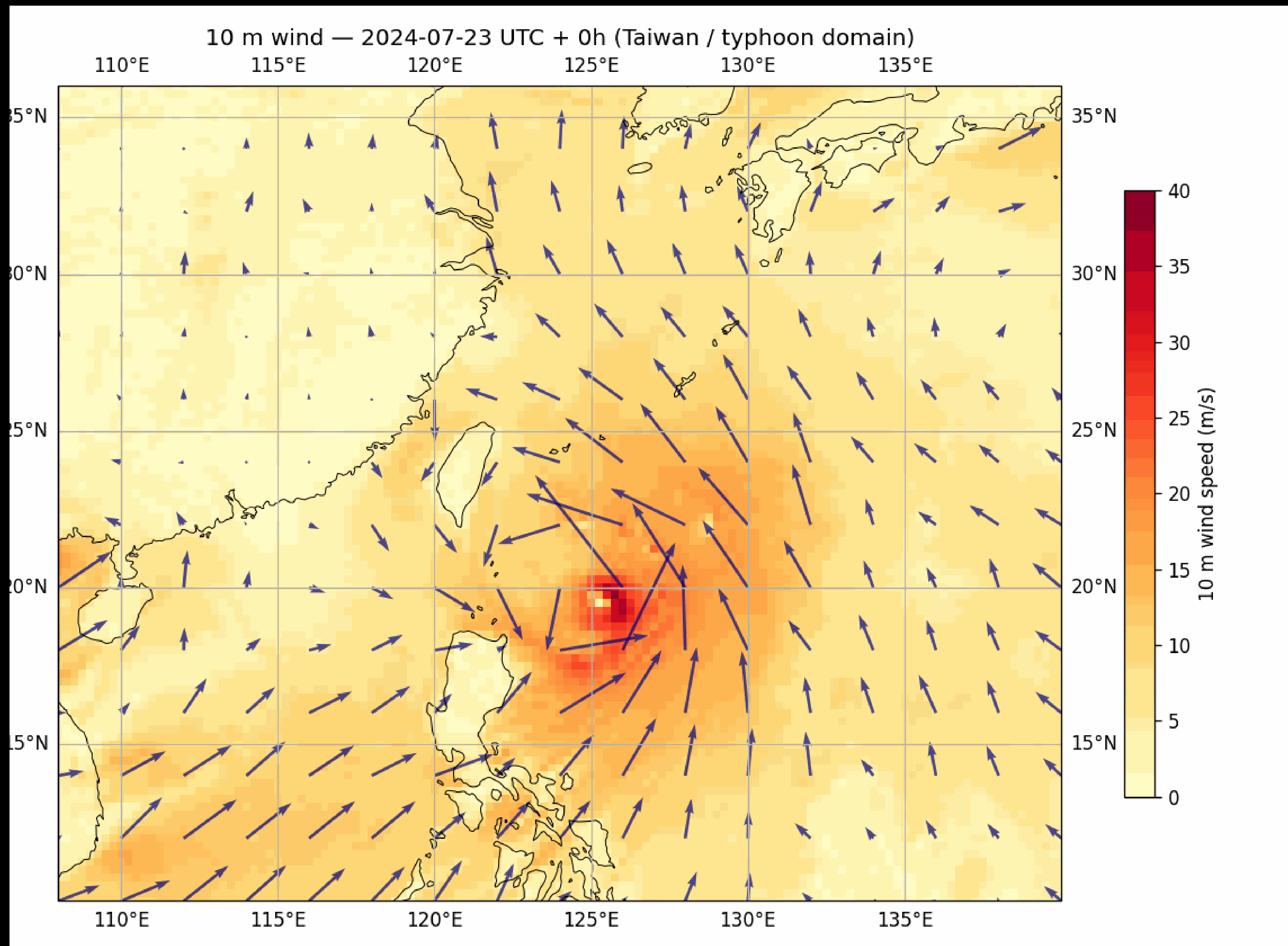
# IO Backend - Create the IO handler
io = ZarrBackend()

nsteps = 4 # 4 steps = 24 hours
io = run.deterministic(["2024-01-01"], nsteps, model, data, io)
```



DEMO – Predict Typhoon with Earth2Studio

Apply Simple Workflow predict Typhoon GAEMI wind



A prompt to cursor:
refer to @examples/01_deterministic_workflow.py , forecast 72 hours u and v based on 2024/07/23, and focus on Taiwan region but big enough to monitor a typhoon track behavior and save as a gif file.

DEMO – Predict Typhoon with Earth2Studio

Apply Simple Workflow predict Typhoon GAEMI wind (u10m)

```
cd ~
```

```
sudo apt-get install python3.12-dev ninja-build nvidia-cuda-toolkit cmake -y
```

```
git clone https://github.com/NVIDIA/earth2studio
```

```
curl -Lsf https://astral.sh/uv/install.sh | sh
```

```
source $HOME/.local/bin/env
```

```
# setup developer Env, reference: https://nvidia.github.io/earth2studio/userguide/developer/overview.html
```

```
cd earth2studio/
```

```
uv venv --python=3.12
```

```
uv sync
```

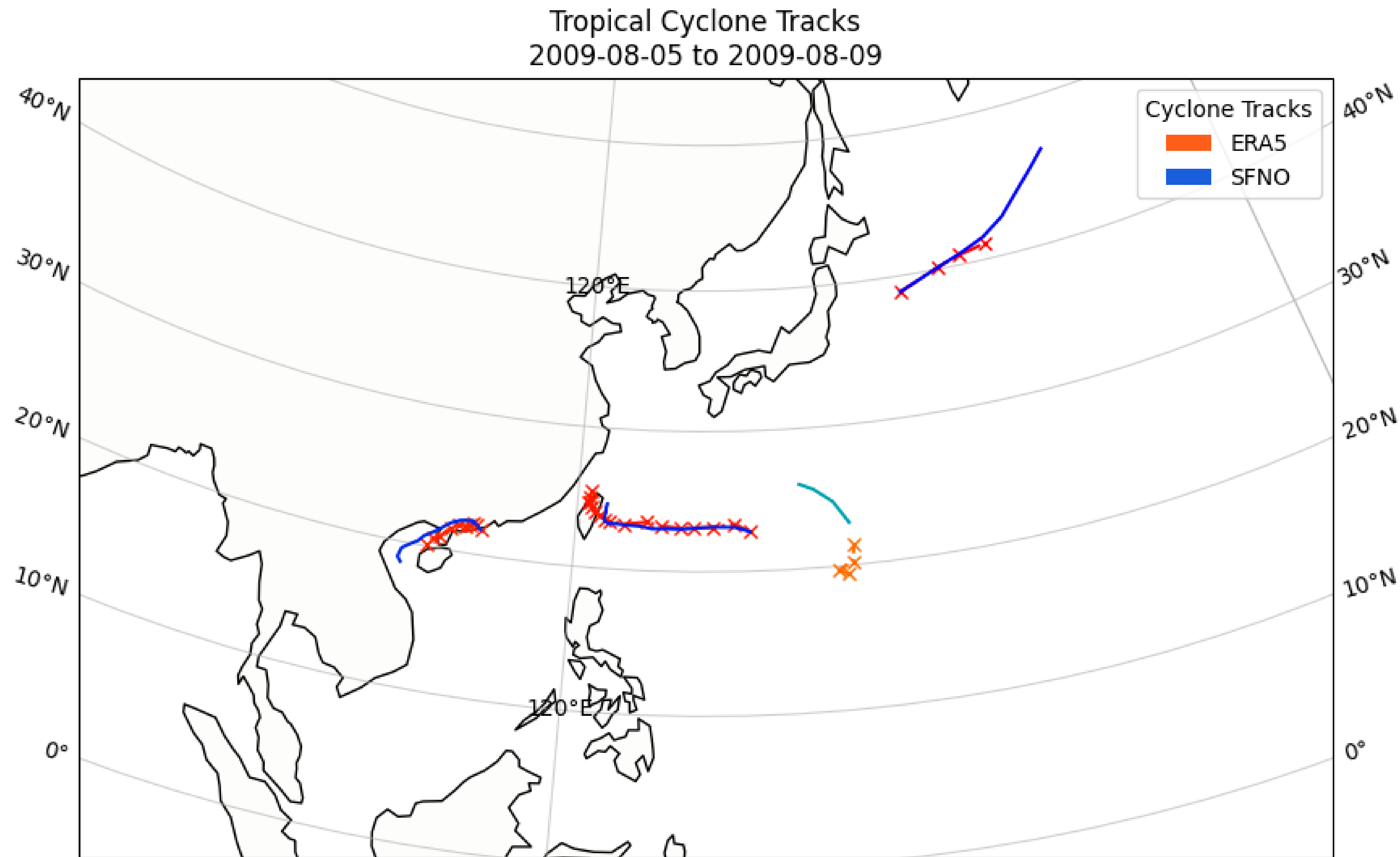
```
uv add cartopy
```

```
uv sync --extra fcn
```

```
uv run python examples/01_deterministic_workflow.py
```

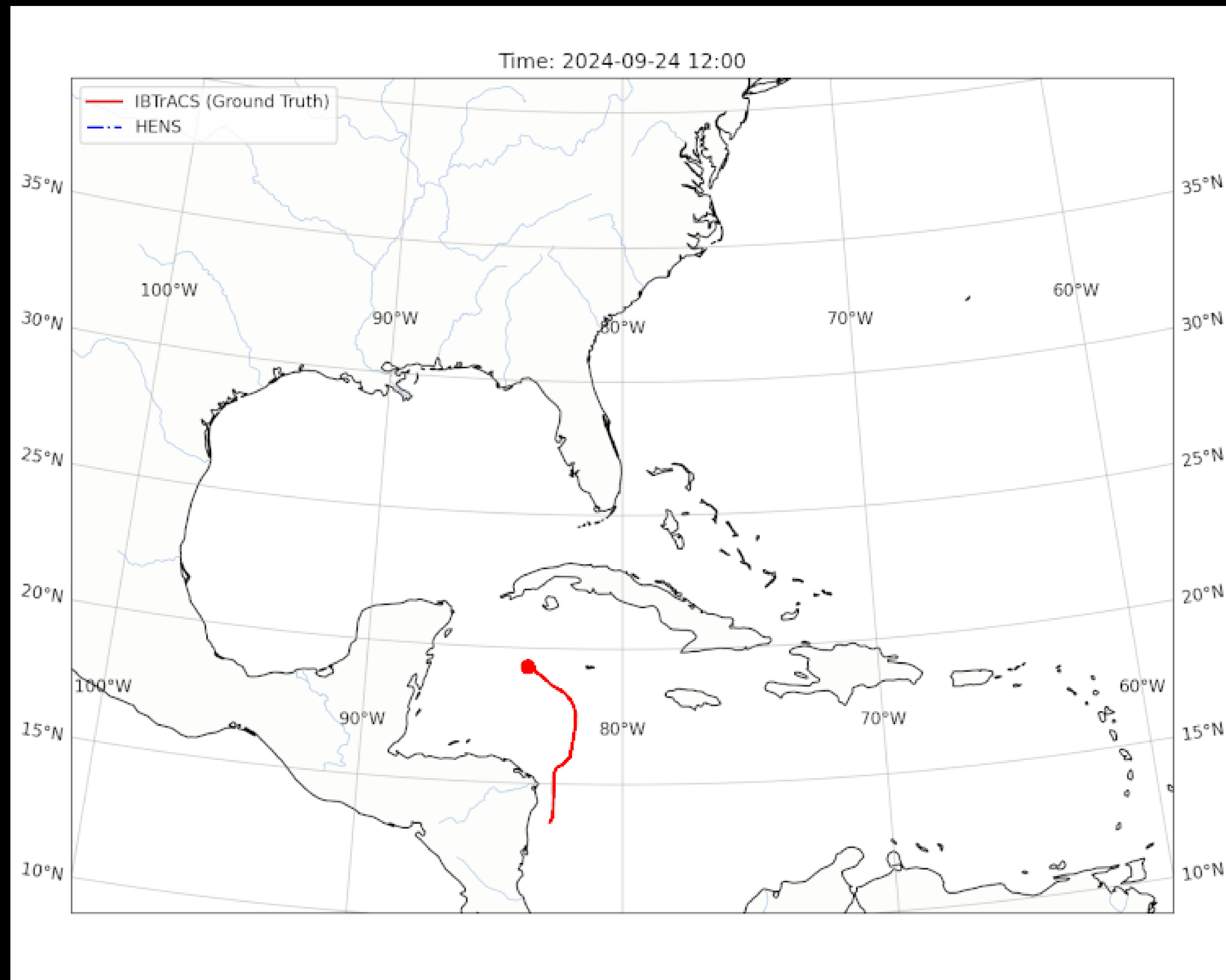

An Example: Tropical Cyclone Tracking

https://nvidia.github.io/earth2studio/examples/13_cyclone_tracking.html



An Example: Huge Ensembles (HENS) Recipe

<https://github.com/NVIDIA/earth2studio/tree/main/recipes/hens>



[Reference]

- Huge Ensembles Part I: Design of Ensemble Weather Forecasts using Spherical Fourier Neural Operators
<https://arxiv.org/pdf/2408.03100>
- Huge Ensembles Part II: Properties of a Huge Ensemble of Hindcasts Generated with Spherical Fourier Neural Operators
<https://arxiv.org/abs/2408.01581>

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³NVIDIA Corporation, Santa Clara, California, USA

⁴National Energy Research Scientific Computing Center (NERSC), LBNL, Berkeley, California, USA

⁵Department of Earth and Atmospheric Sciences, Indiana University, Bloomington, Indiana, USA

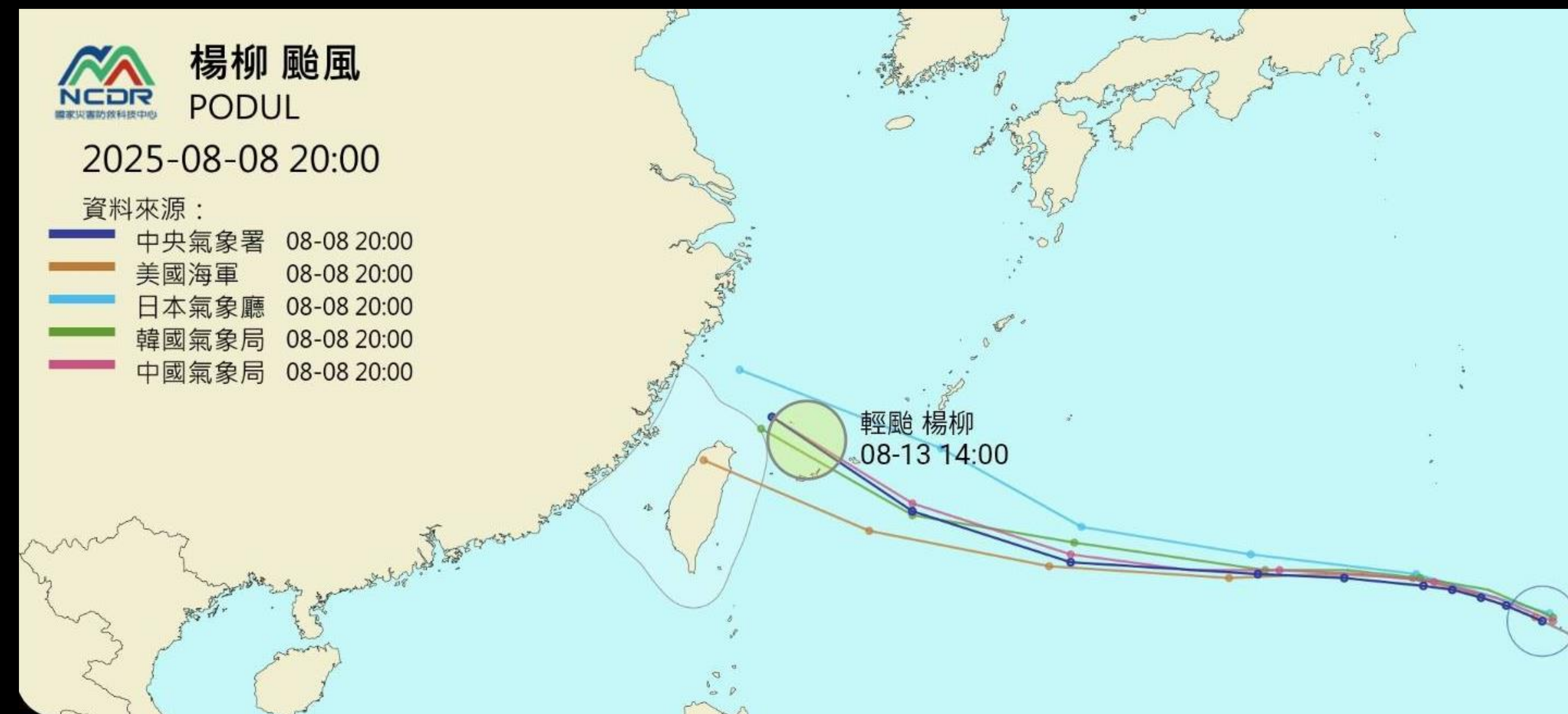
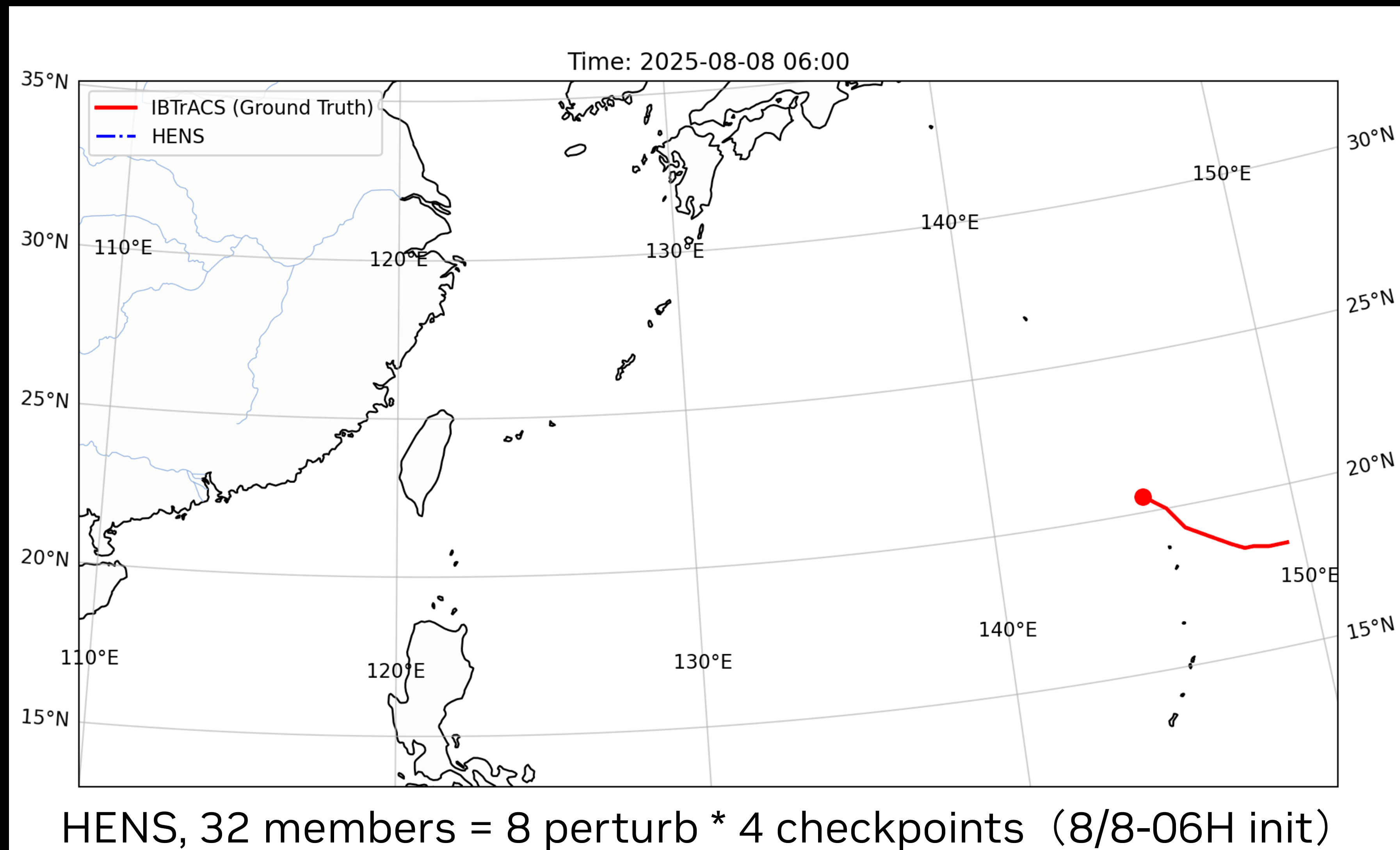
⁶Department of Earth System Science, University of California, Irvine, USA

*These authors contributed equally to this work.

Correspondence: Ankur Mahesh (ankur.mahesh@berkeley.edu)

An Example: Run Earth2Studio HENS on 2025/08 Typhoon Podul

Generate 32 members of 5-days Forecast in 10 minutes with single GPU



More Examples

<https://nvidia.github.io/earth2studio/examples/index.html>

Earth2Studio

main

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Section Navigation

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Huge Ensembles (HENS) Checkpoints

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Tropical Cyclone Tracking

Running DLESyM Inference

CBottle Data Generation and Infilling

CBottle Super Resolution

IO Backend Performance

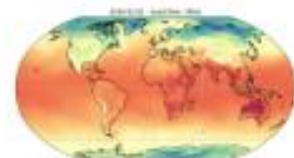
Ensemble Forecasting with Downscaling

Creating a Local Data Source

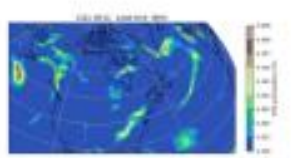
StormScope Satellite and Radar Nowcasting

Extending Earth2Studio

Running Examples



Running Deterministic Inference



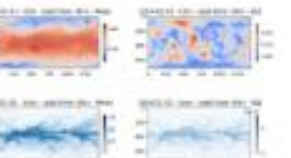
Running Diagnostic Inference



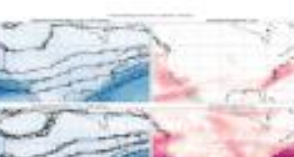
Running Ensemble Inference



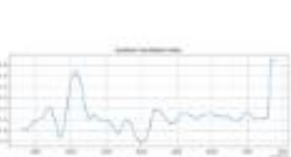
Generative Downscaling



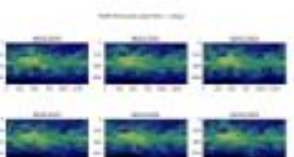
Single Variable Perturbation Method



Model Hook Injection: Perturbation



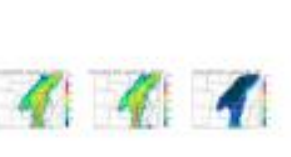
Statistical Inference



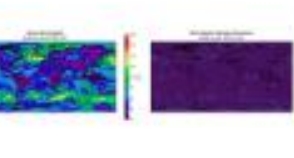
Distributed Manager Inference



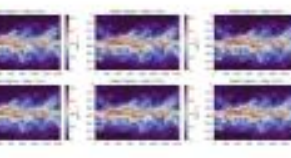
Running StormCast Inference



Running StormCast Ensemble Inference



Huge Ensembles (HENS) Checkpoints



Temporal Interpolation



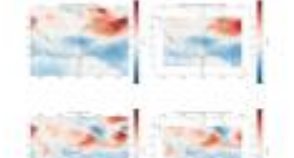
Tropical Cyclone Tracking



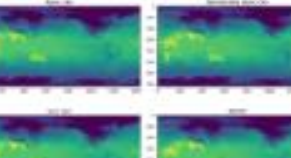
Running DLESyM Inference



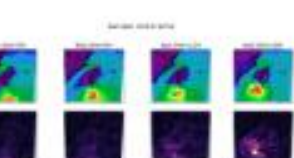
CBottle Data Generation and Infilling



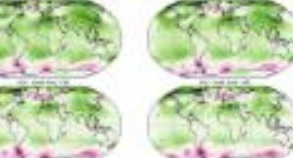
CBottle Super Resolution



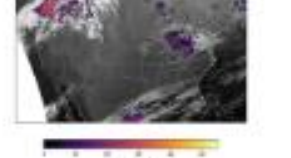
IO Backend Performance



Ensemble Forecasting with Downscaling



Creating a Local Data Source



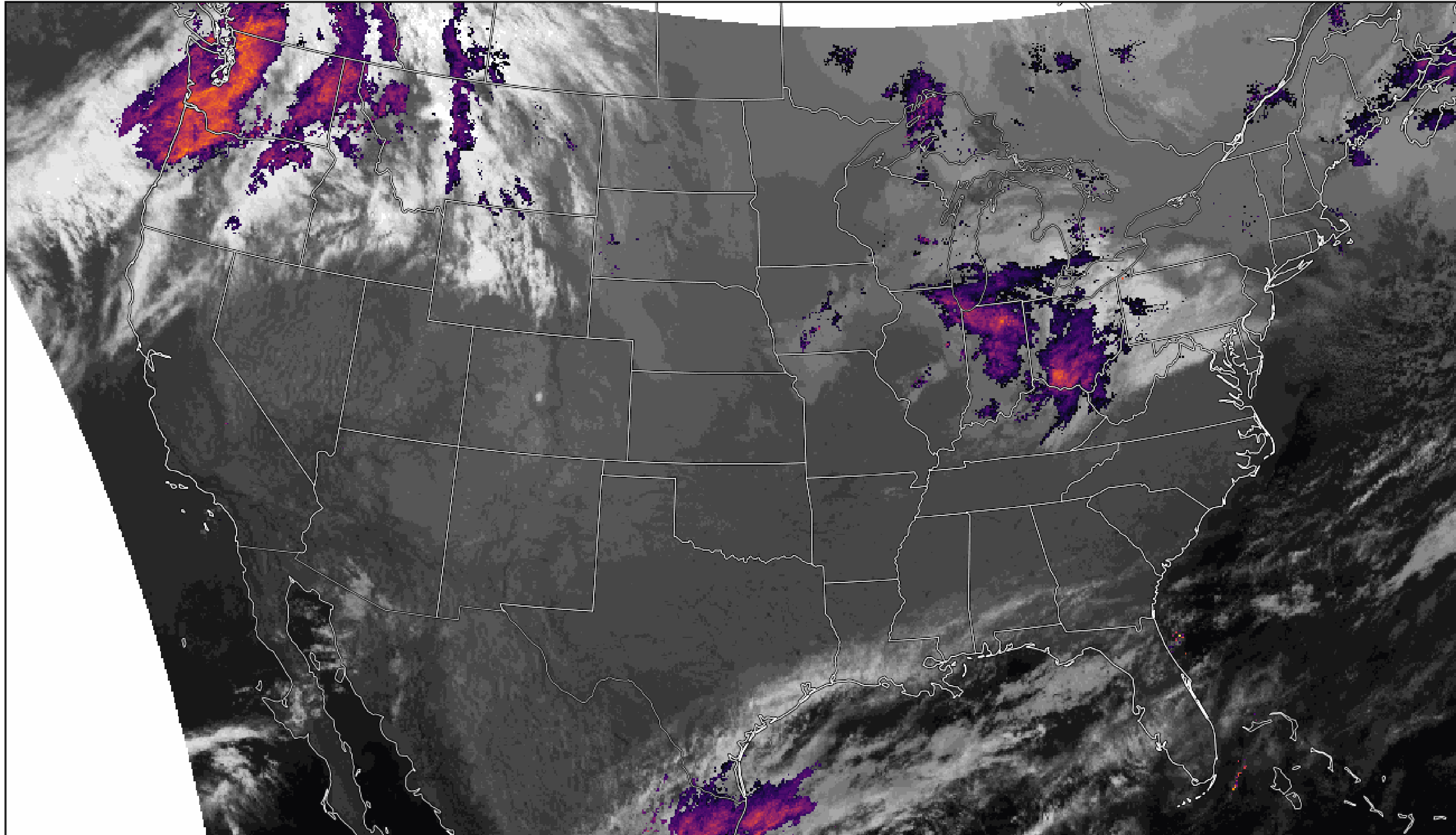
StormScope Satellite and Radar Nowcasting



DEMO – Inference StormScope with Earth2Studio

Use hourly model

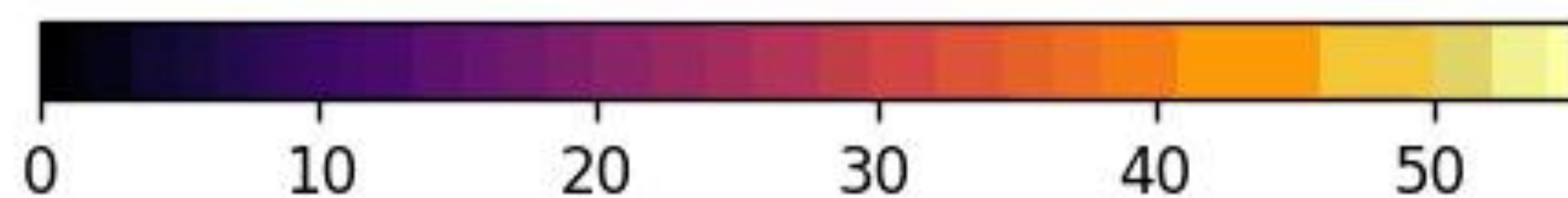
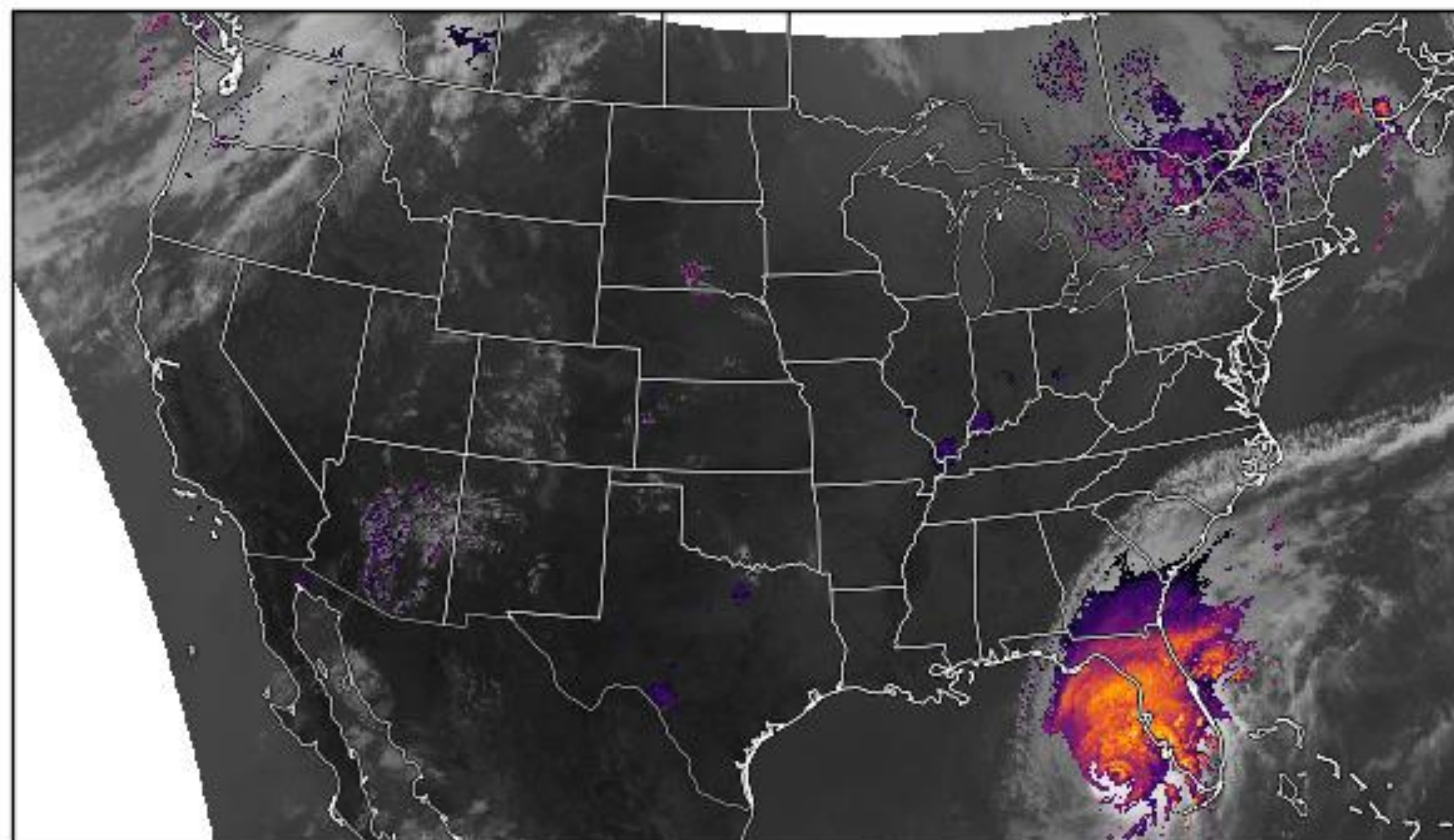
Predicted GOES abi13c with MRMS overlay from 2023-12-05 12:00:00 UTC (lead 1:00:00 min) — frame 1/20



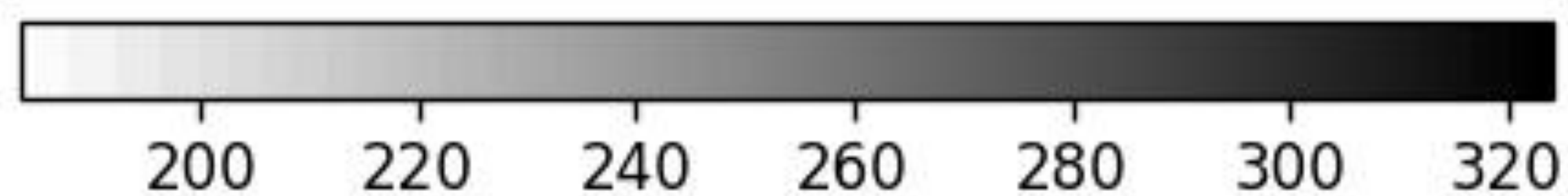
DEMO – Inference StormScope with Earth2Studio

A Prompt: refer to [@earth2studio/examples/20_stormscope_goes_example.py](#), forecast 2 hours nowcasting on 2024 Hurricane Milton with 10 minutes temporal resolution and save animation as a fig file.

2024 Hurricane Milton nowcast — 2024-10-09T18:00 UTC (lead +0 min)



MRMS Reflectivity [dBZ]



GOES Clean IR 10.35um [K]

DEMO – Inference StormScope with Earth2Studio

Train CorrDiff on HRRR-Mini dataset

```
cd ~
```

```
sudo apt-get install python3.12-dev ninja-build nvidia-cuda-toolkit cmake -y
```

```
git clone https://github.com/NVIDIA/earth2studio
```

```
curl -Lsf https://astral.sh/uv/install.sh | sh
```

```
source $HOME/.local/bin/env
```

```
# might take > 20 minutes to build lib earth2grid
```

```
cd earth2studio/
```

```
uv venv --python=3.12
```

```
uv sync
```

```
uv add cartopy
```

```
uv sync --extra stormscope
```

```
uv run python examples/20_stormscope_goes_example.py
```




Thank You!